

8-1.框图

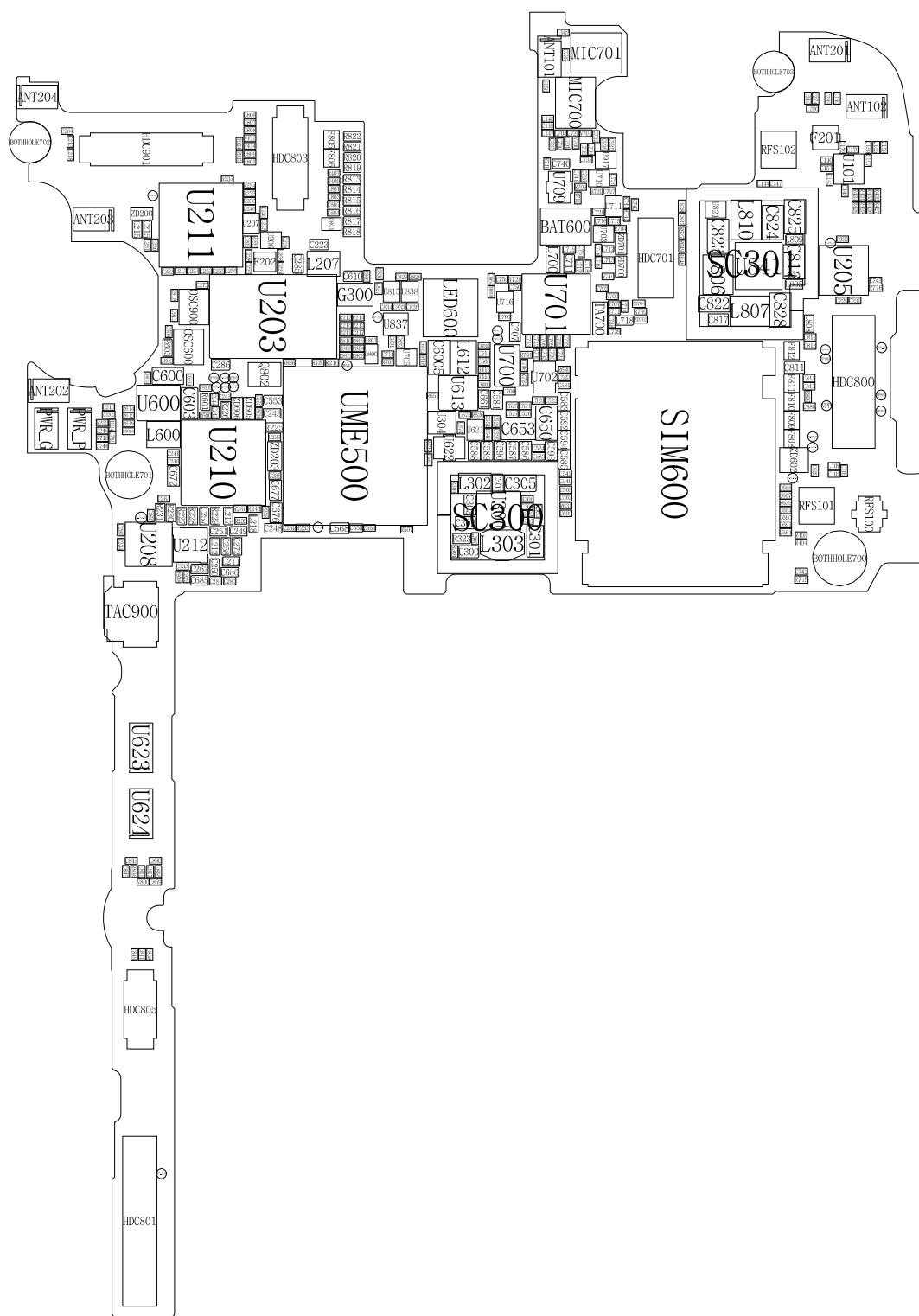
The diagram illustrates the system architecture of the MSM8974. The central component is the **CP + AP MSM8974**, which is connected to various peripherals:

- Input/Output and Storage:** Includes a 16MHz Crystal, Wacom Digitizer Controller, OCTA (display), VT CAM, Mega CAM, NAND FLASH, T-FLASH, and SIM.
- Audio and Sensors:** Features a 32.768KHz Crystal, Micro Controller, Audio Processor [U704], Audio Codec WCD9320, and various sensors (6 Axis, Magnetic, Barometer, MAIN MIC, SUB MIC, 3rd MIC, EAR MIC, RCV, SPK, EAR).
- Power Management:** Includes PMIC PM8841, PMIC PM8941, and IF PMIC MAX77804, which is connected to a FLASH LED and a motor (M).
- Communication:** Supports multiple wireless standards:
 - Cellular:** LTE B1/21 SUB ANT, WB1 SUB ANT, GPS ANT, LTE B3/19 SUB ANT, WB6/19 SUB ANT, and LTE B1/3/19/21 WB1/5/6/19 2G ANT.
 - Local Wireless:** BT/WiFi ANT, ISDB-Tmm ANT, and Felica/NFC ANT.
- Other Components:** Includes a 27.12MHz Crystal, Felica/NFC IC, ISDB-Tmm Tuner Module, BT/WLAN/ANT+ Module, Filter, LNA, FEM, RF SW, SW, MHL Transmitter, USB CONNECTOR, Transceiver WTR1605L, and MMPS.

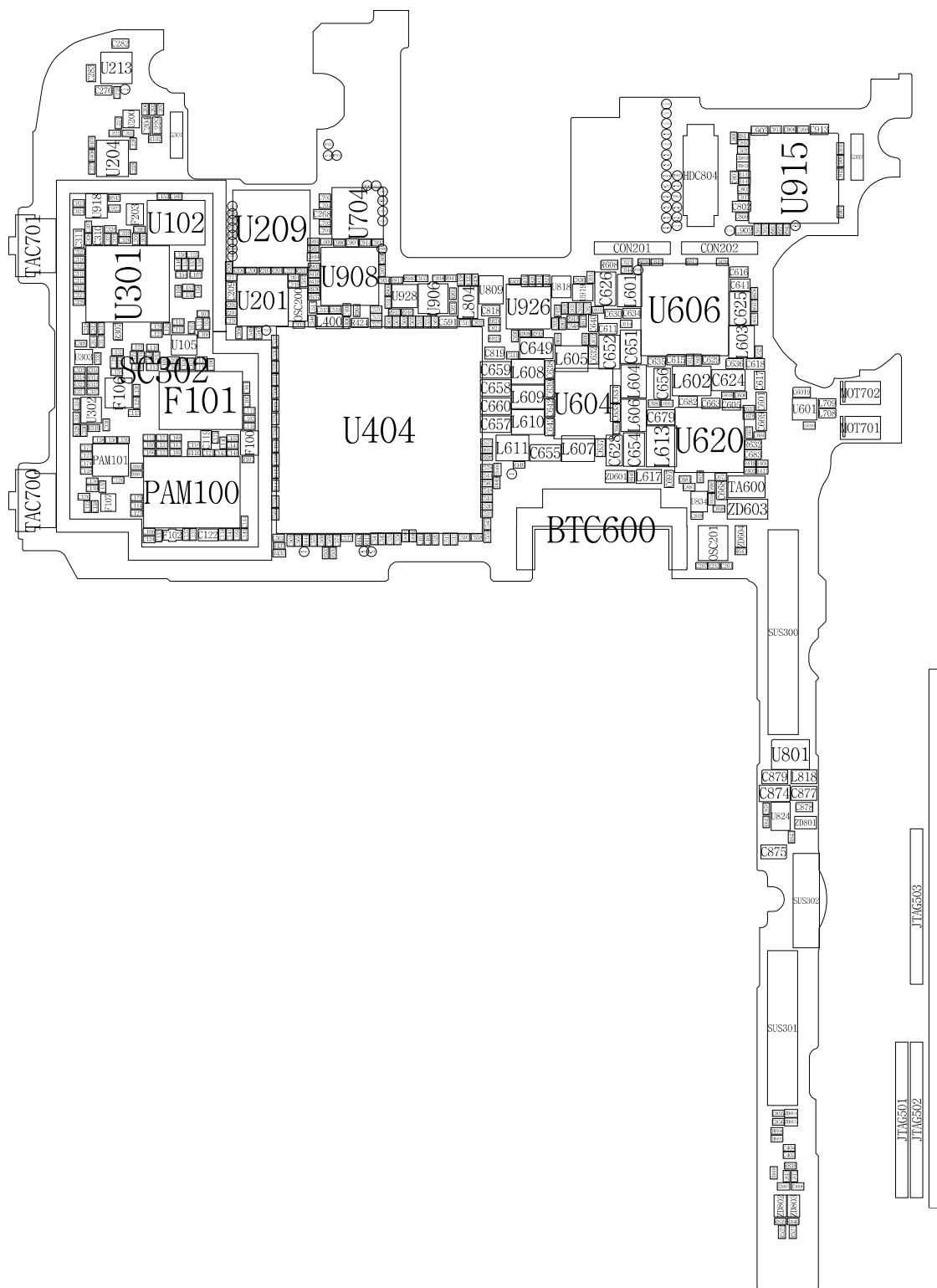
The diagram shows the interconnections between these components, including data buses like UART, I2C, SPI, and USB3.0, as well as various control lines.

8-2. 主板图

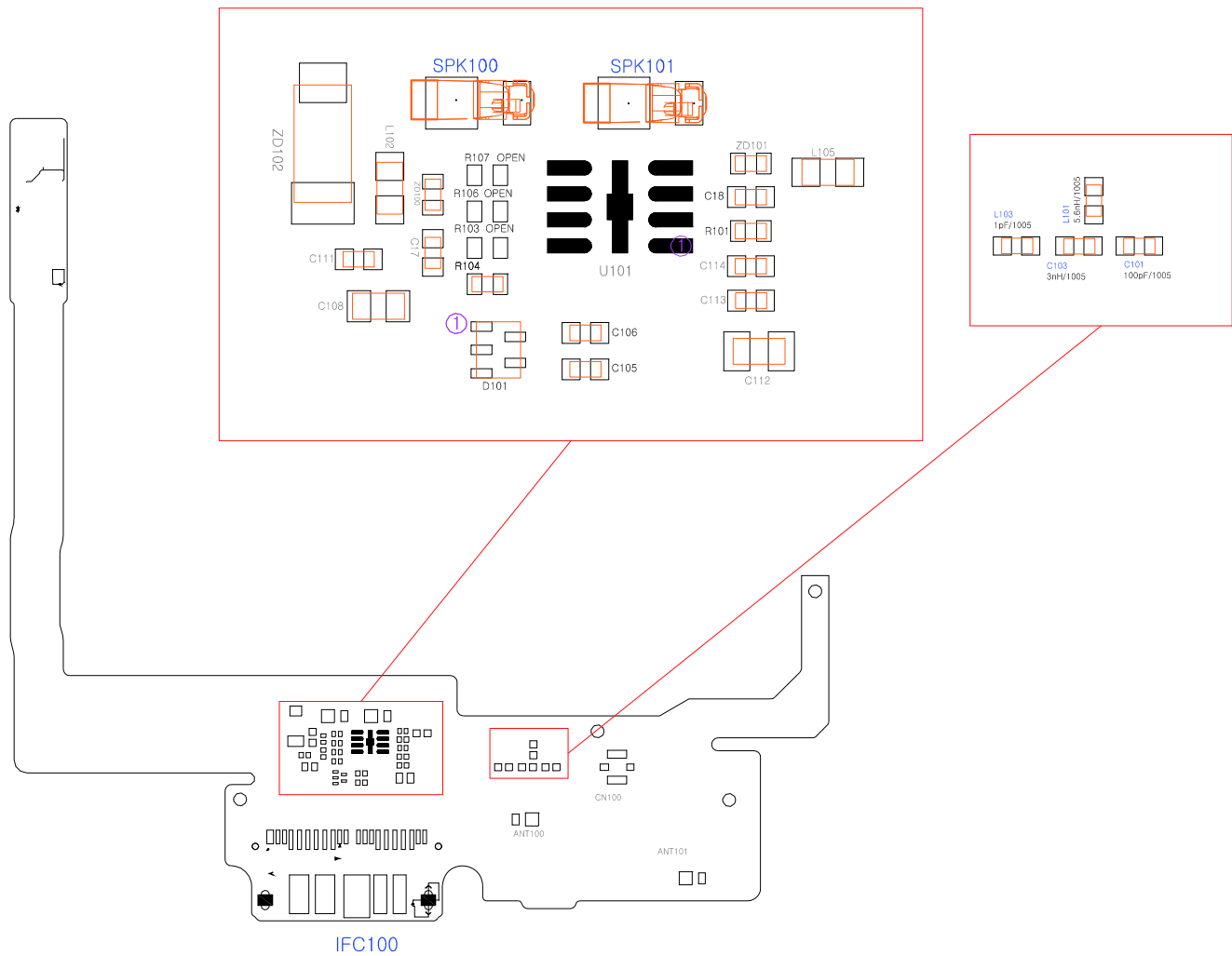
8-2-1.主板顶部



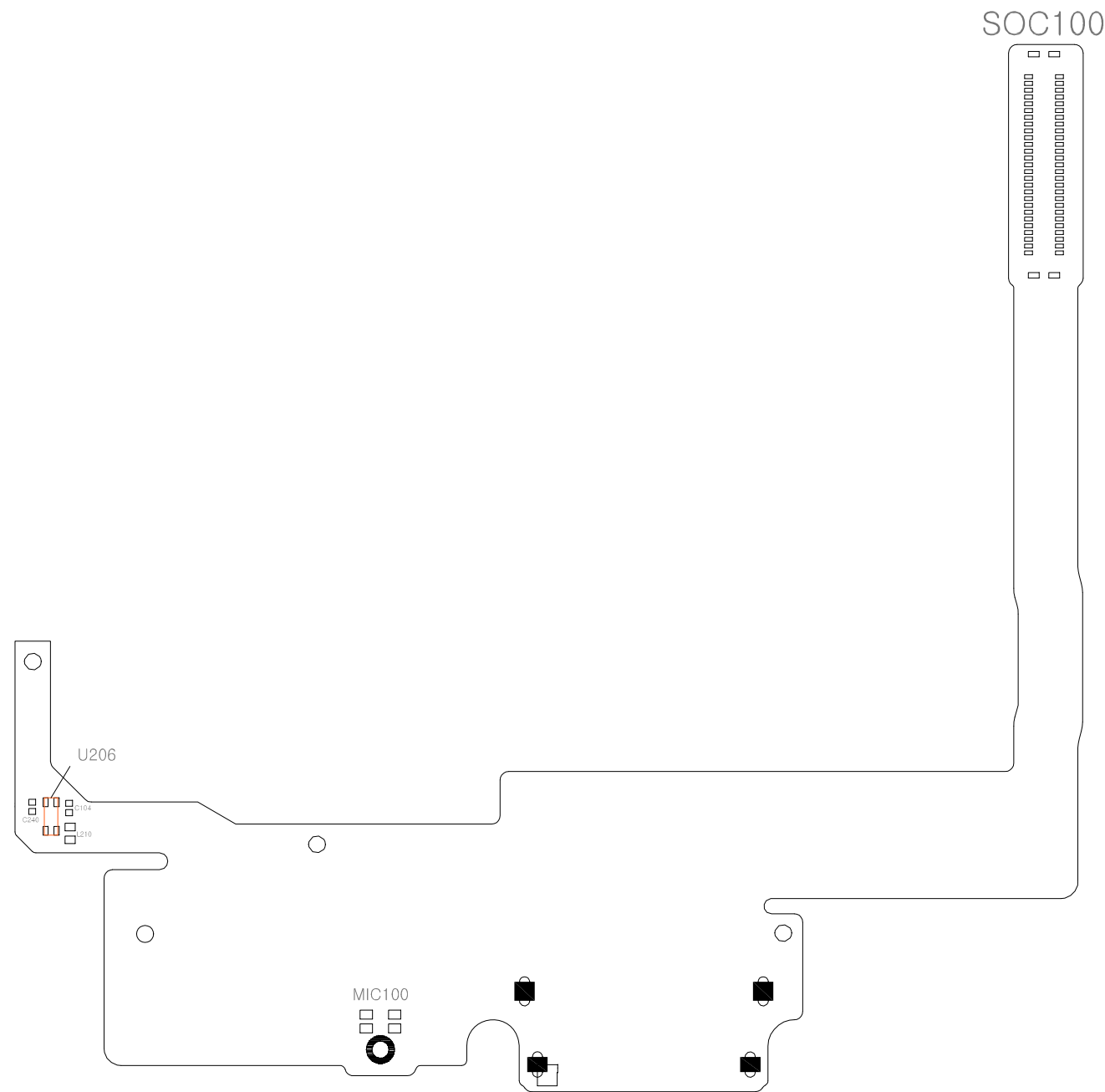
8-2-2.底部



8-2-3.副板顶部

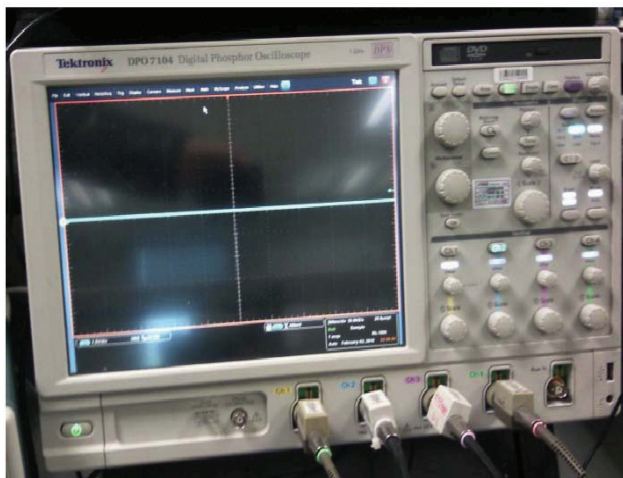


8-2-4.副板底部



8-3.故障排除流程图

设备



↑示波器



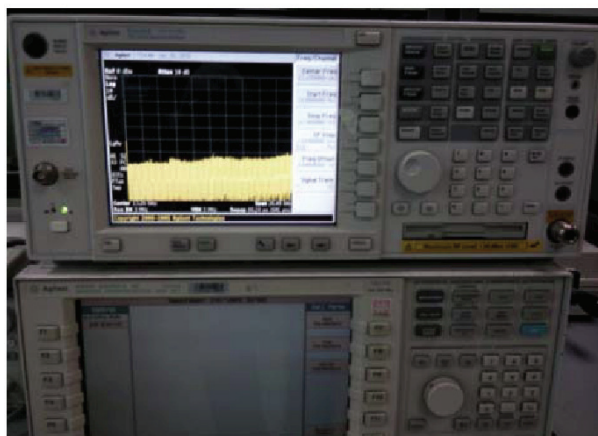
↑数字万用表



↑电源供电器



↑+字螺丝刀, ESD 安全镊子

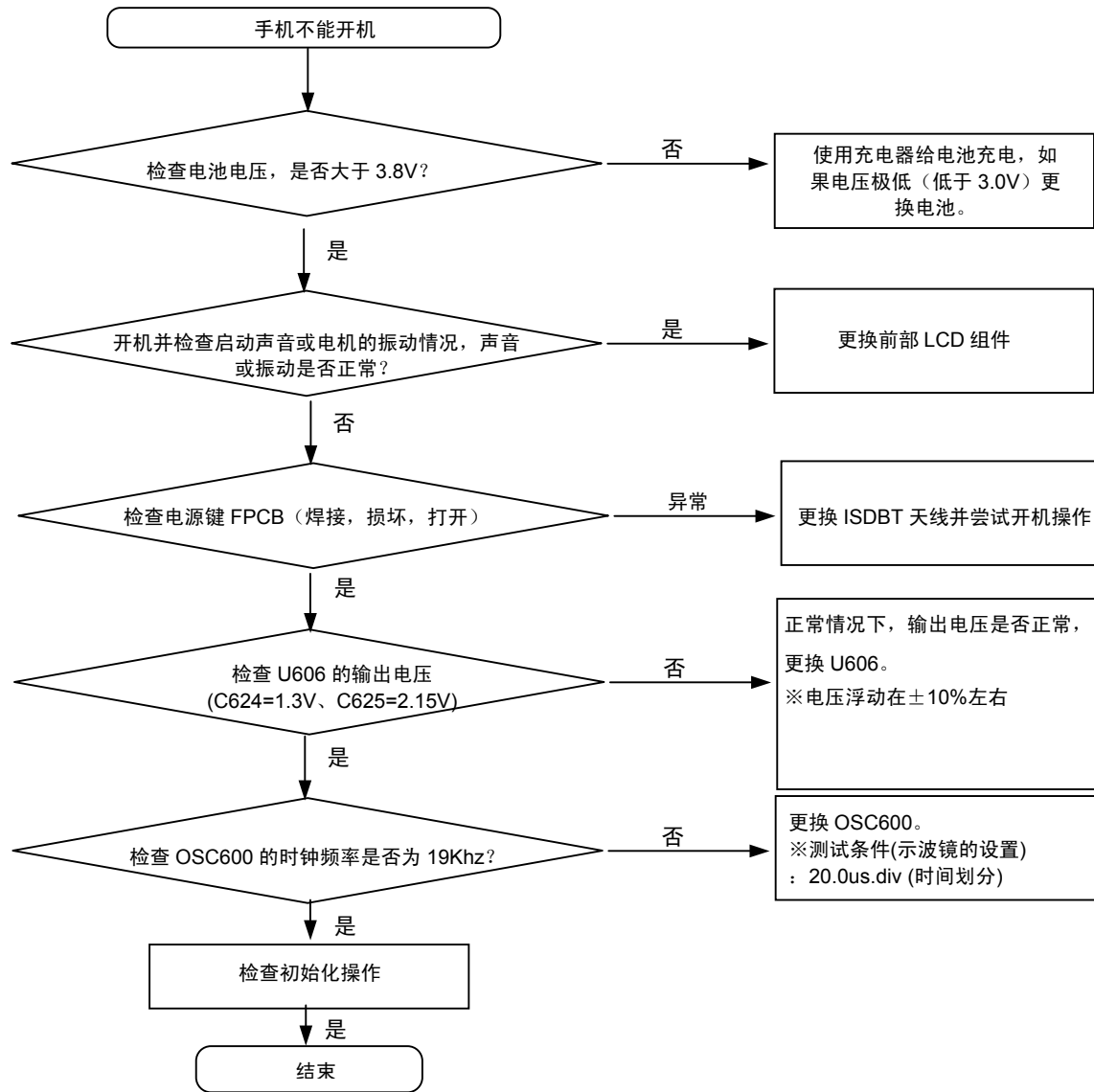


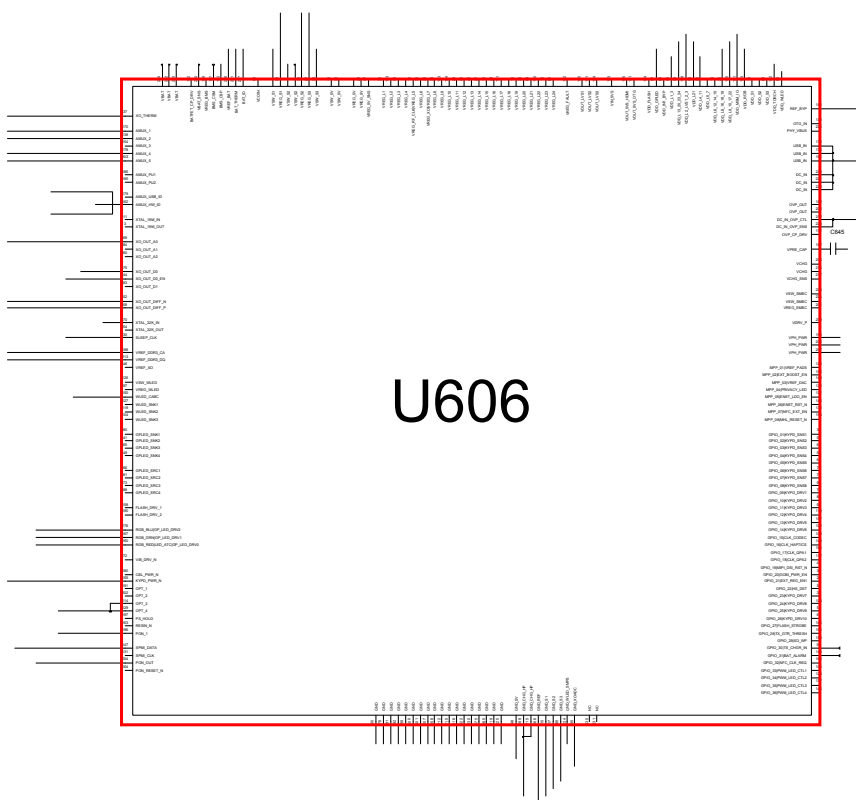
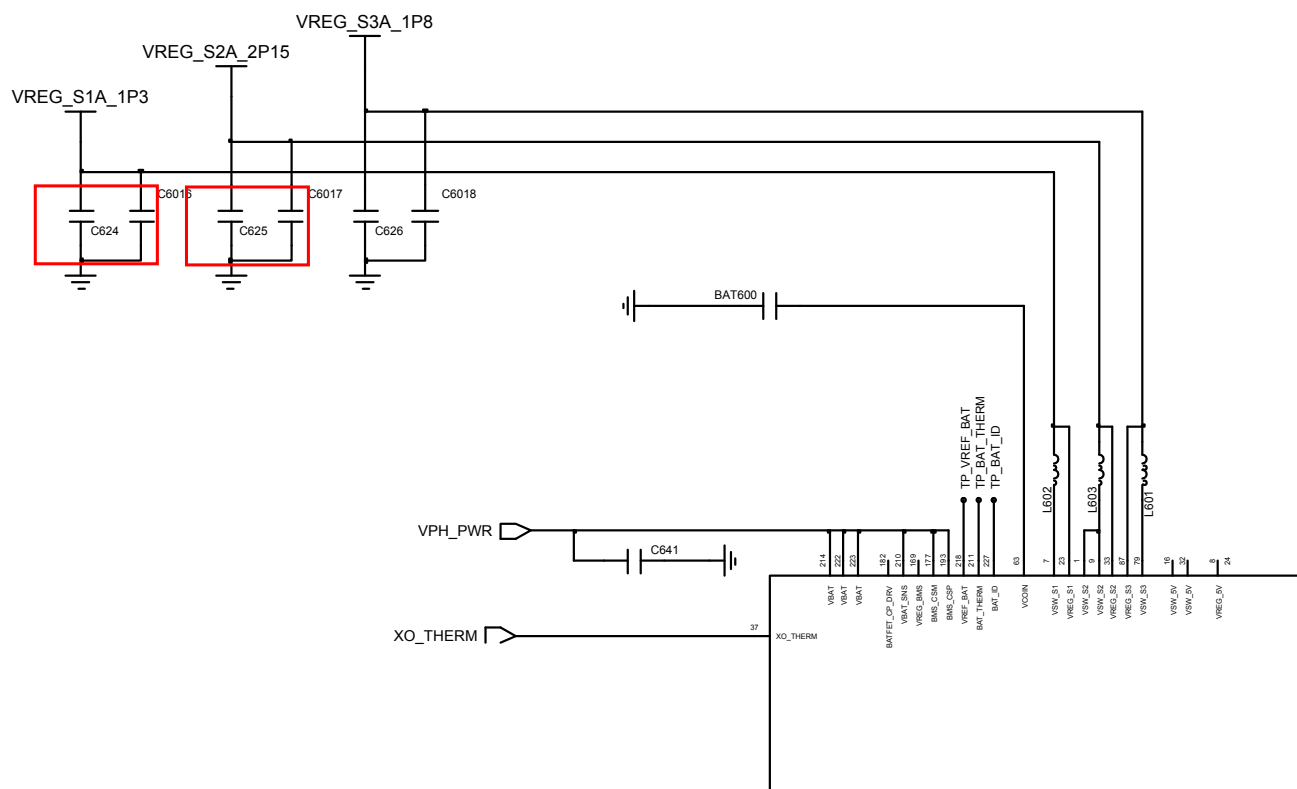
↑8960 和光谱分析器

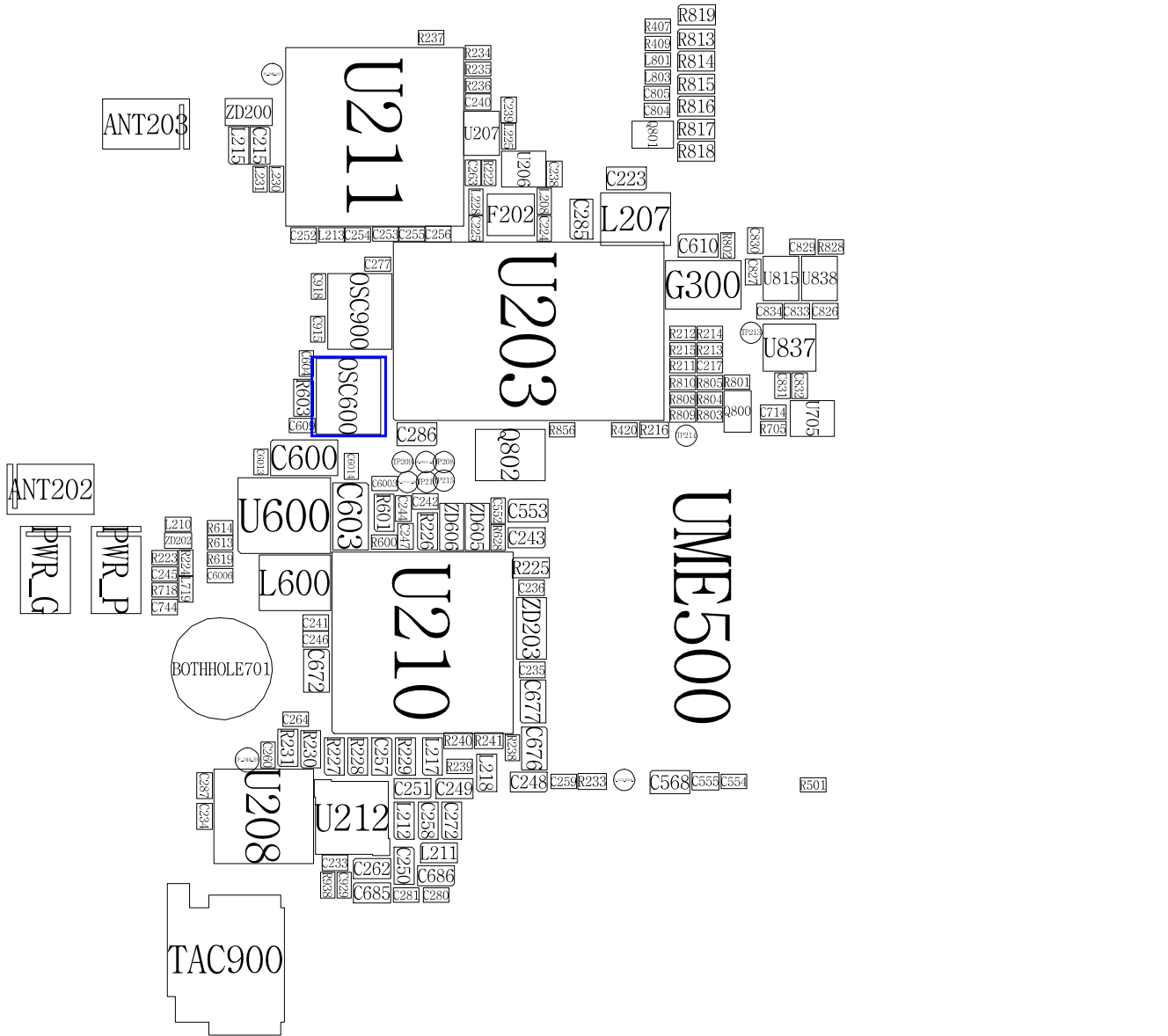


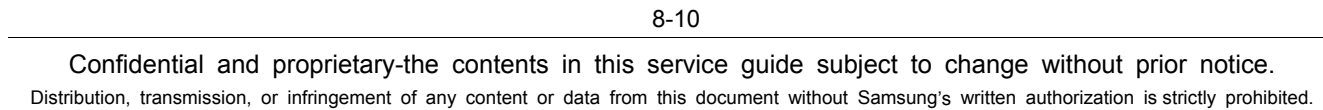
↑焊接熨斗

8-3-1. 开机

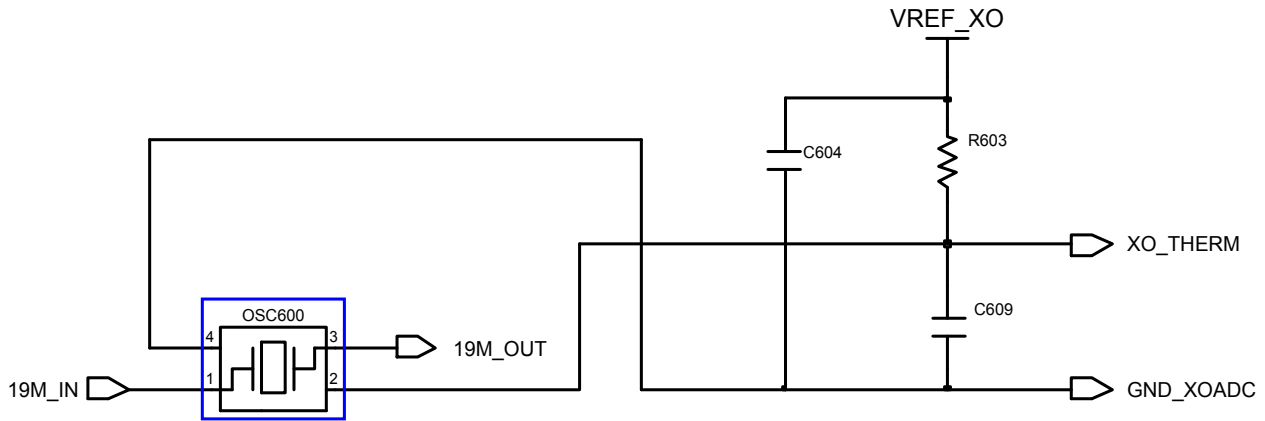
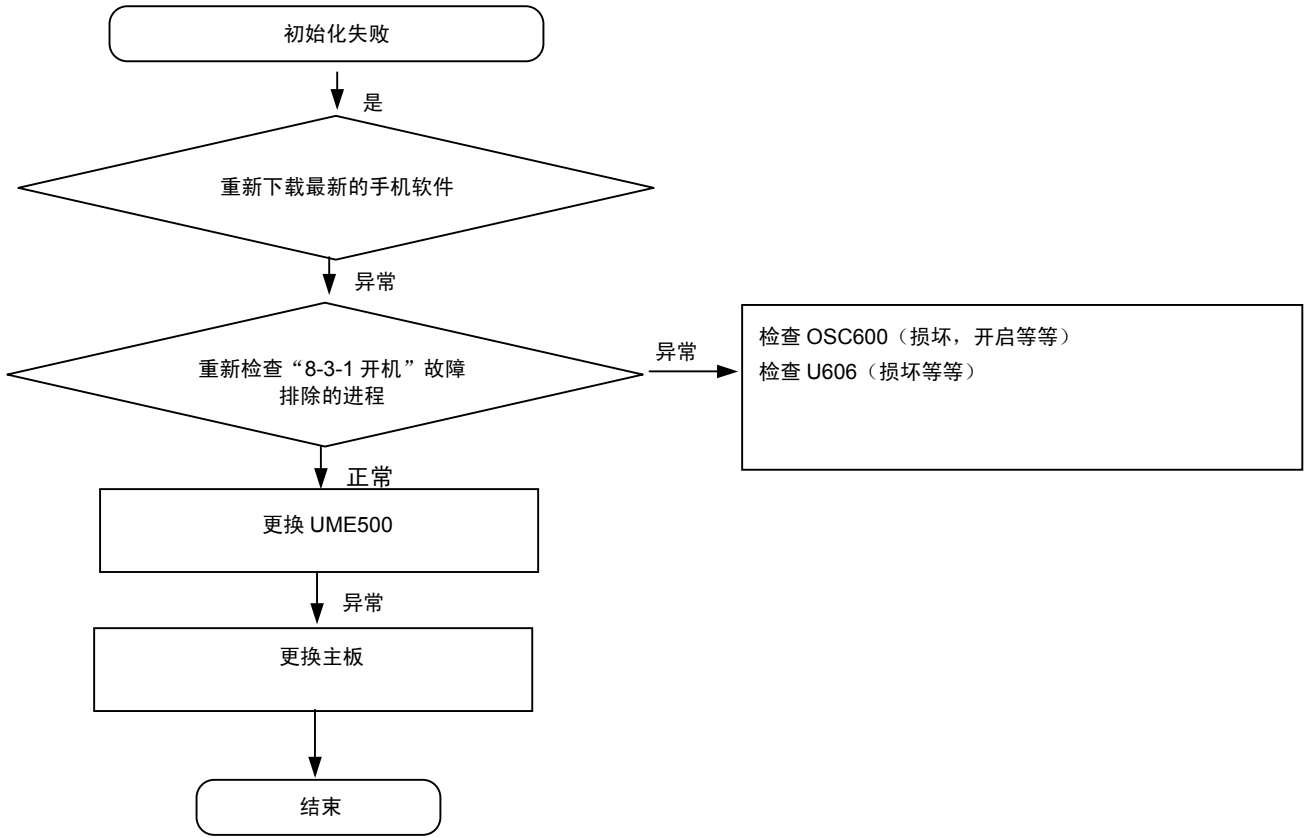


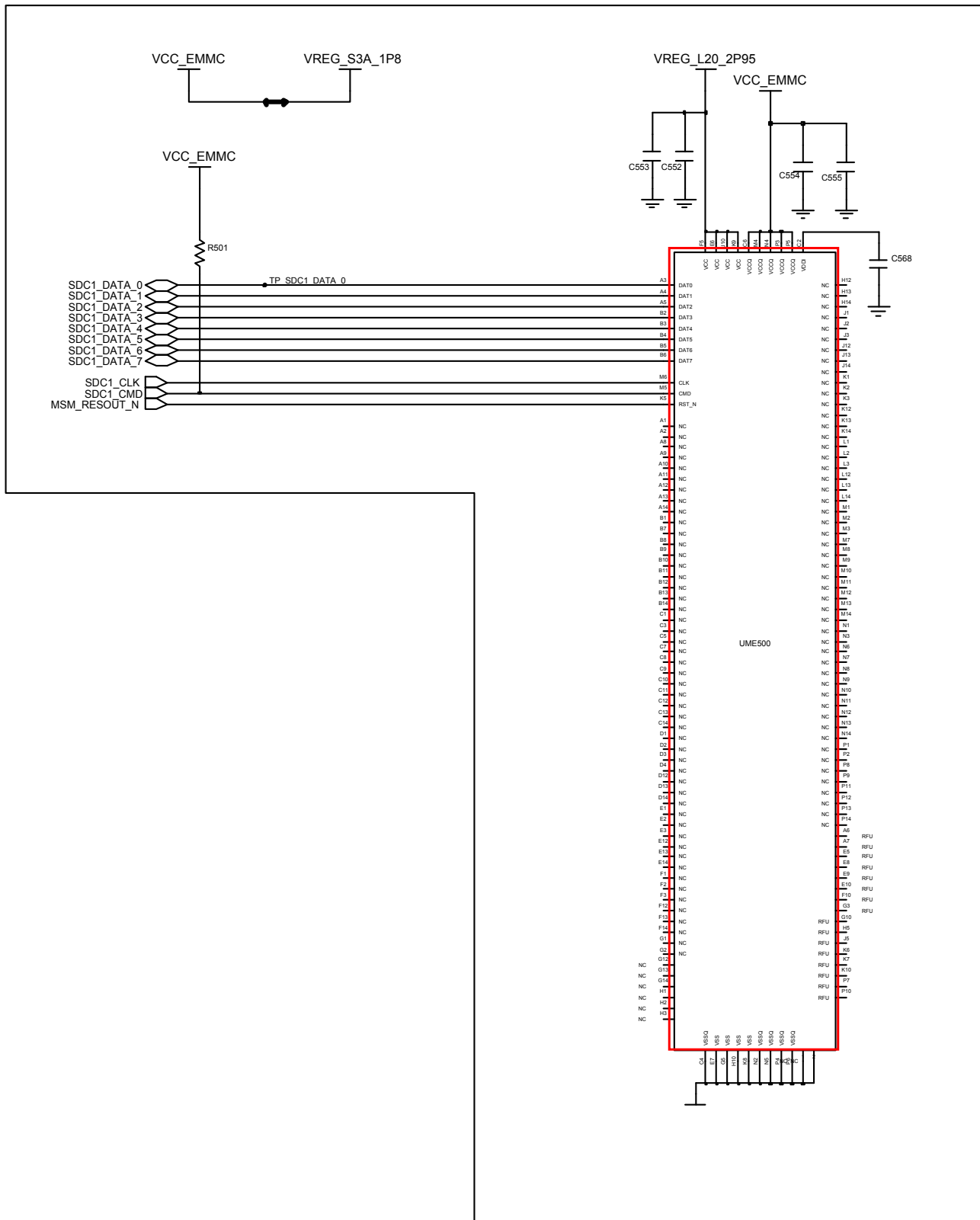


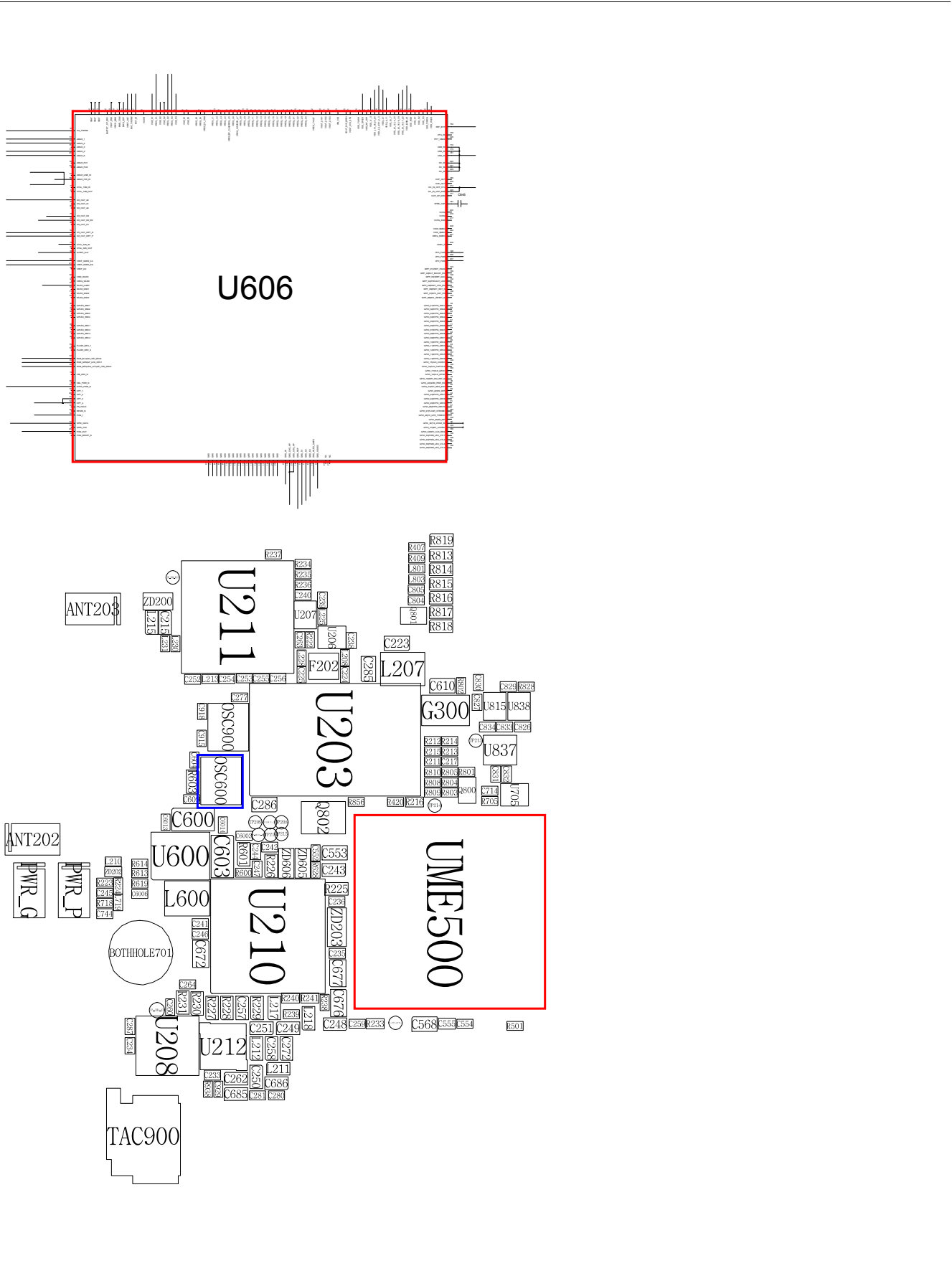


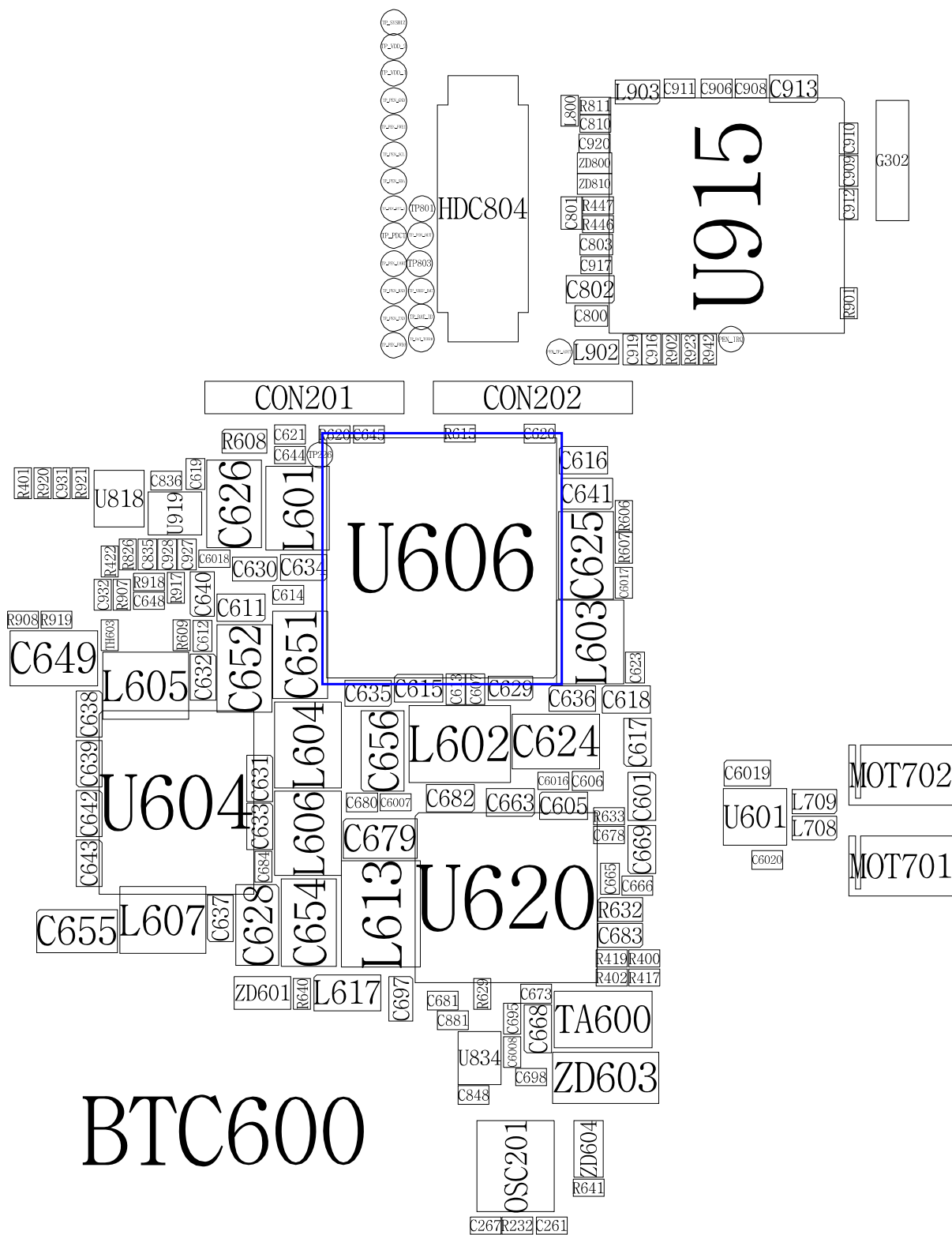


8-3-2. 初始化

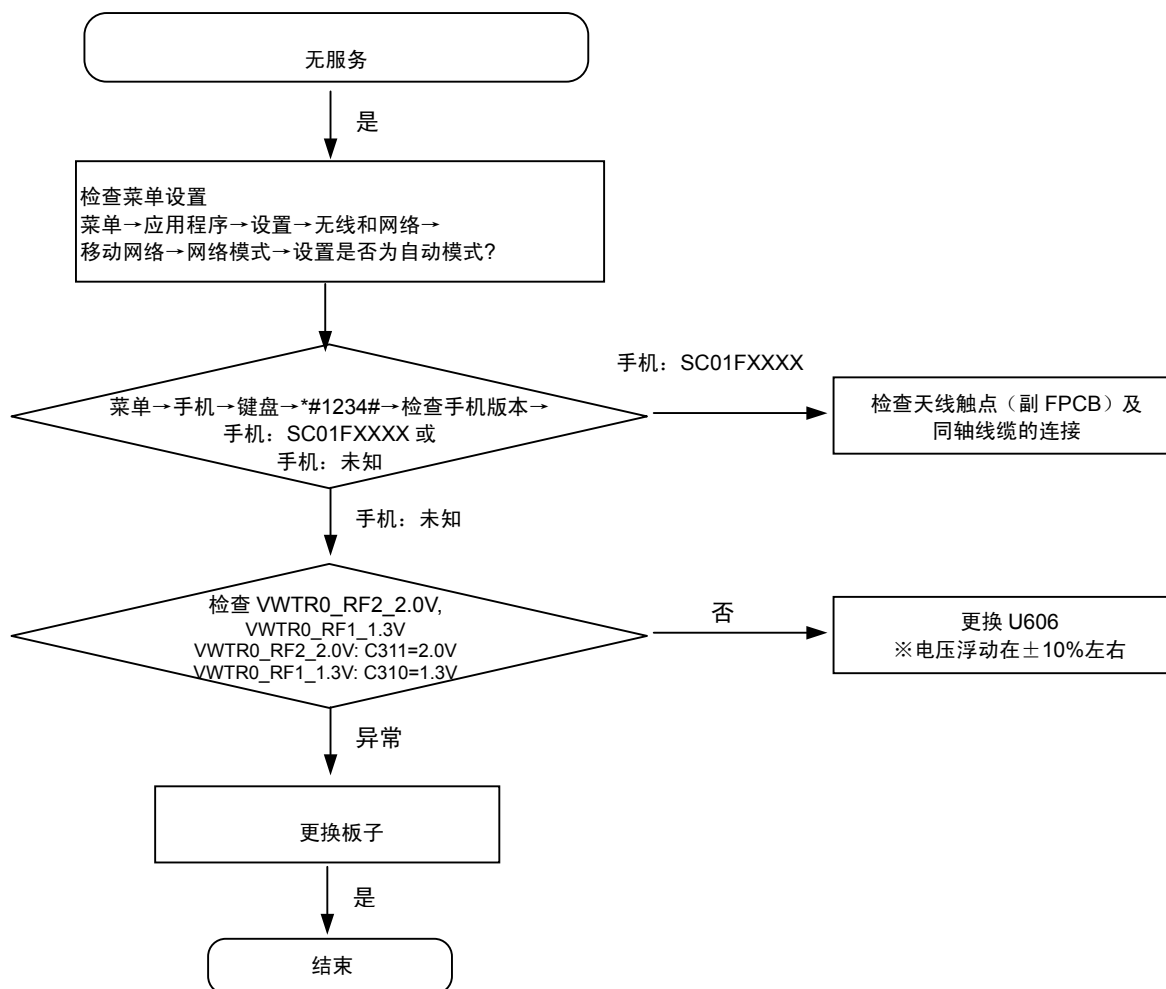


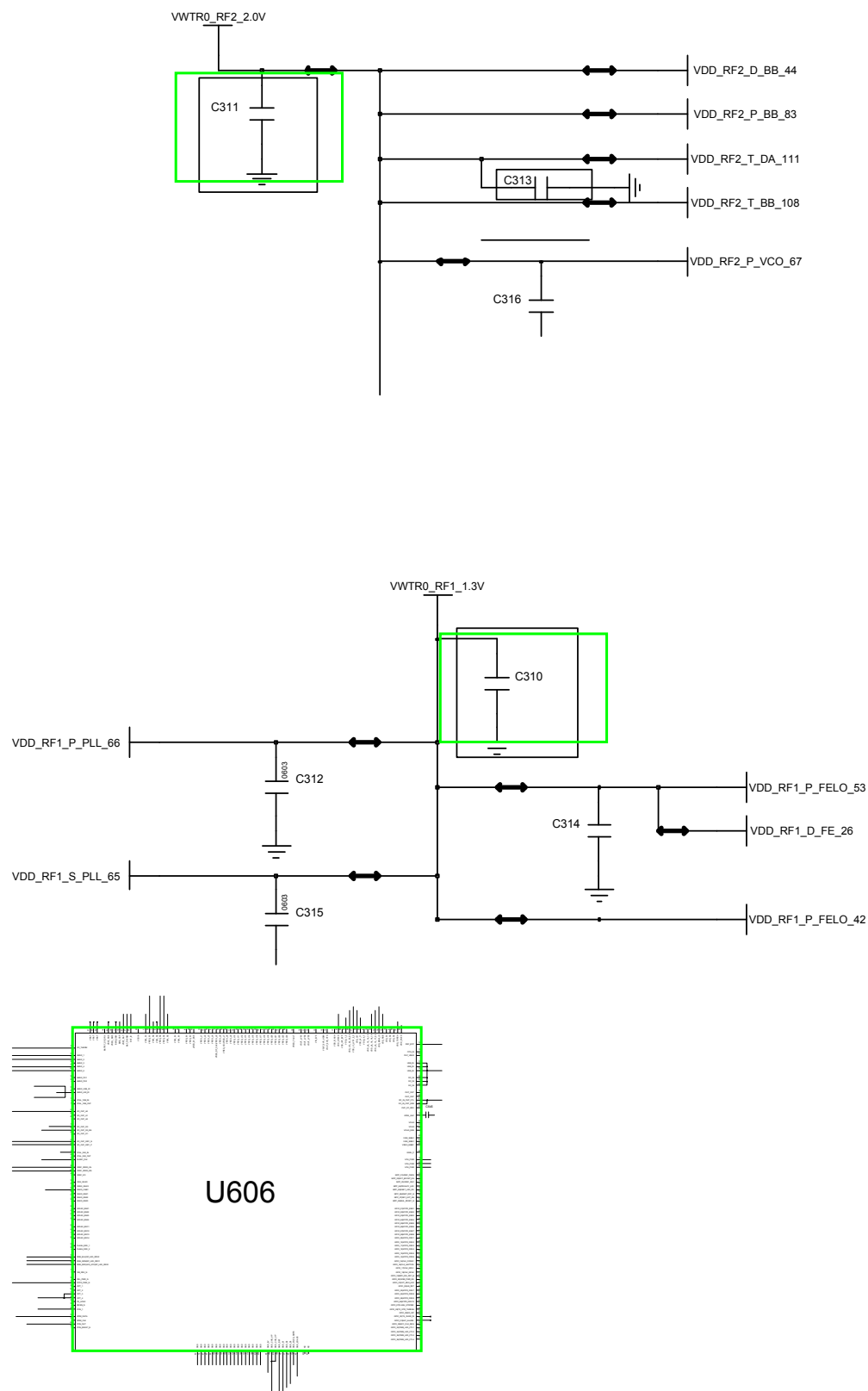


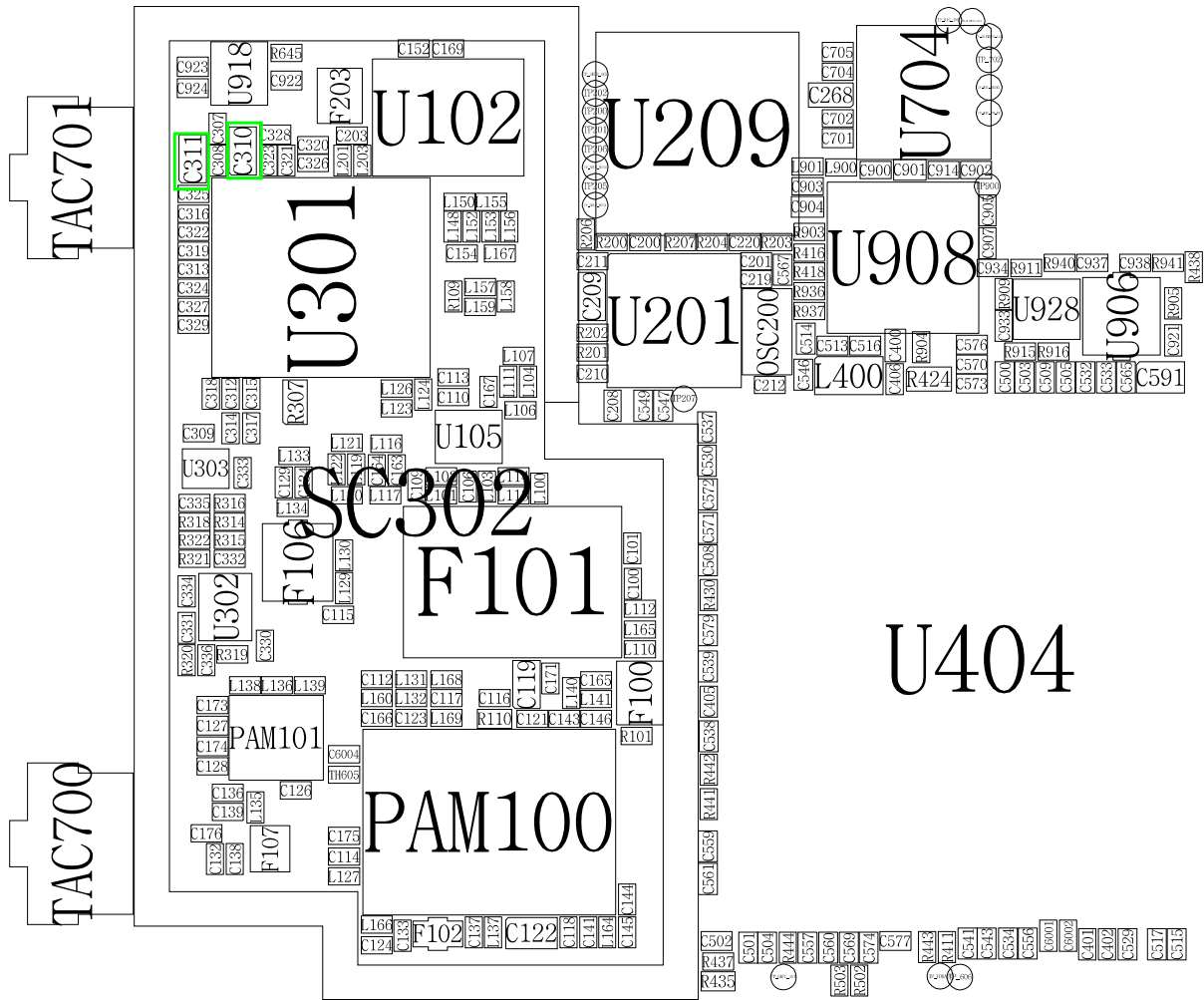


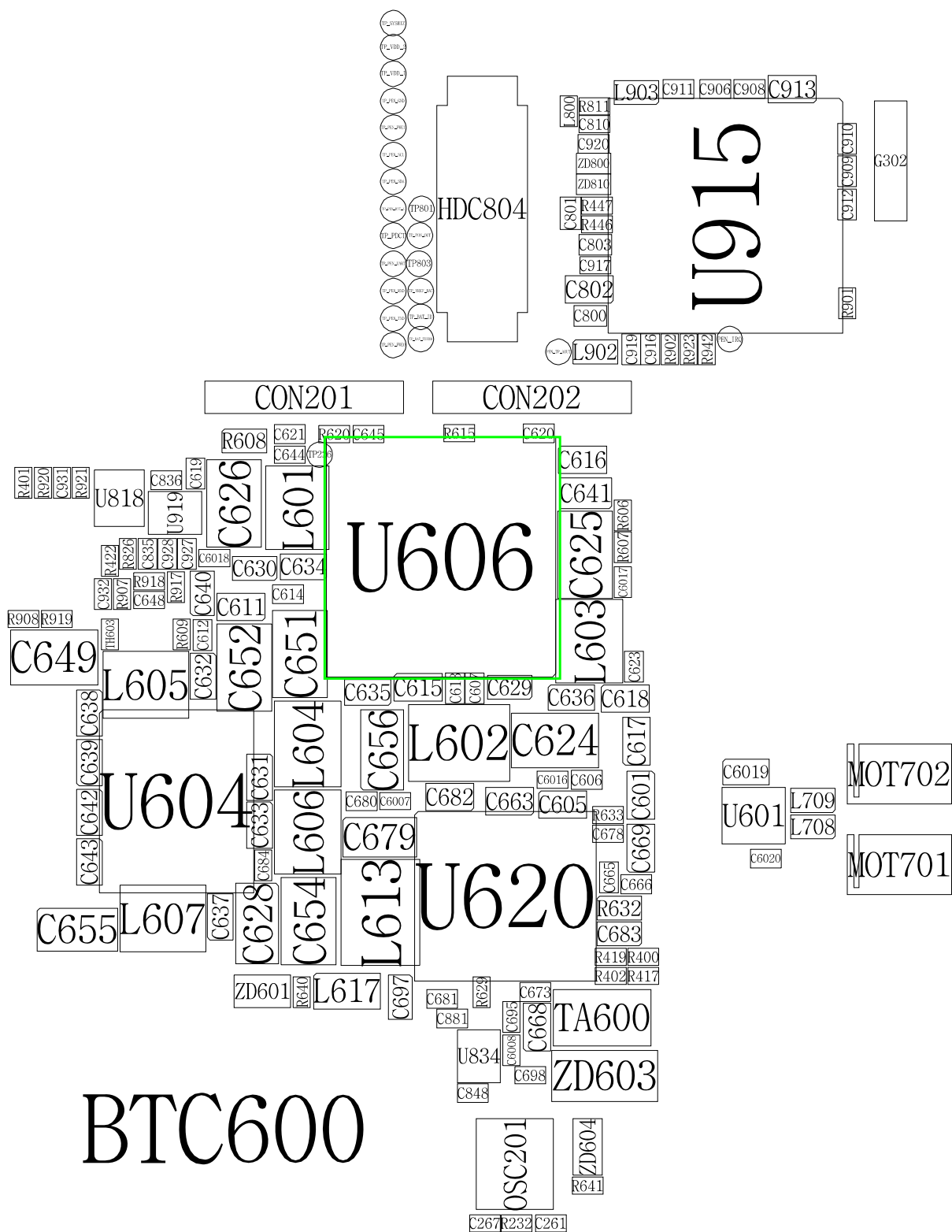


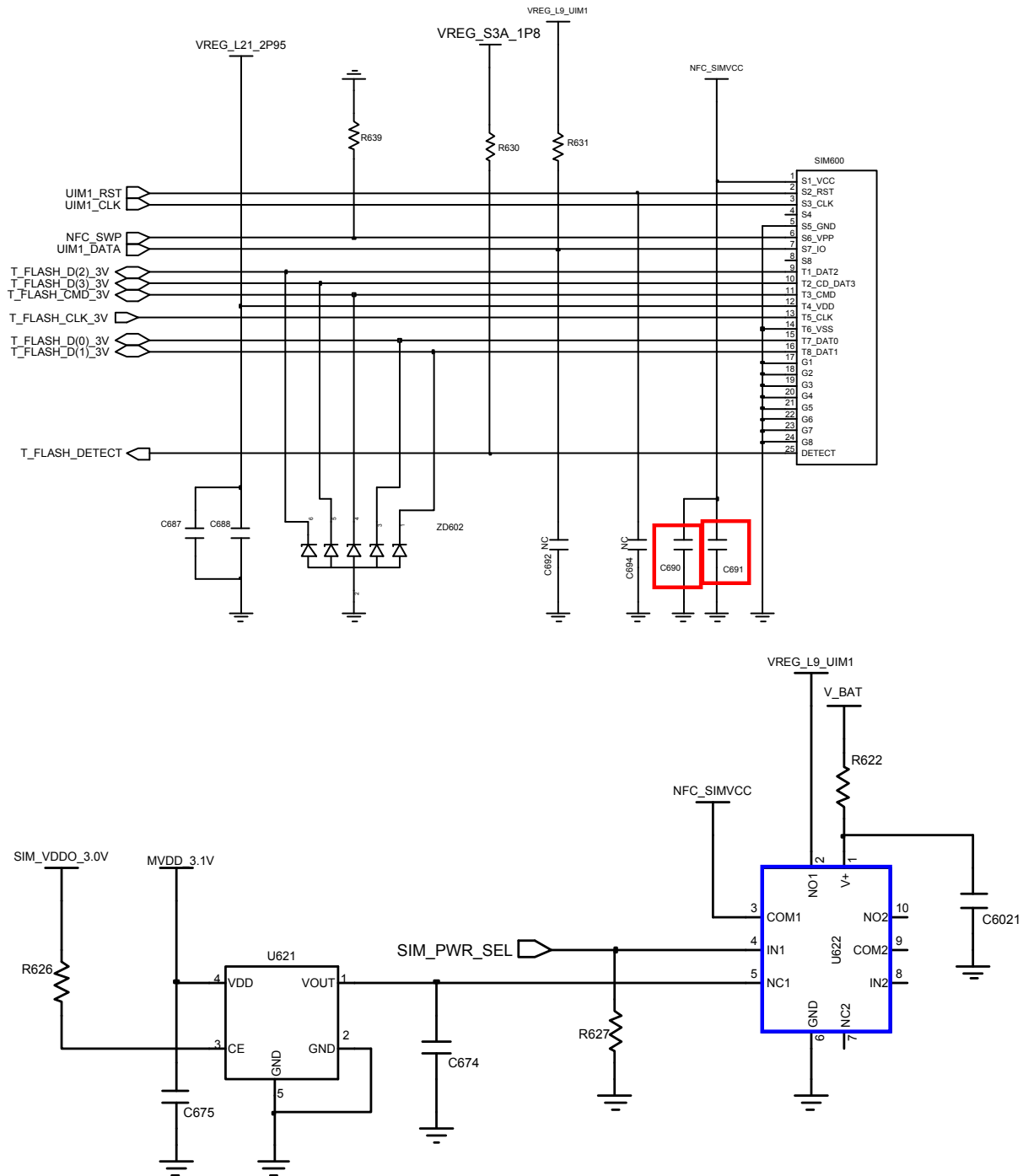
8-3-3.无服务

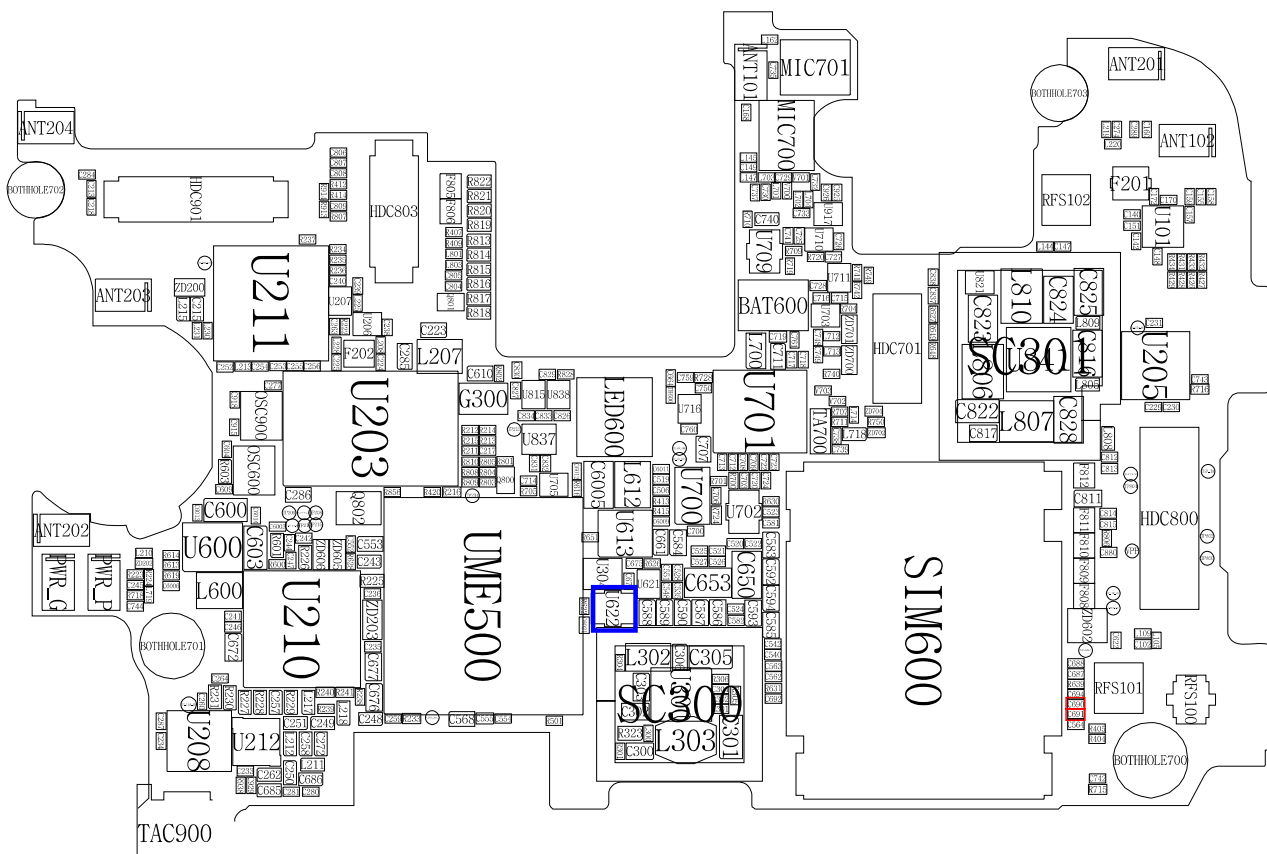
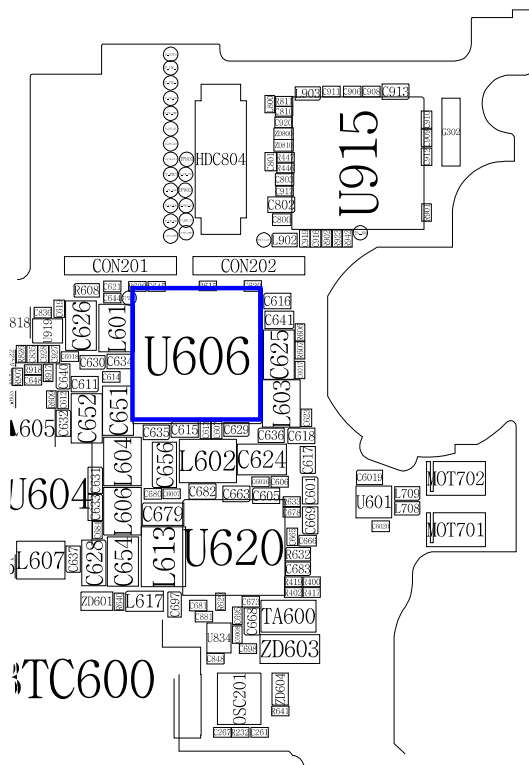




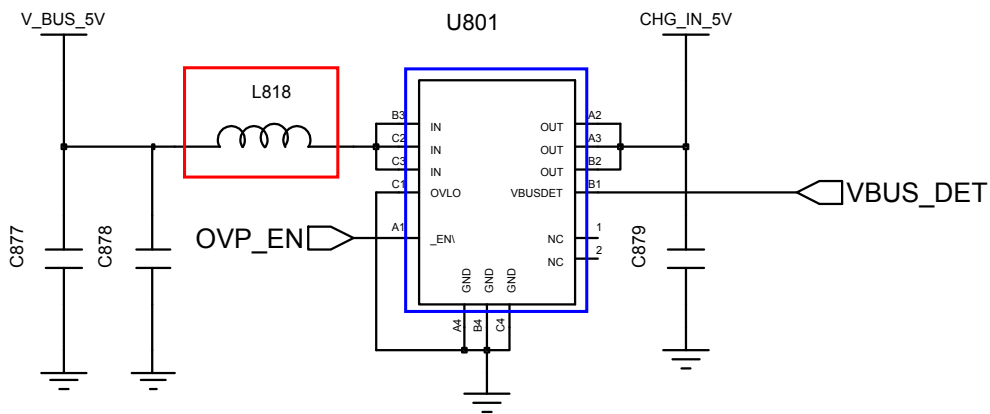
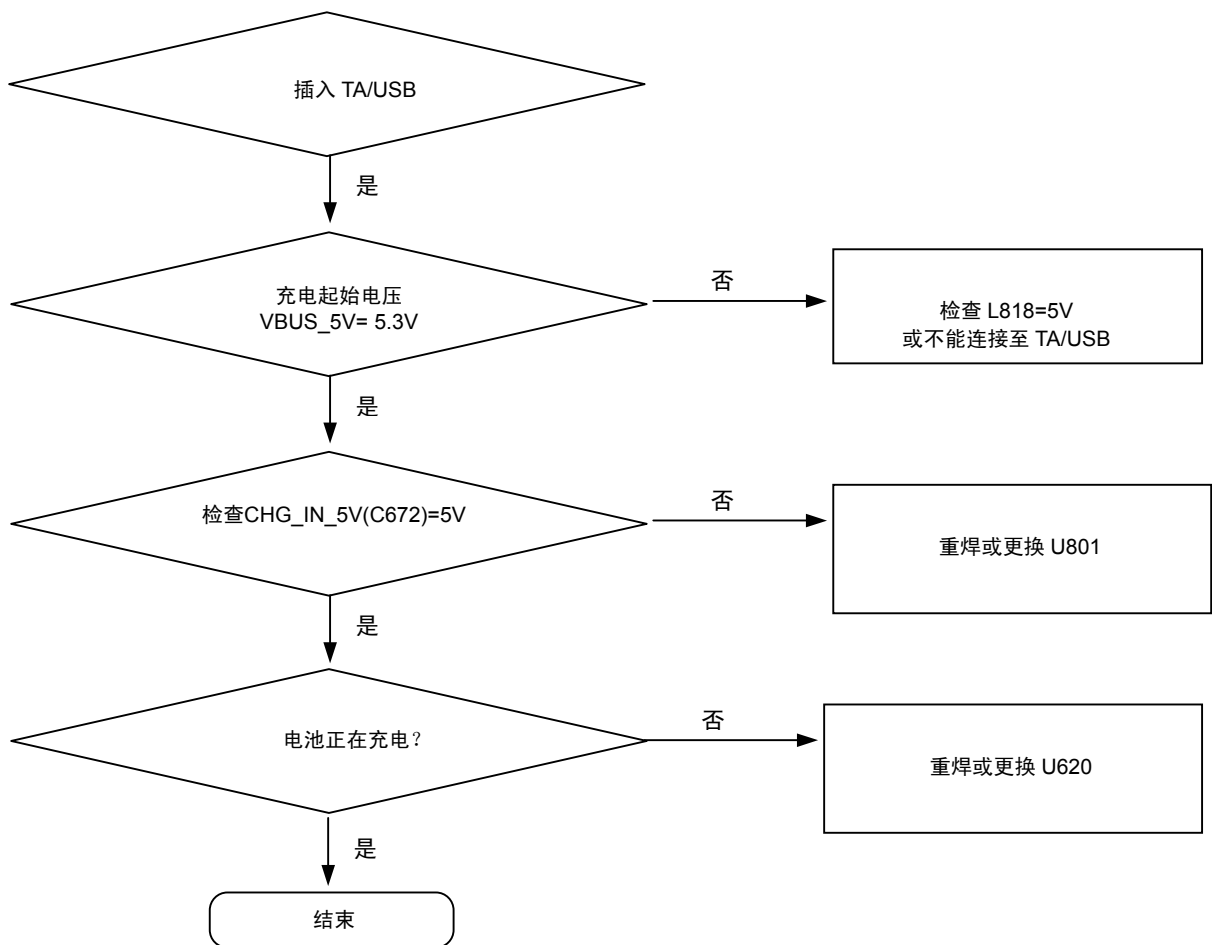


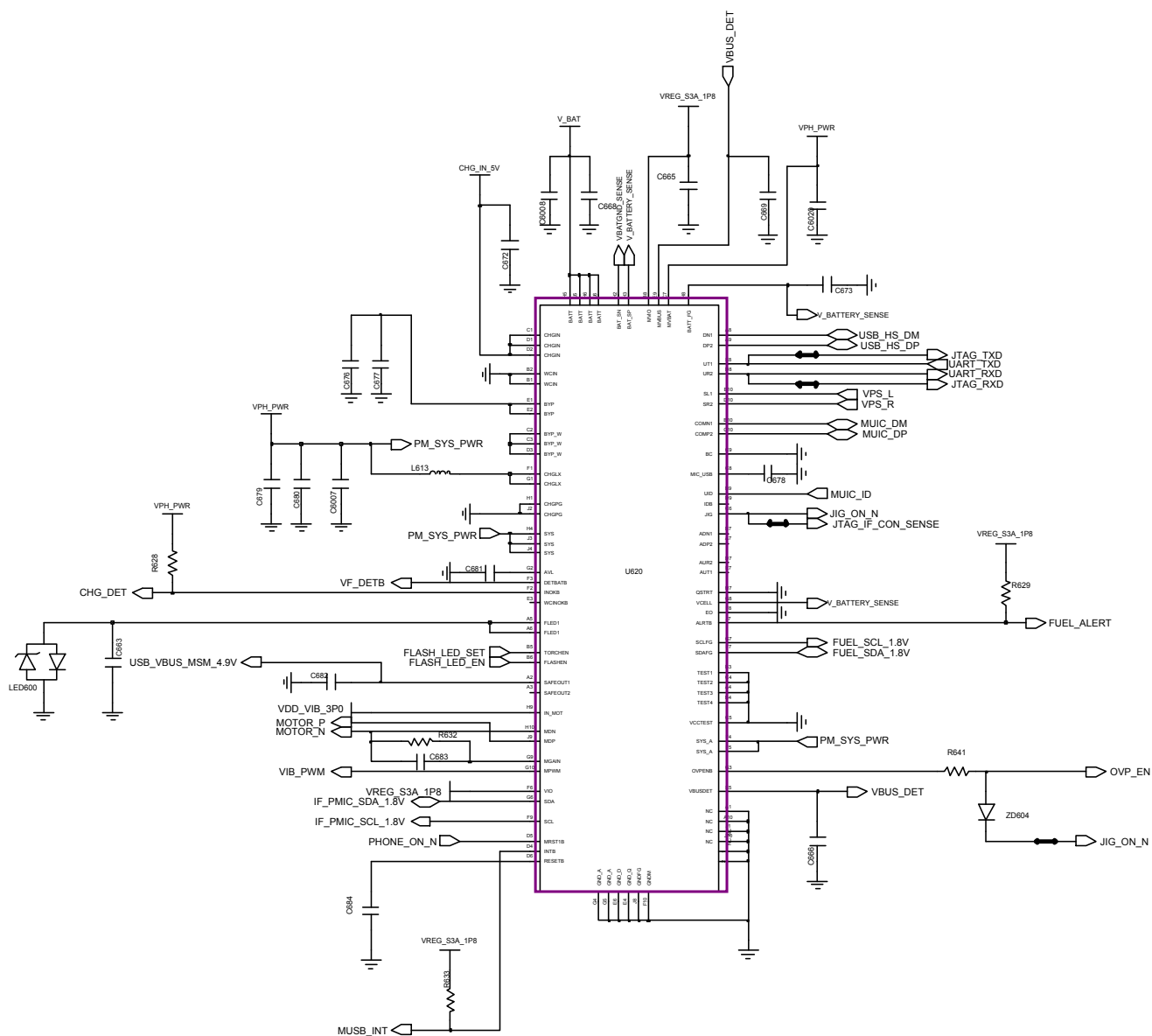


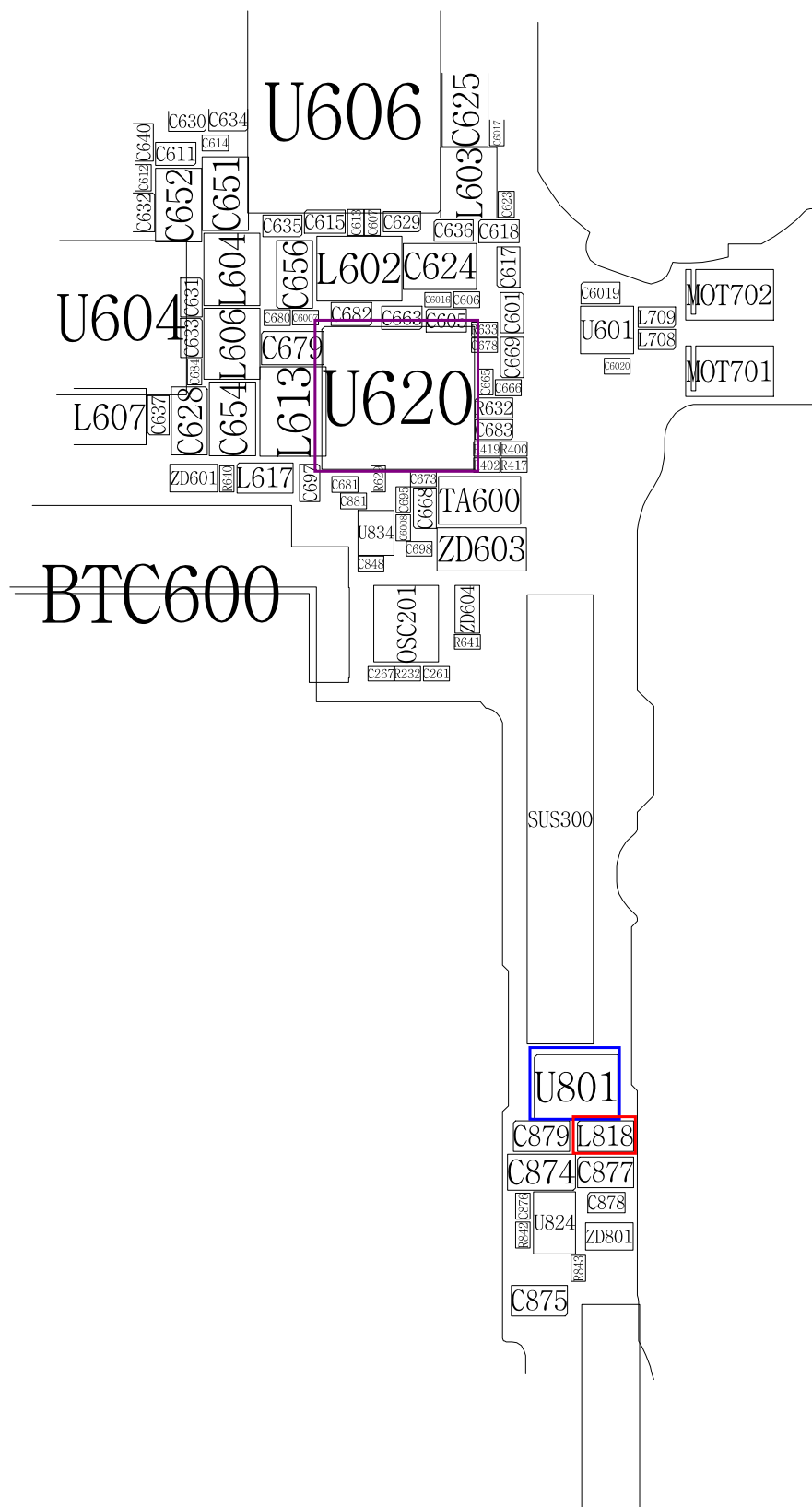




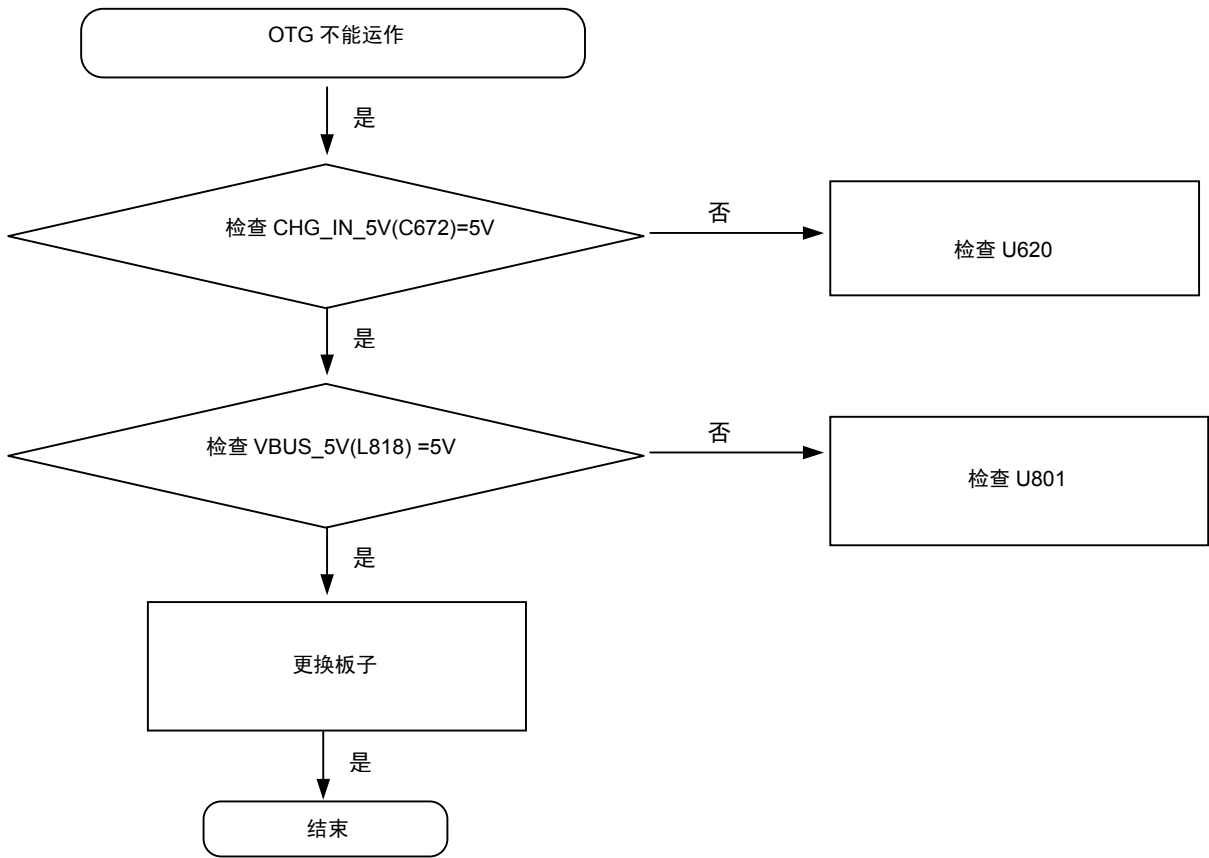
8-3-5.充电部分

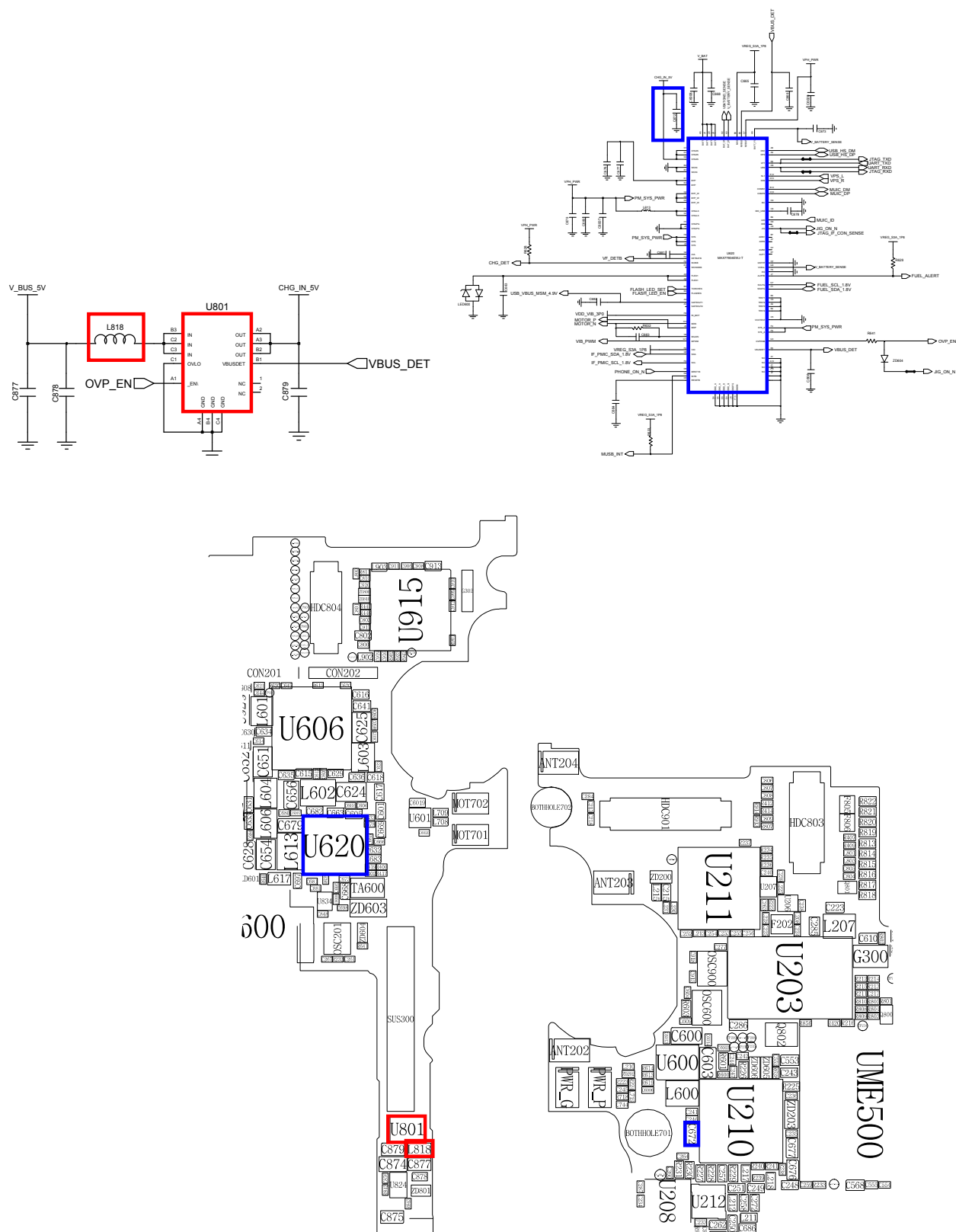




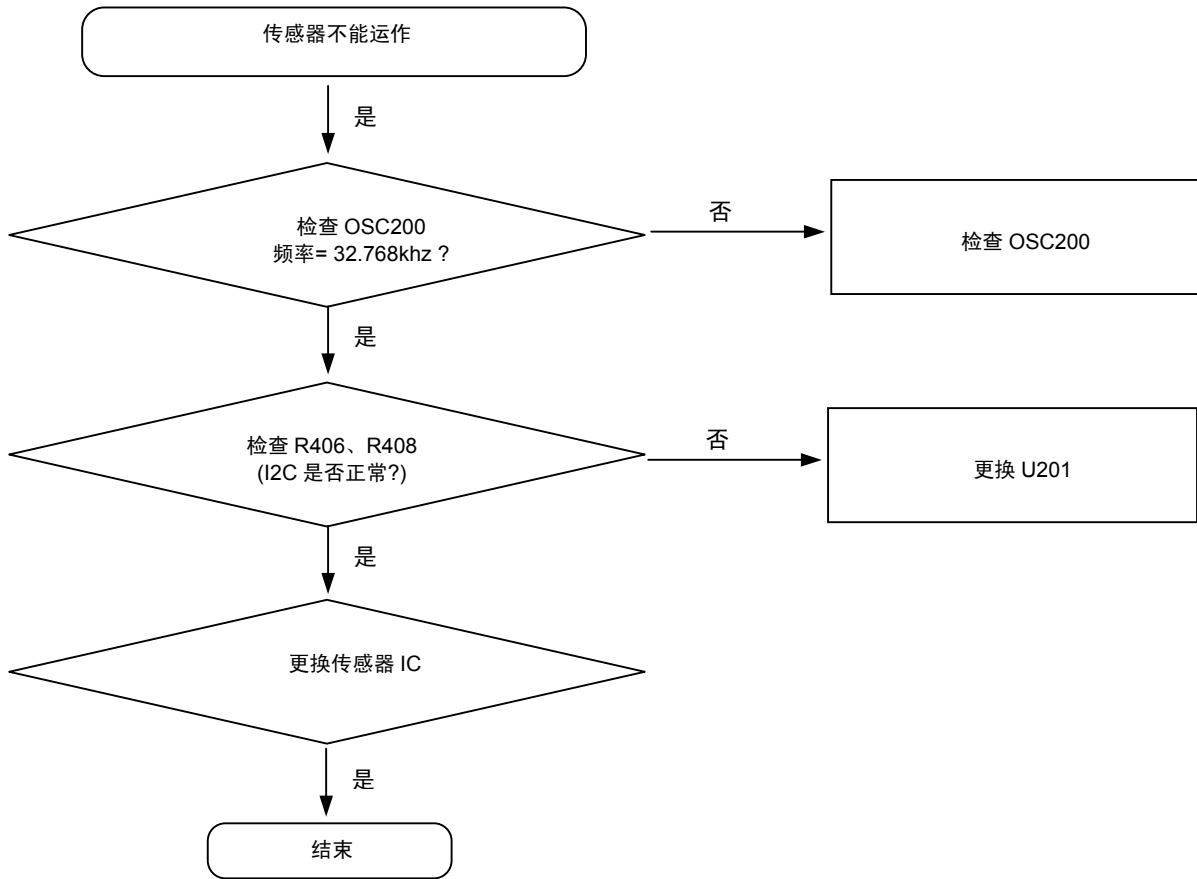


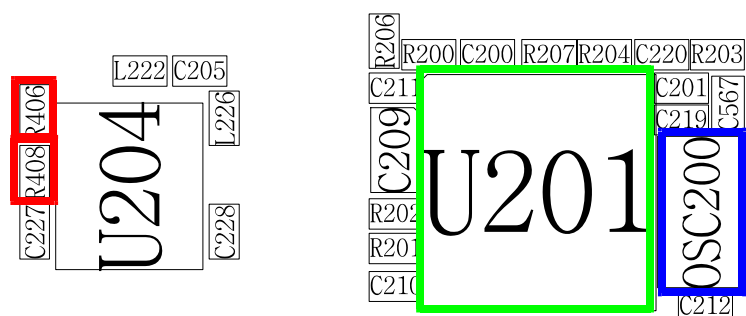
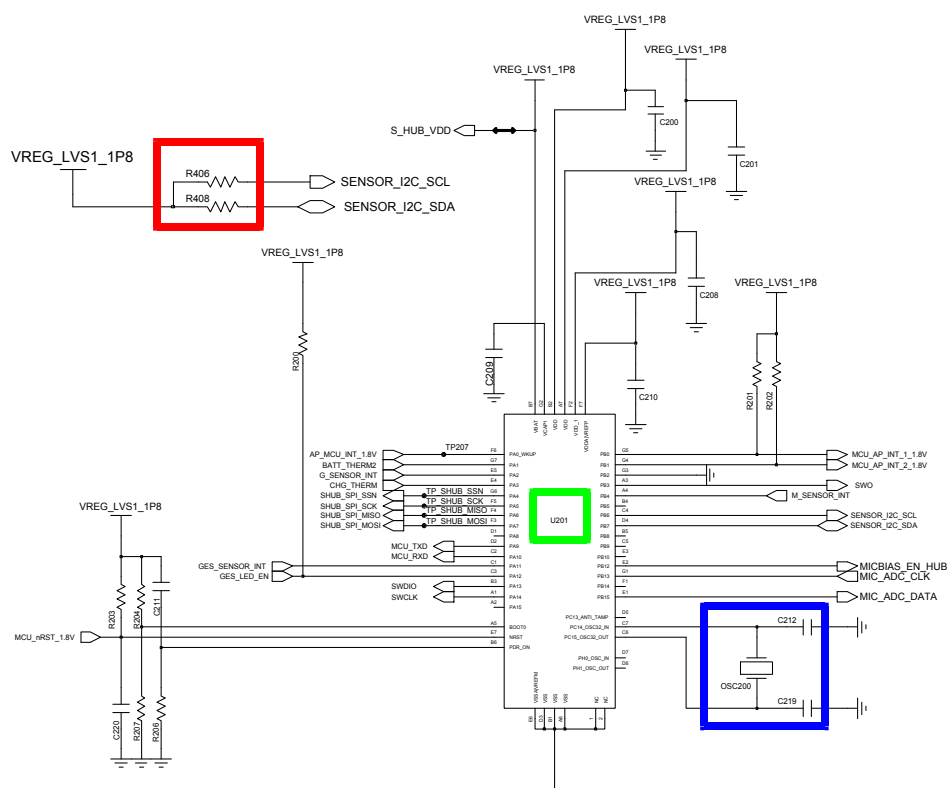
8-3-6.OTG



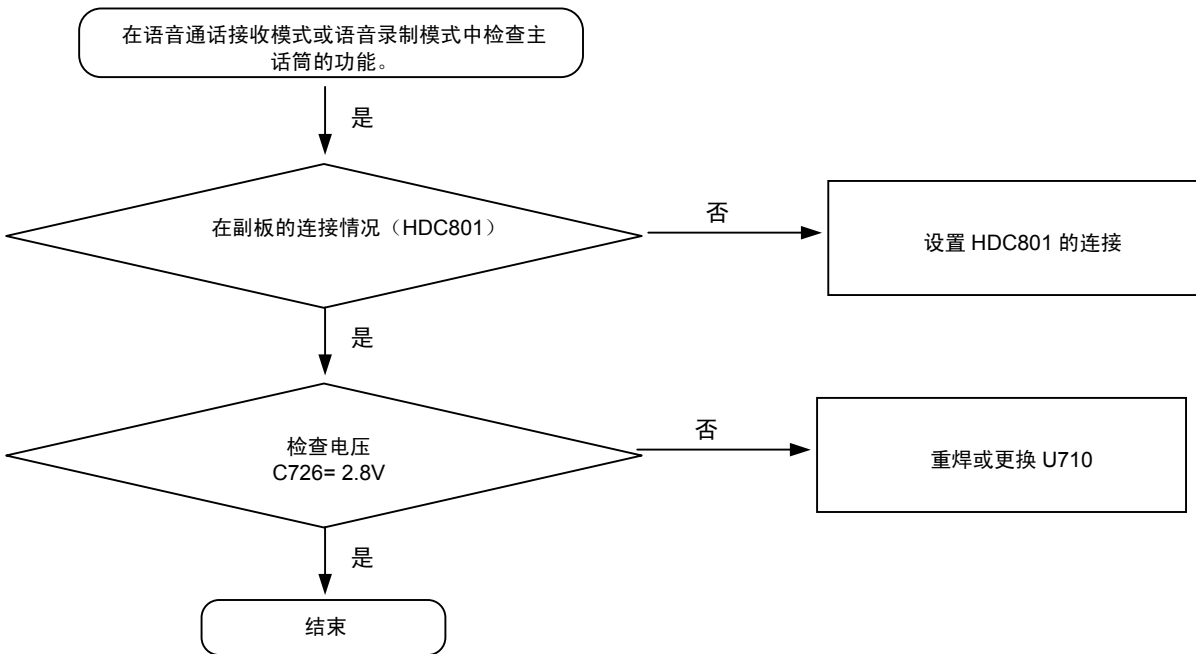


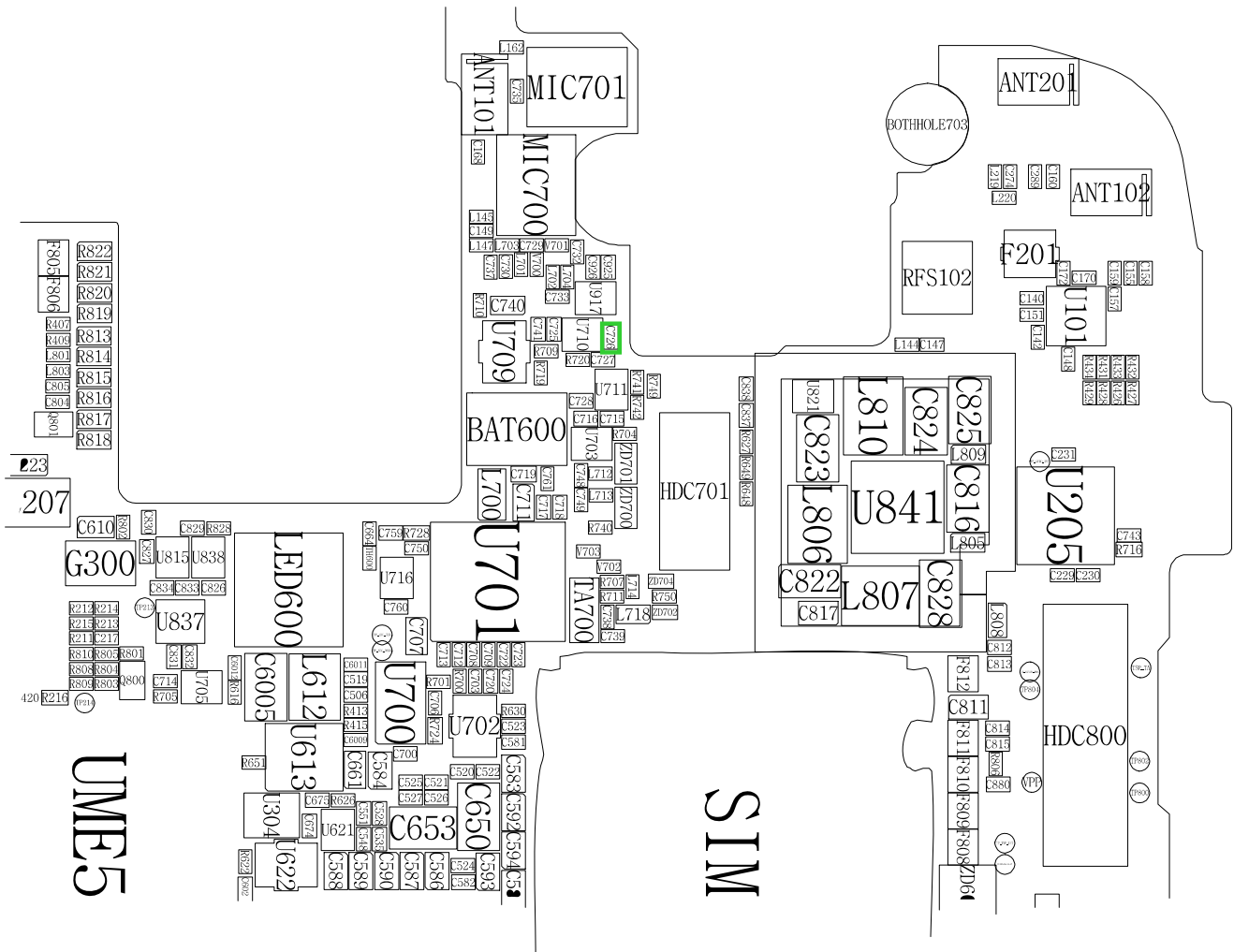
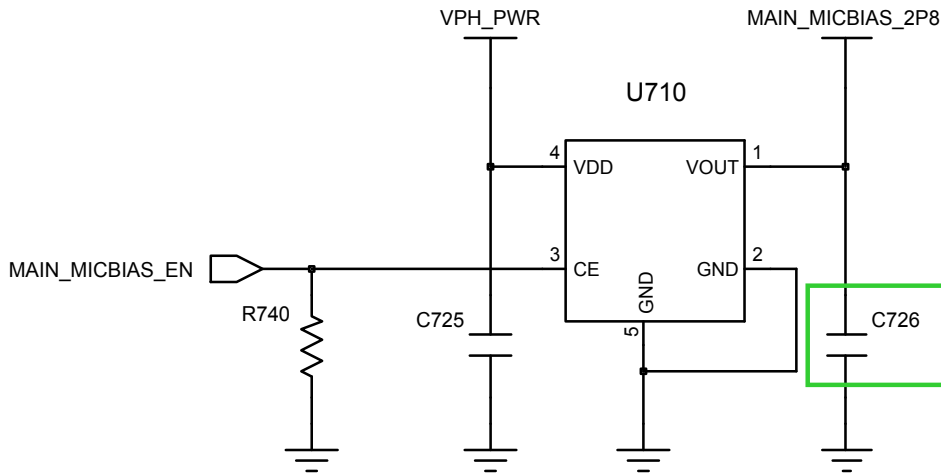
8-3-7.传感器

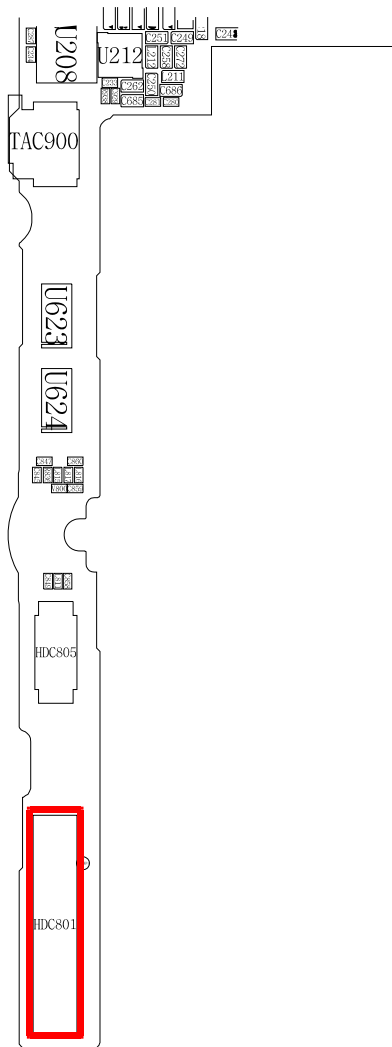




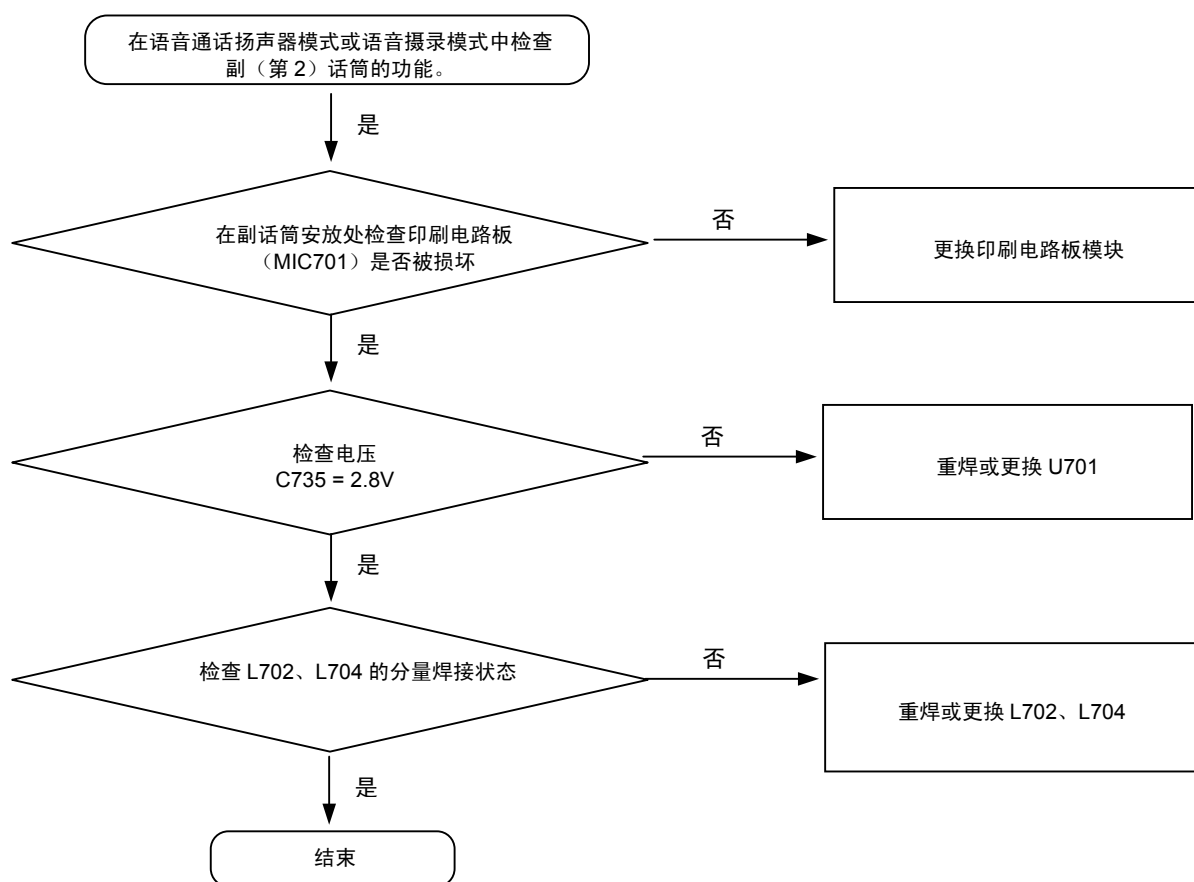
8-3-8-1. 话筒部分-主话筒

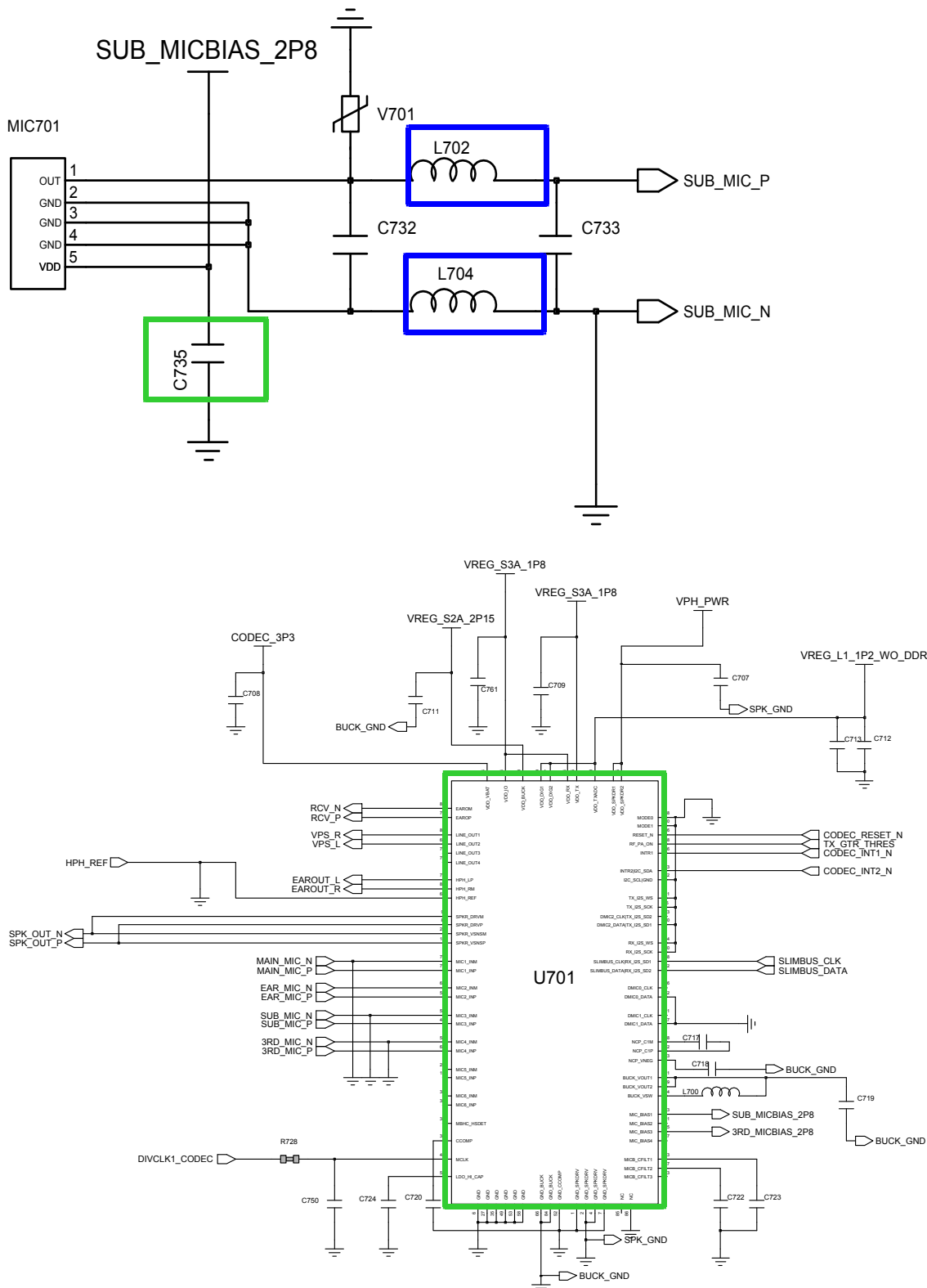


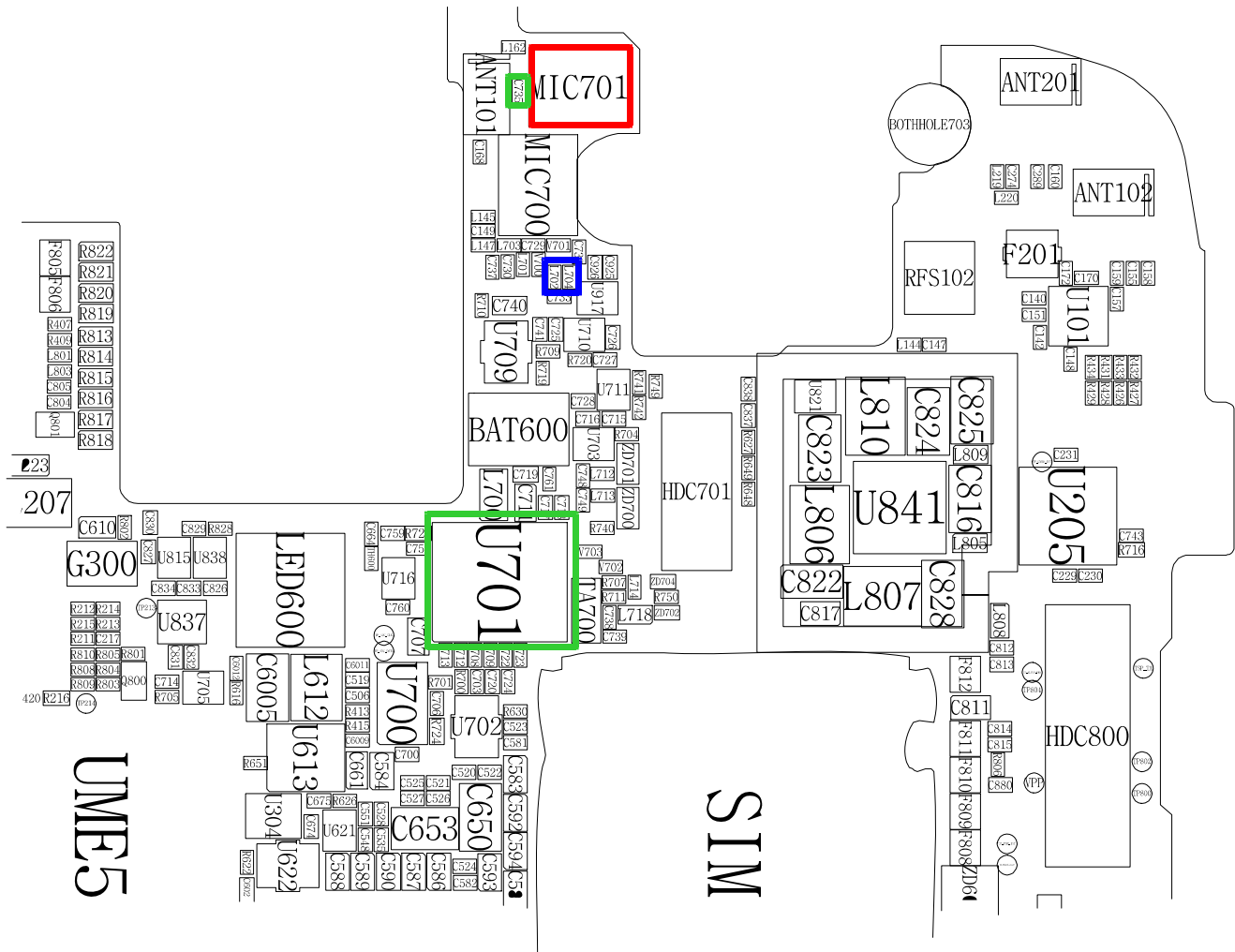




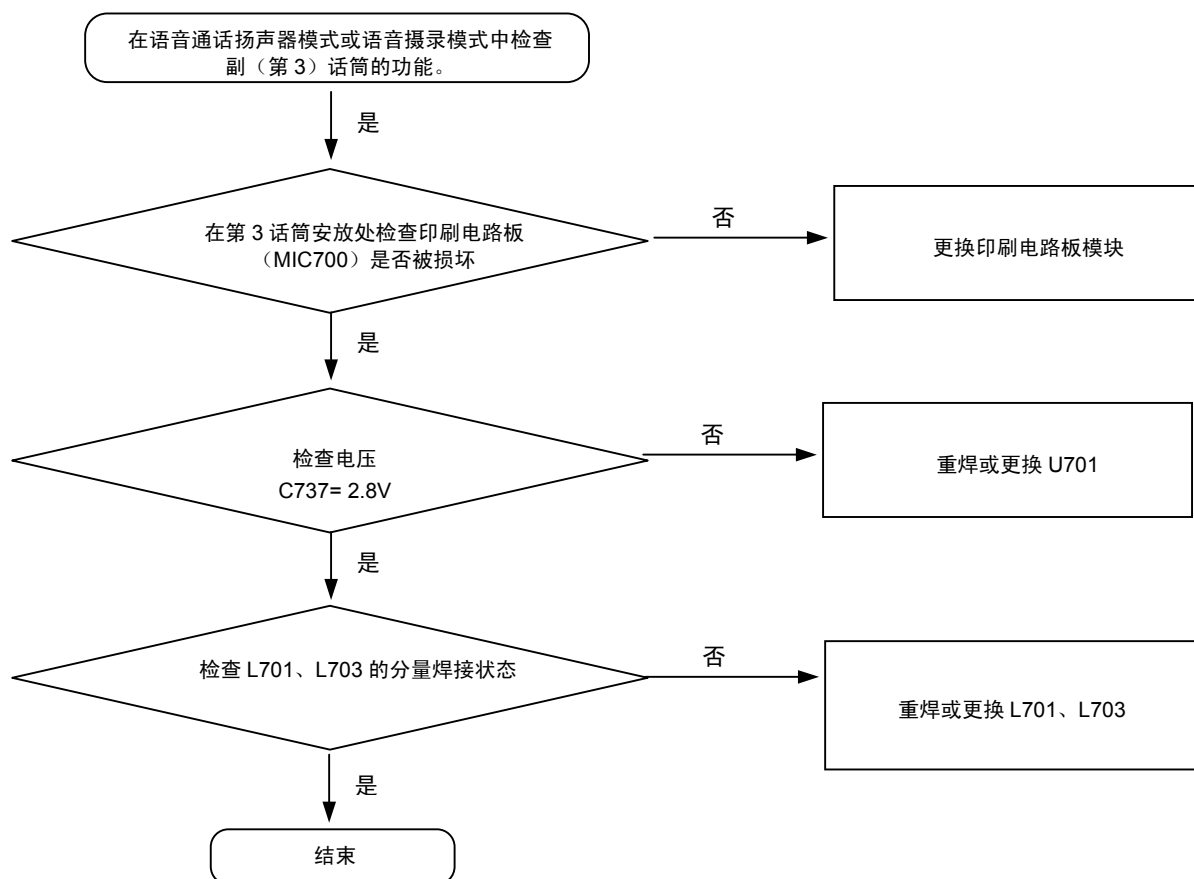
8-3-8-2. 话筒部分-副(第 2)话筒

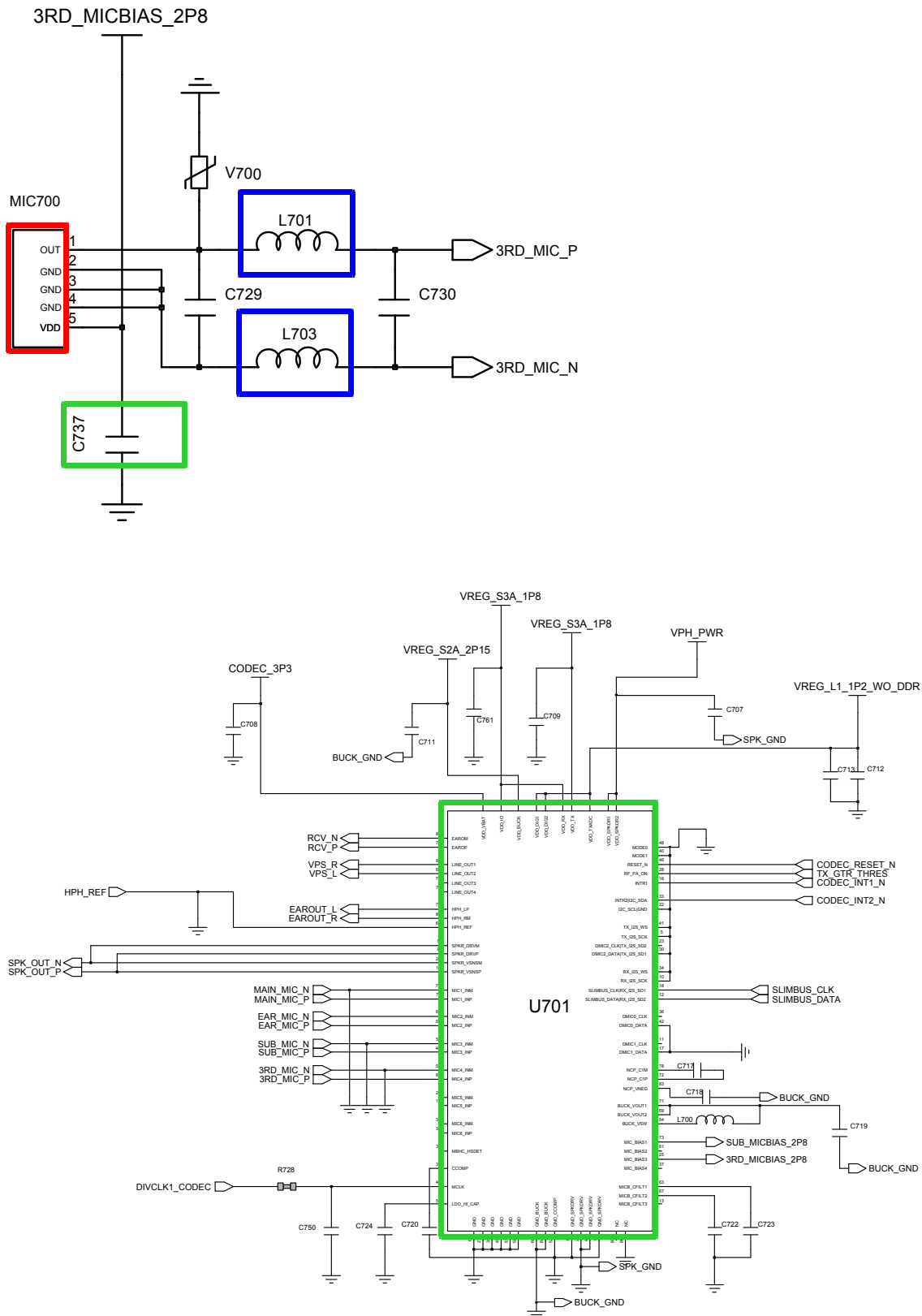




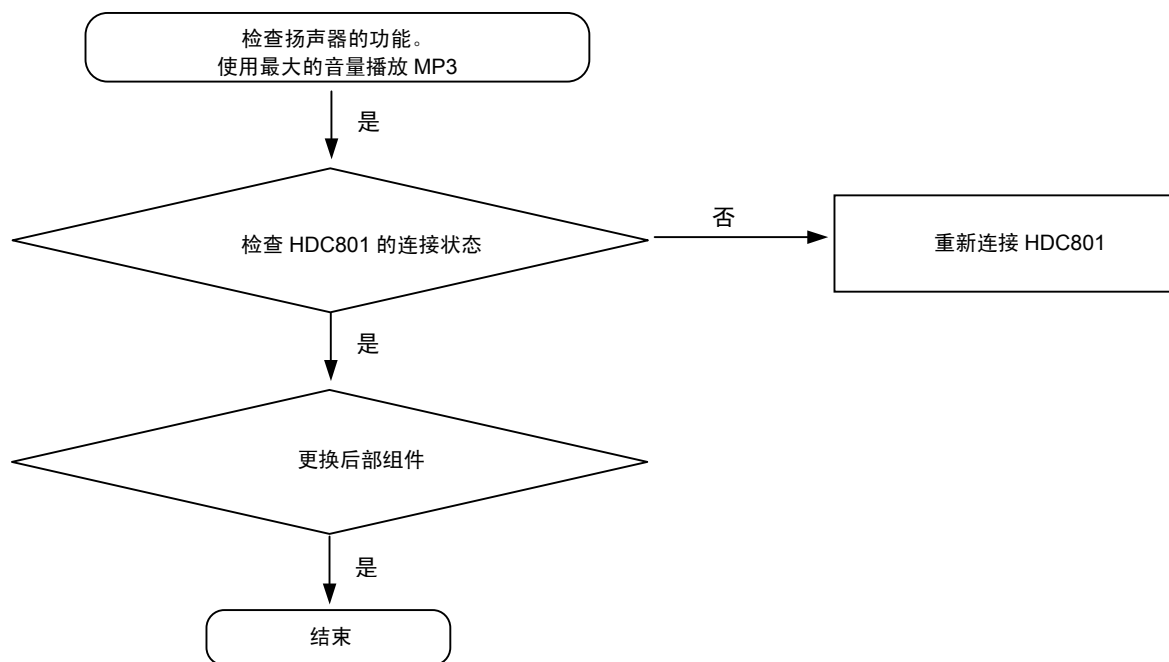


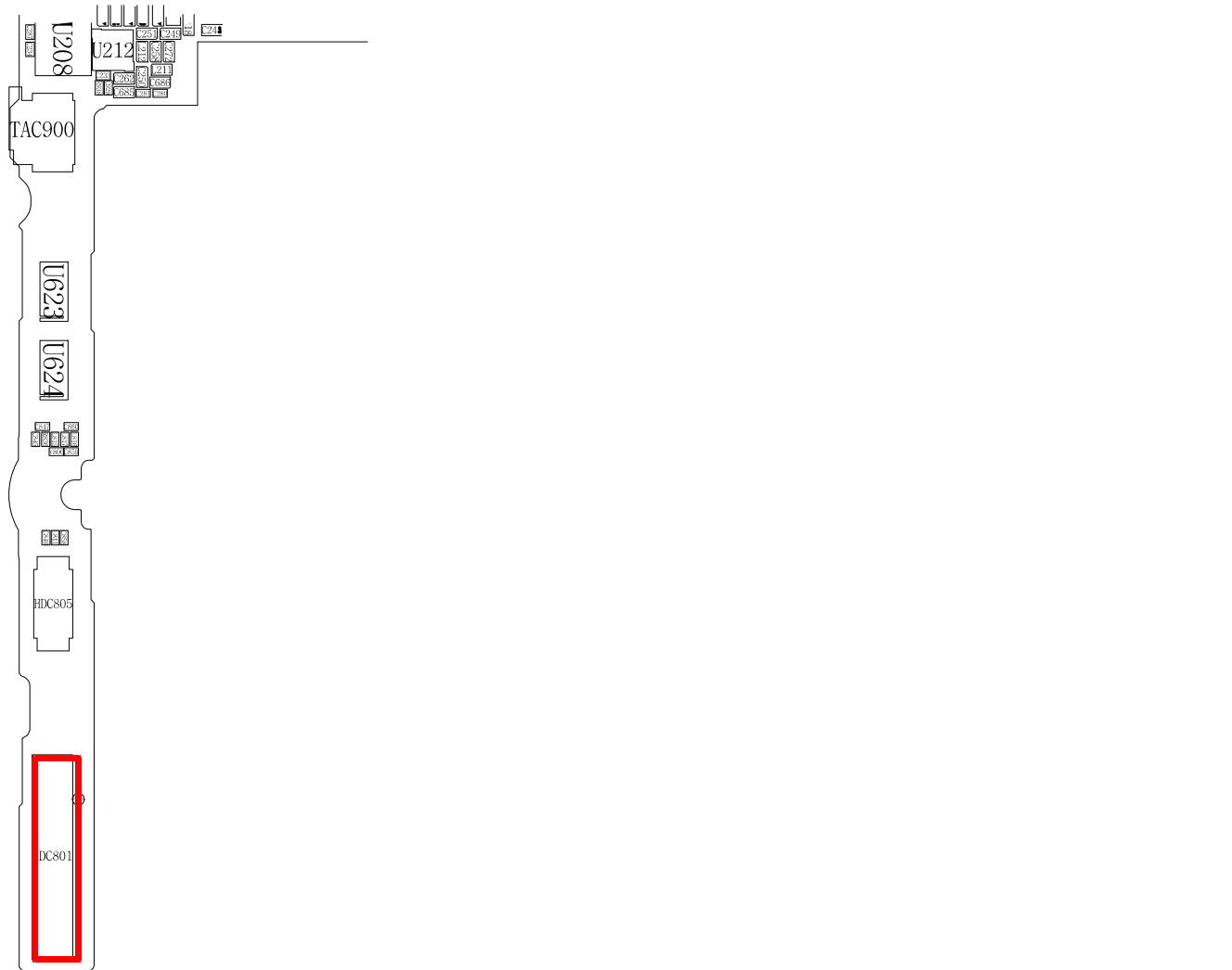
8-3-8-3. 话筒部分-副(第 3)话筒



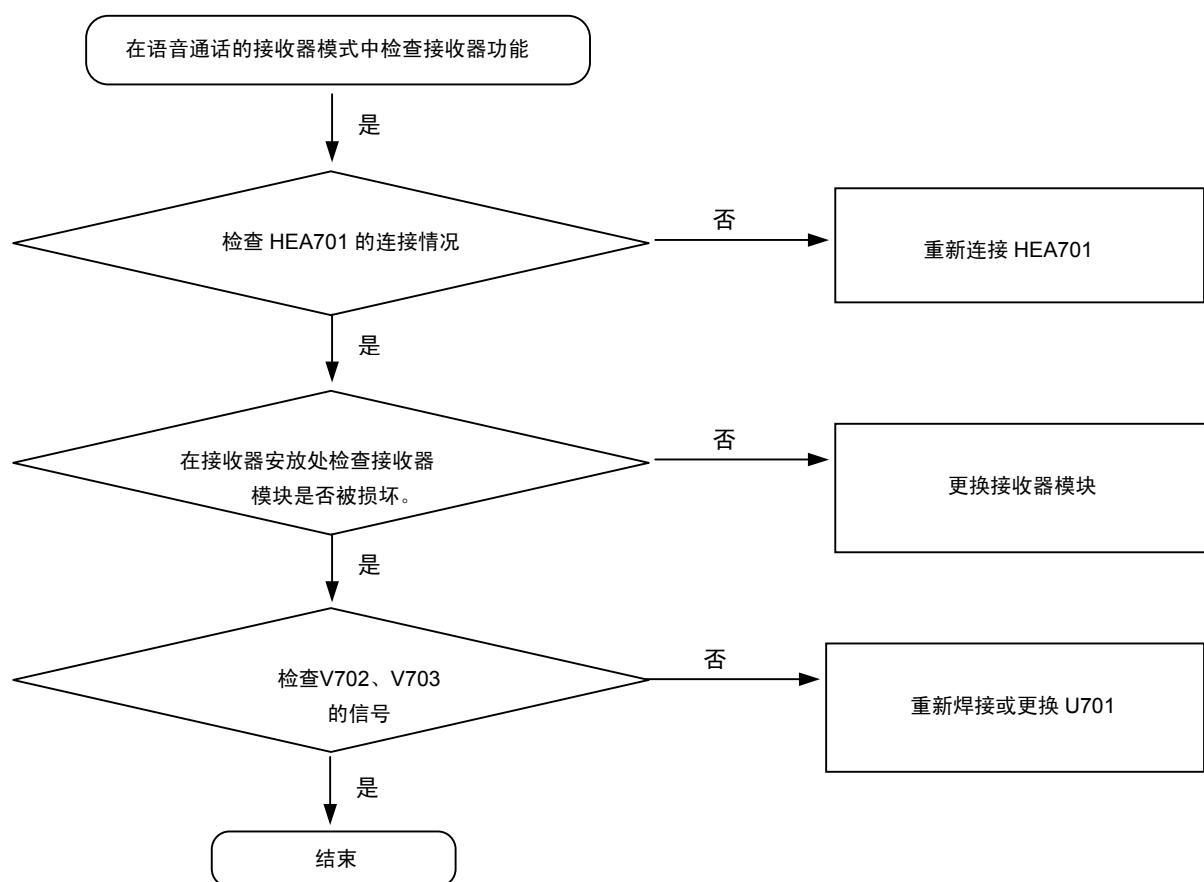


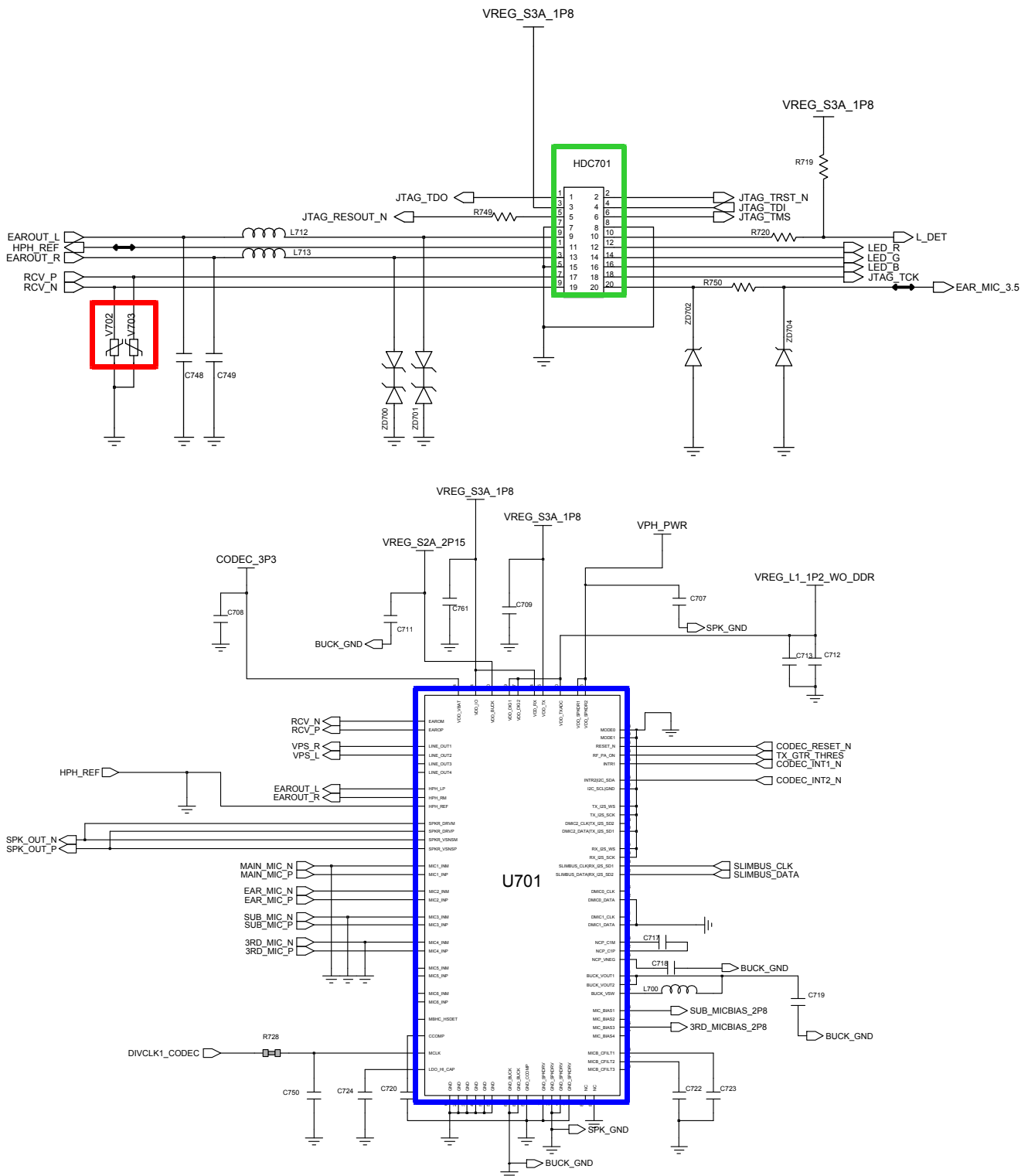
8-3-9.扬声器部分



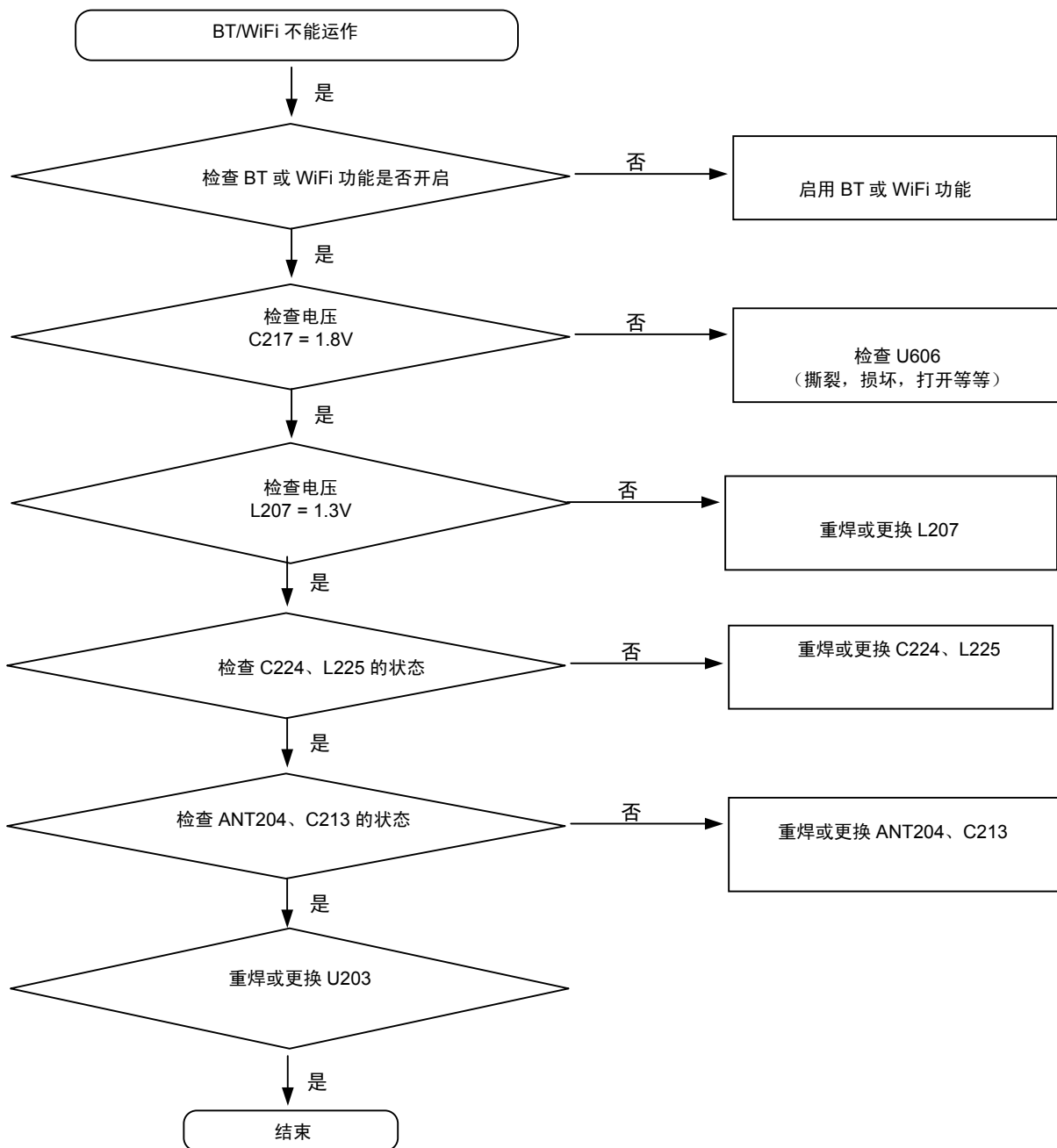


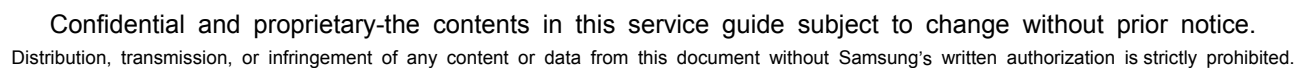
8-3-10.接收器部分



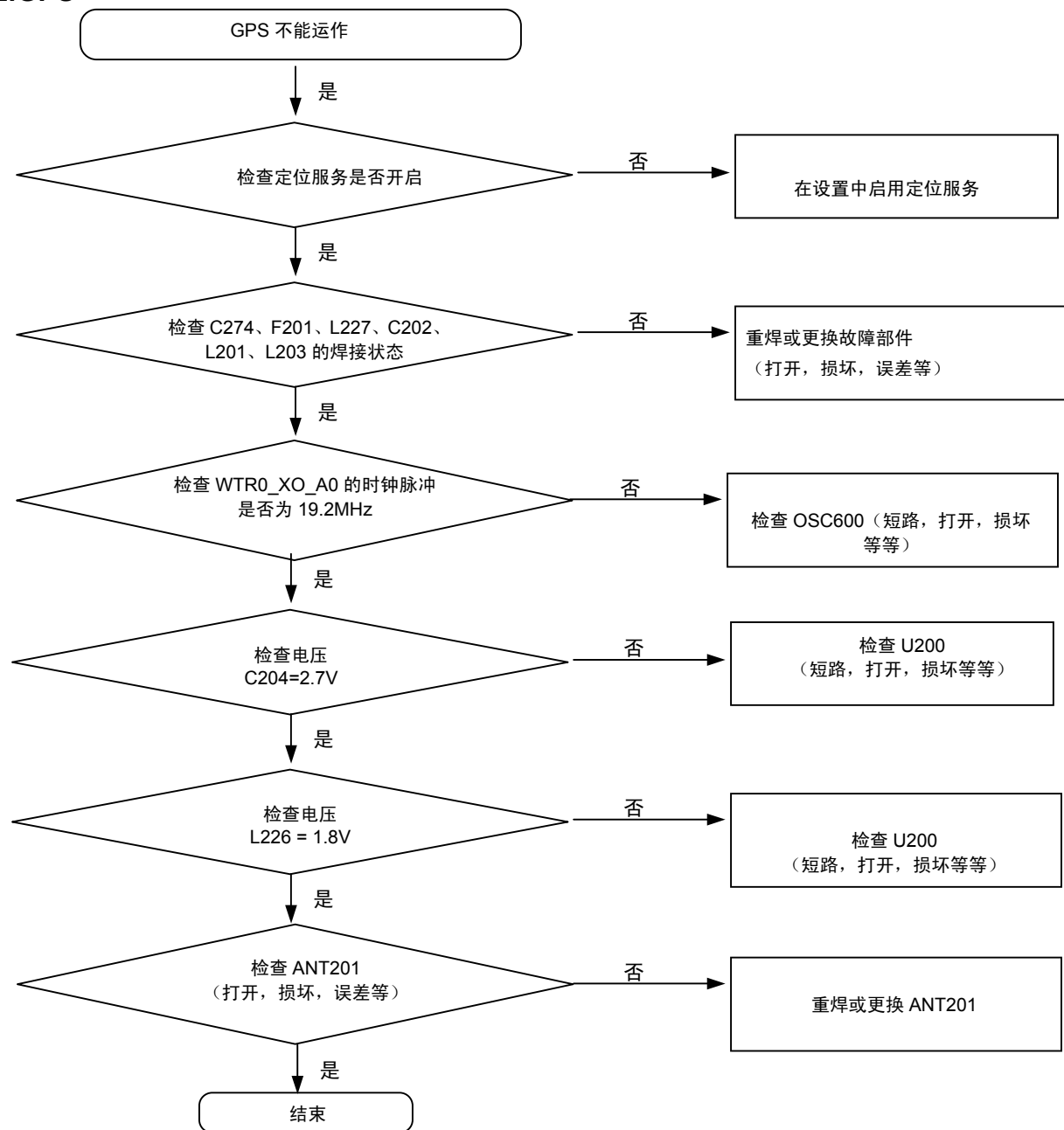


8-3-11.BT/WIFI

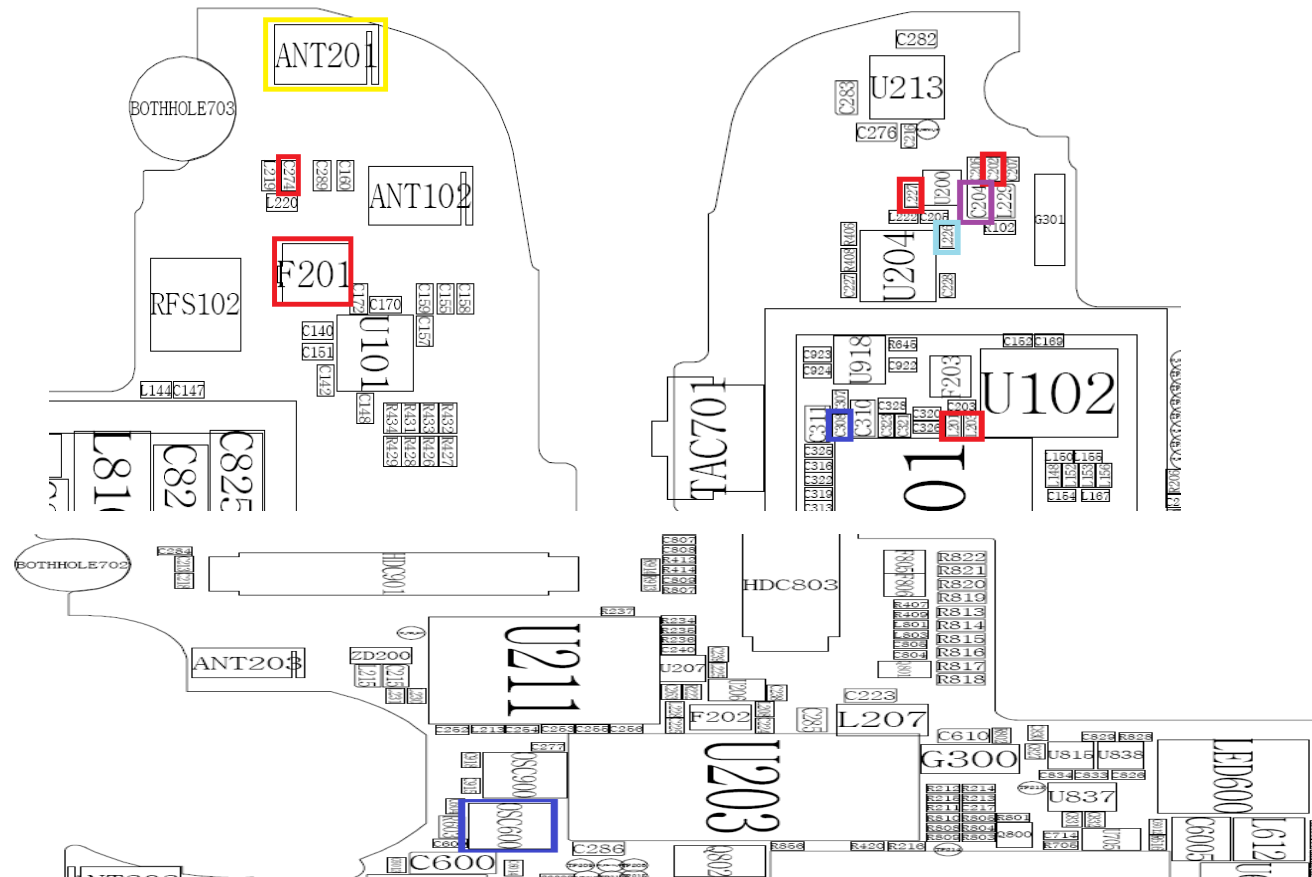
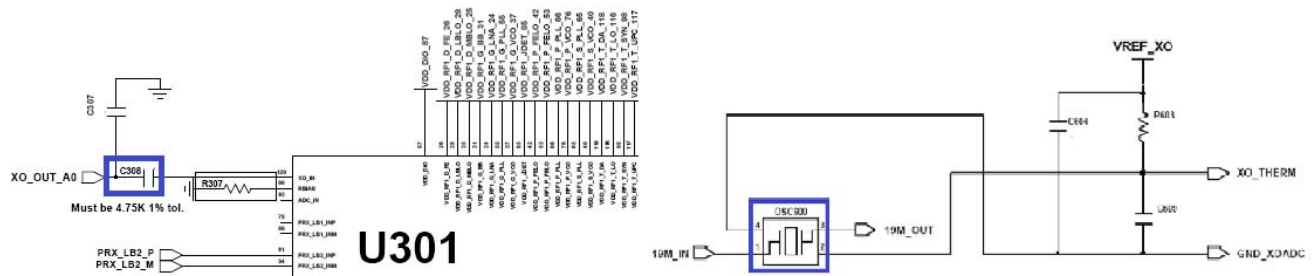
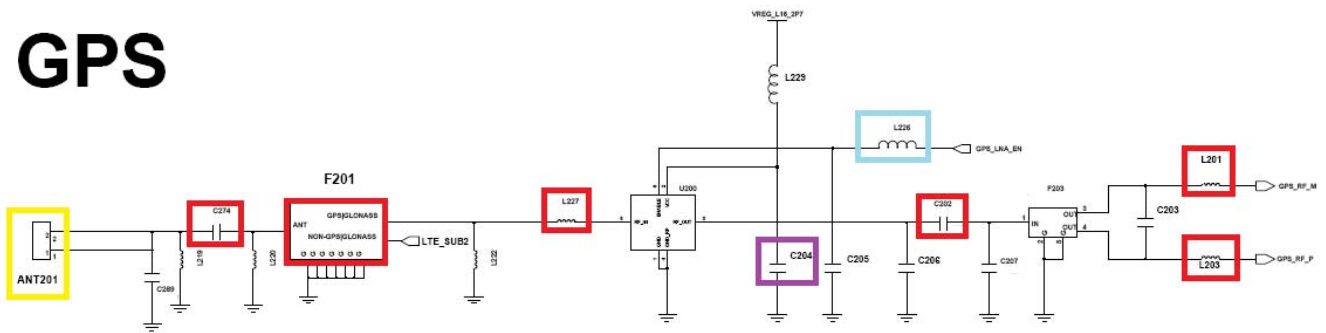




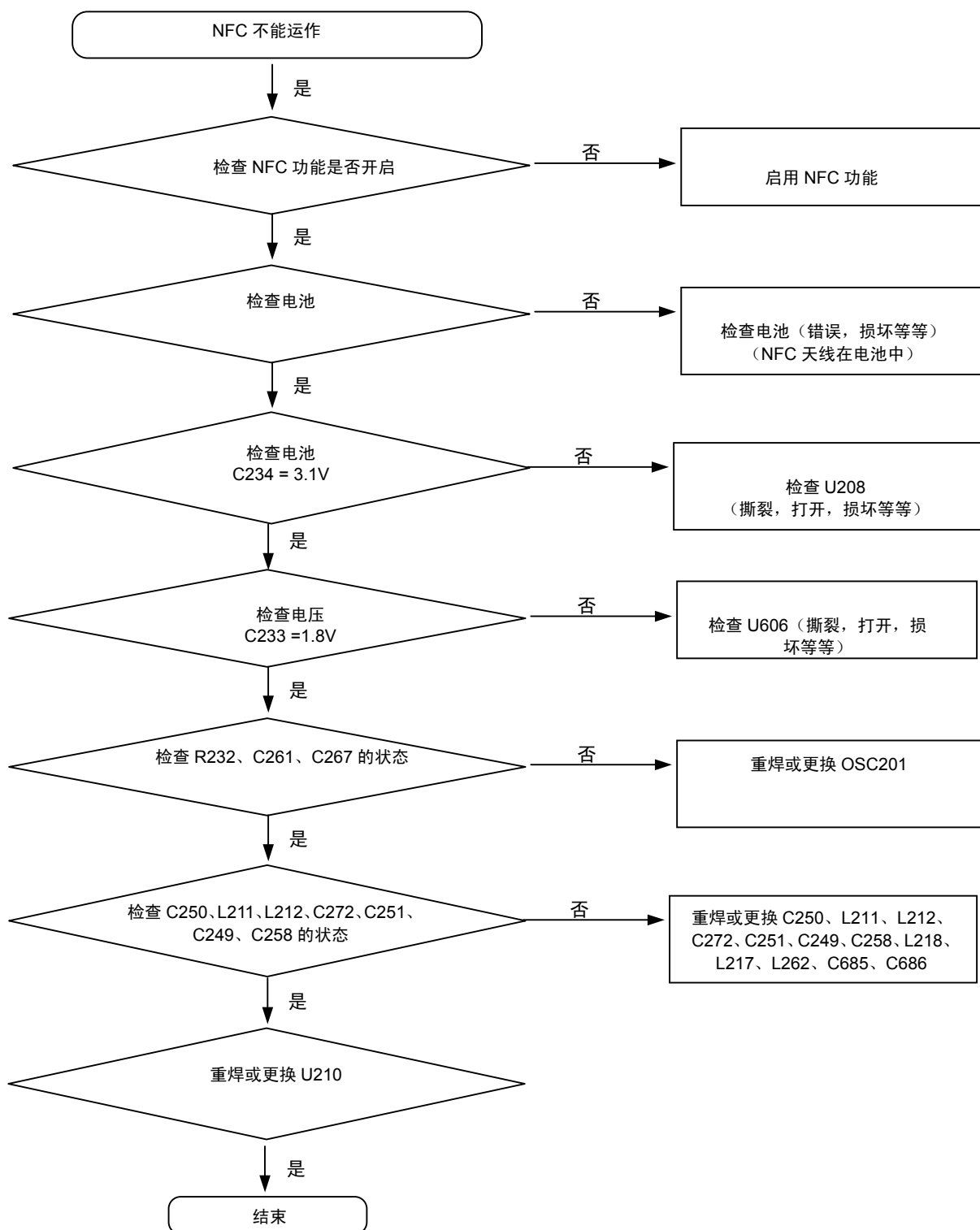
8-3-12.GPS

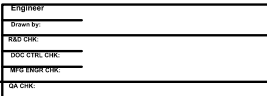


GPS

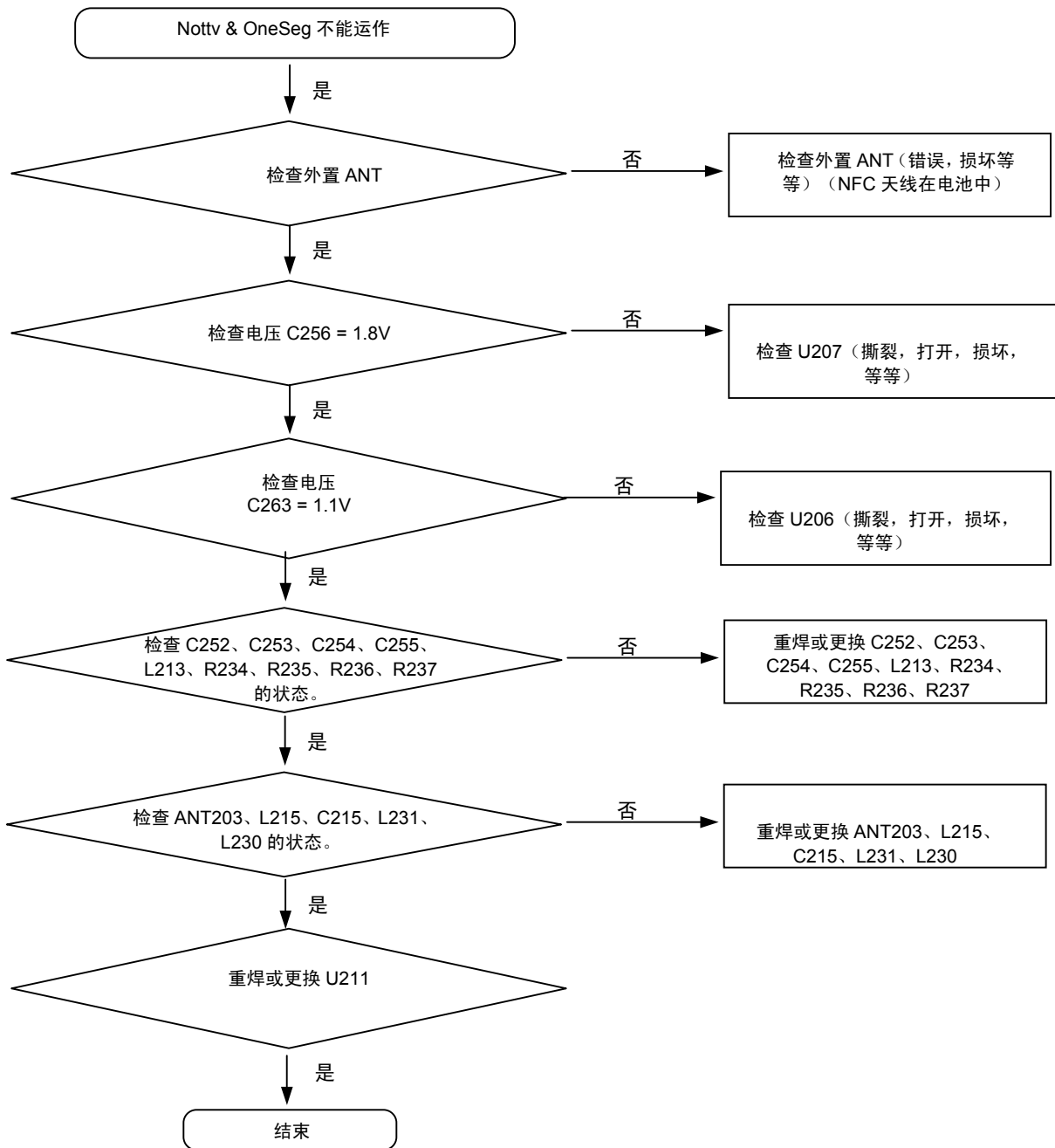


8-3-13.NFC/FeliCa

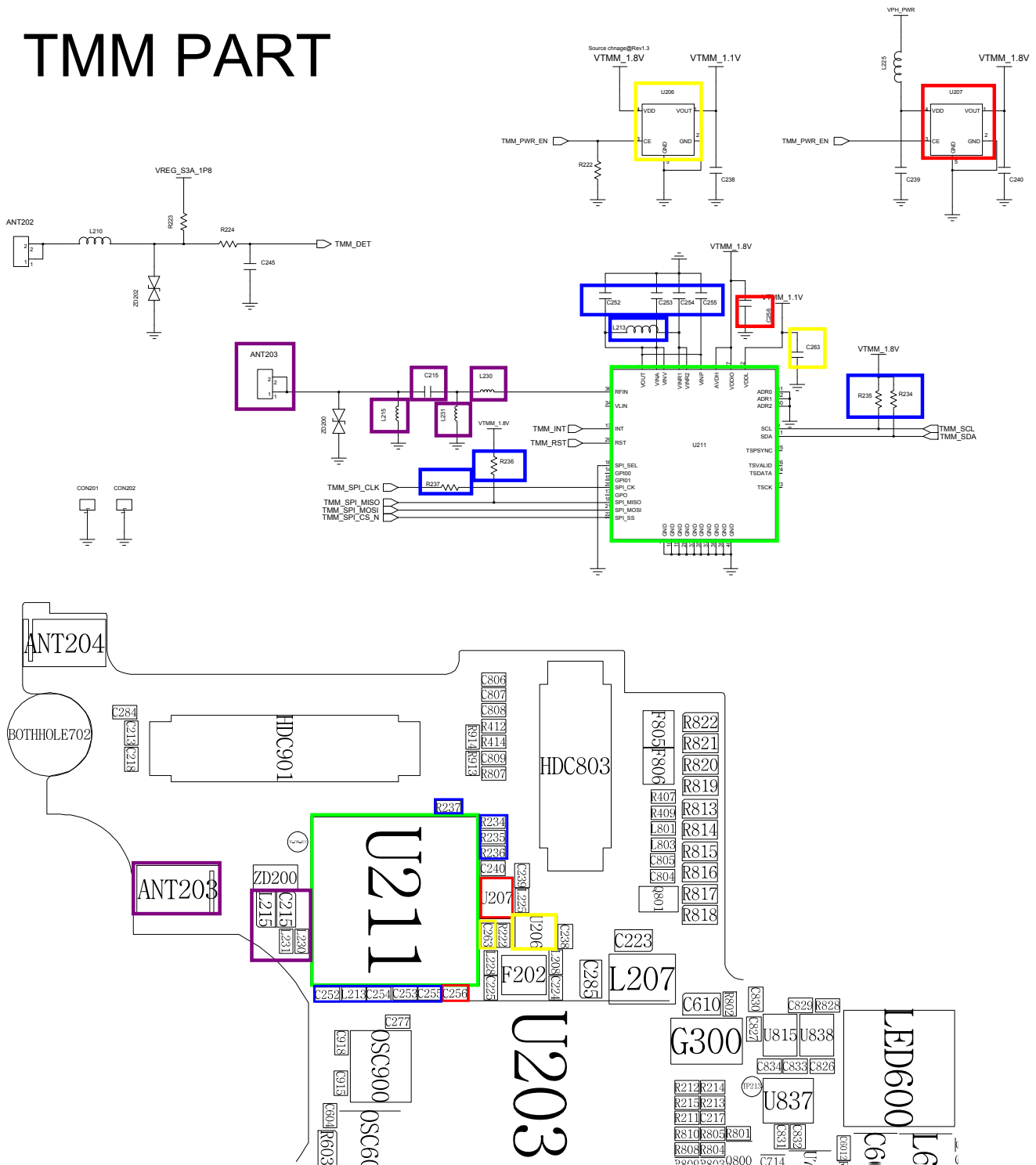




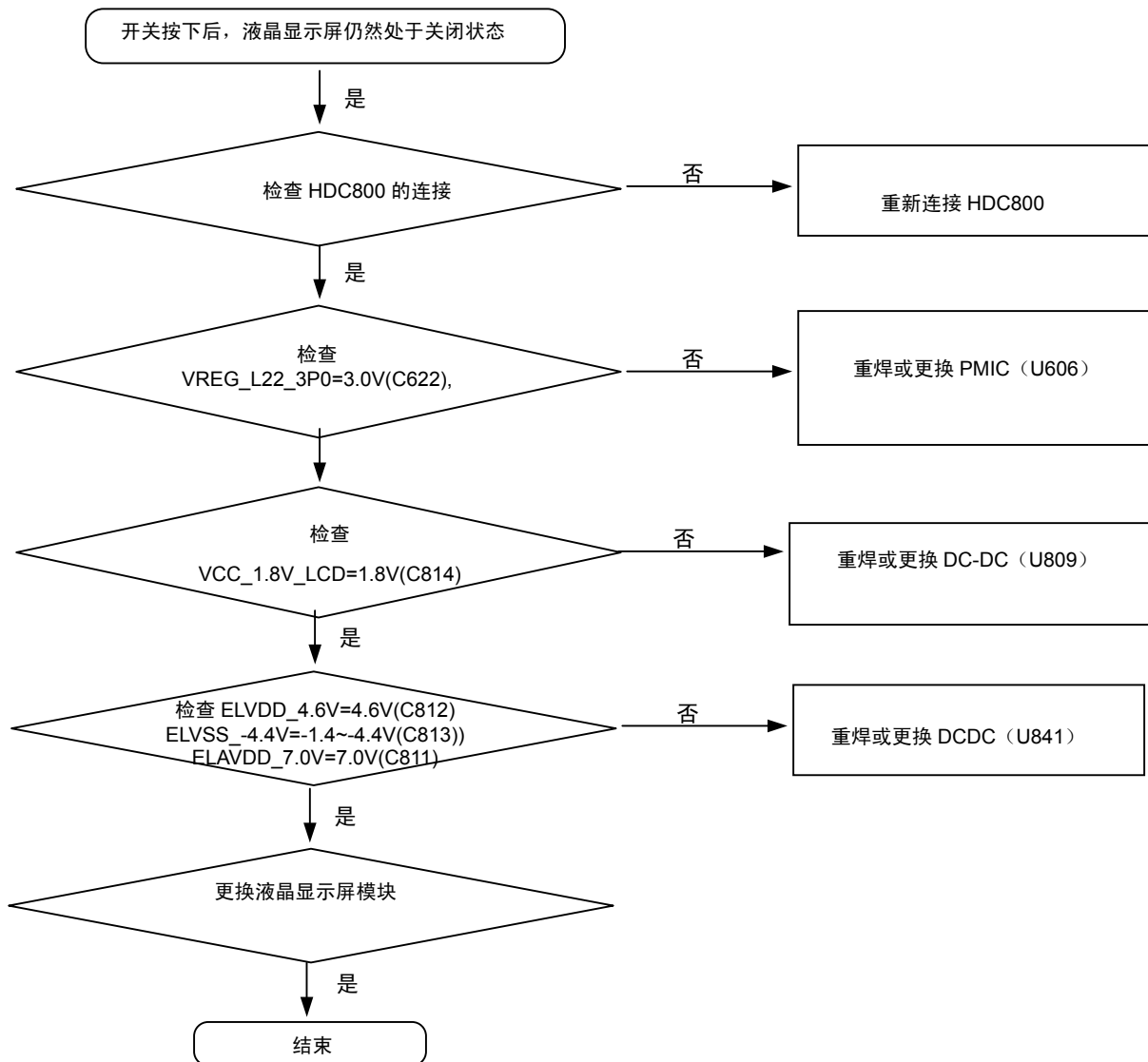
8-3-13-1.Nottv & OneSeg

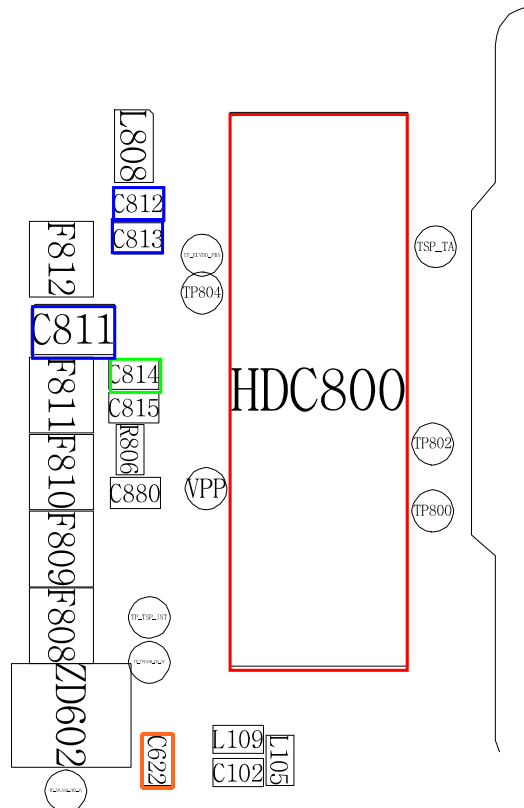
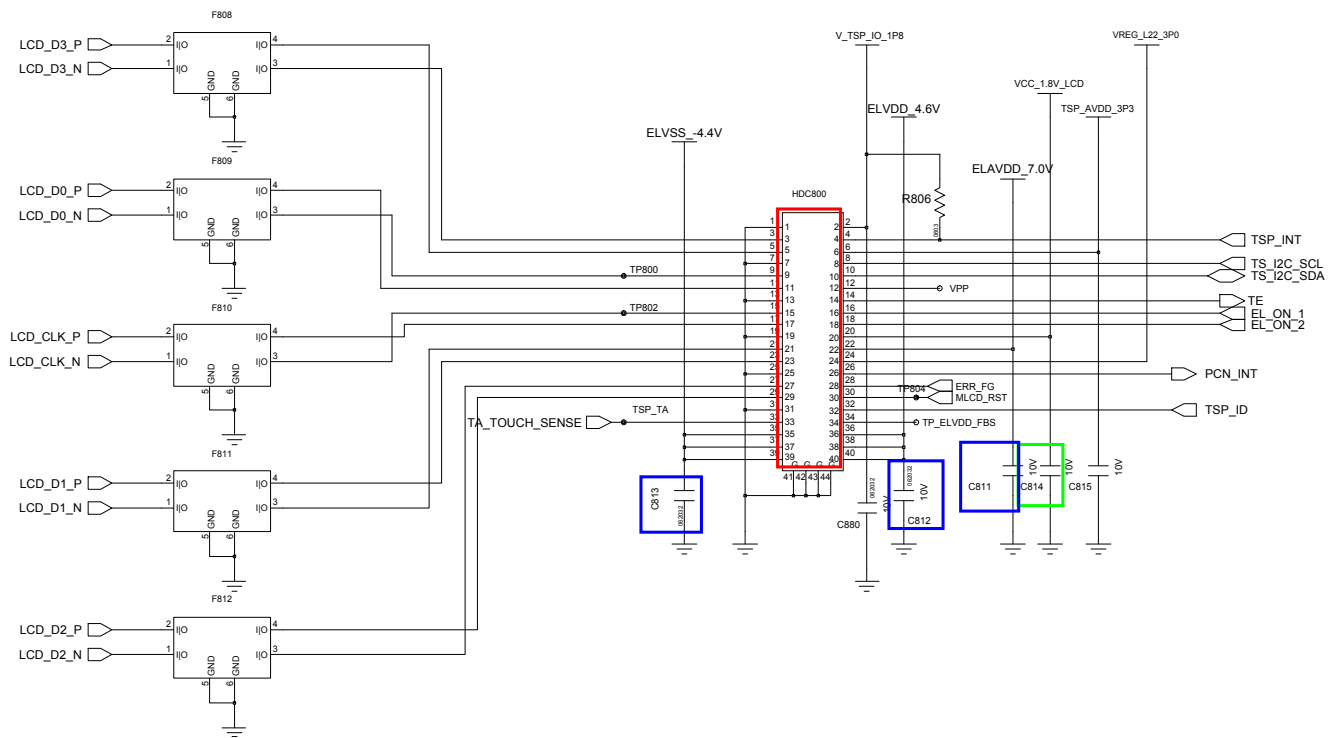


TMM PART

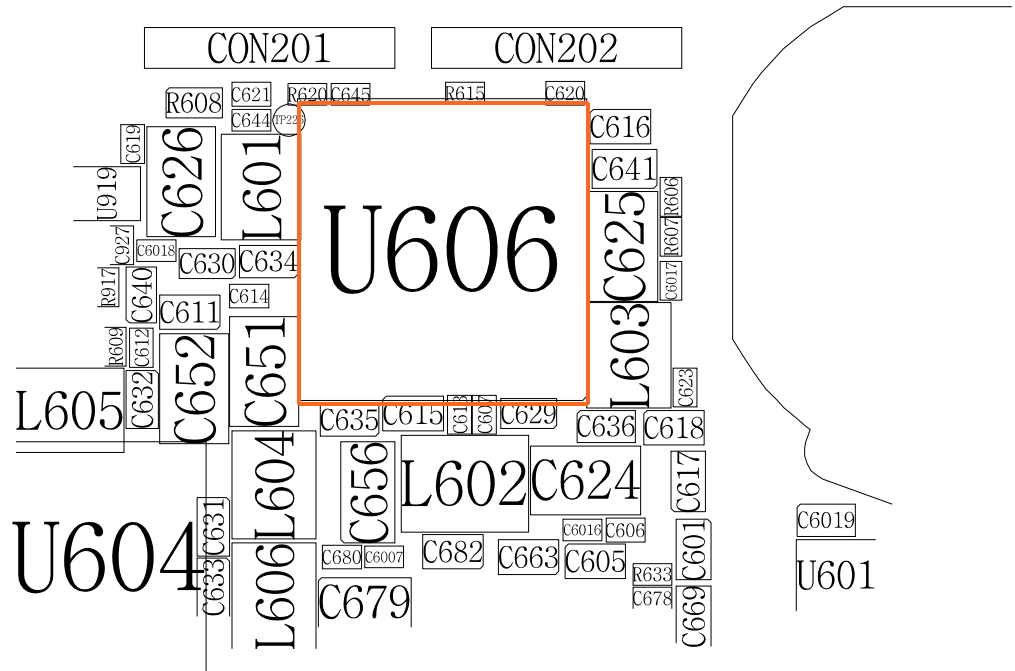


8-3-14.液晶显示屏

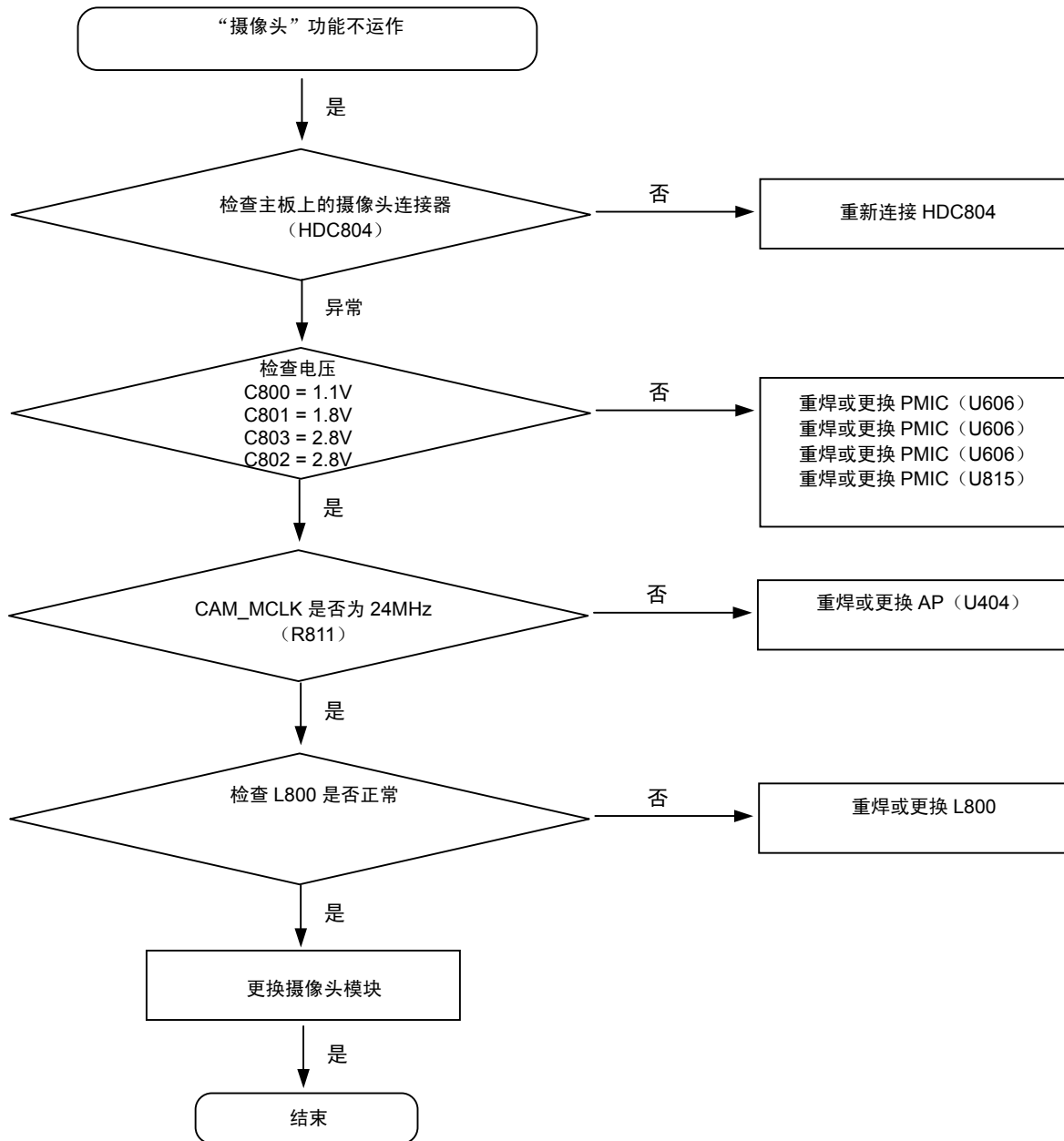


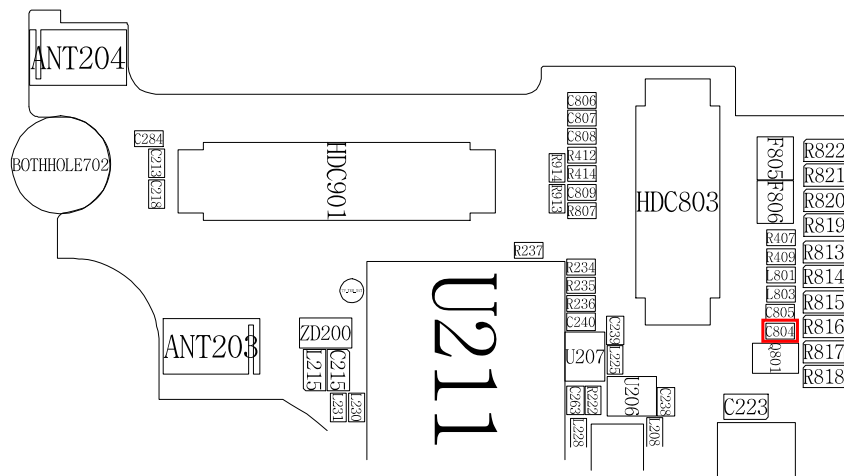
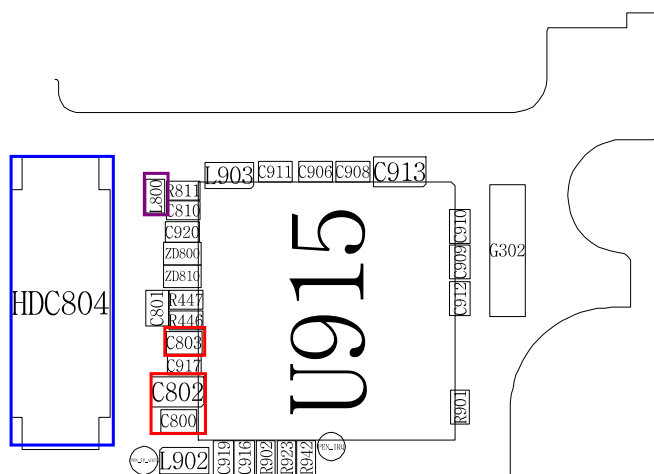
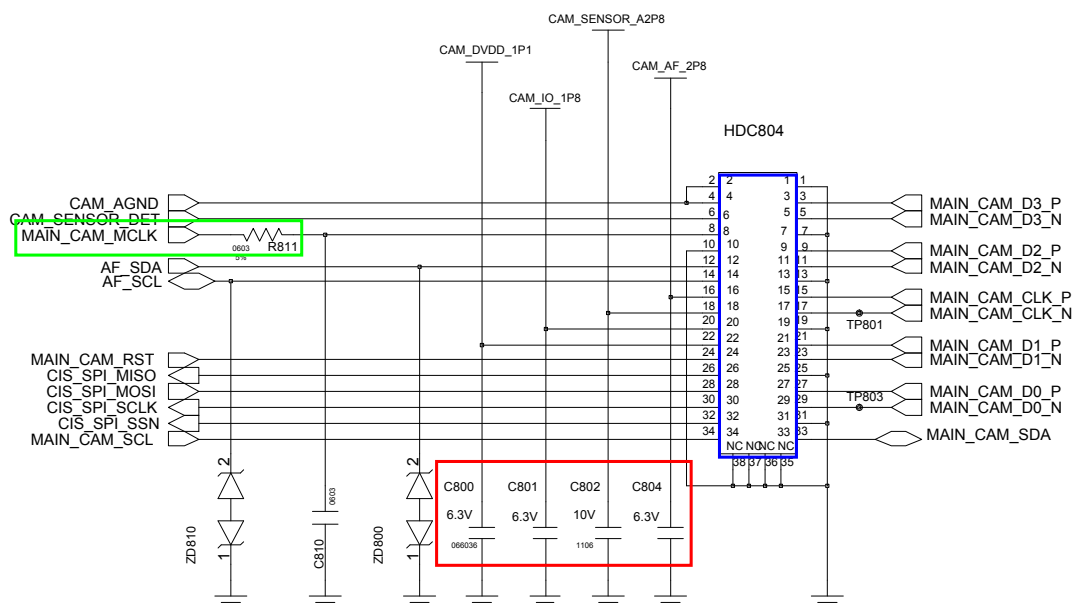


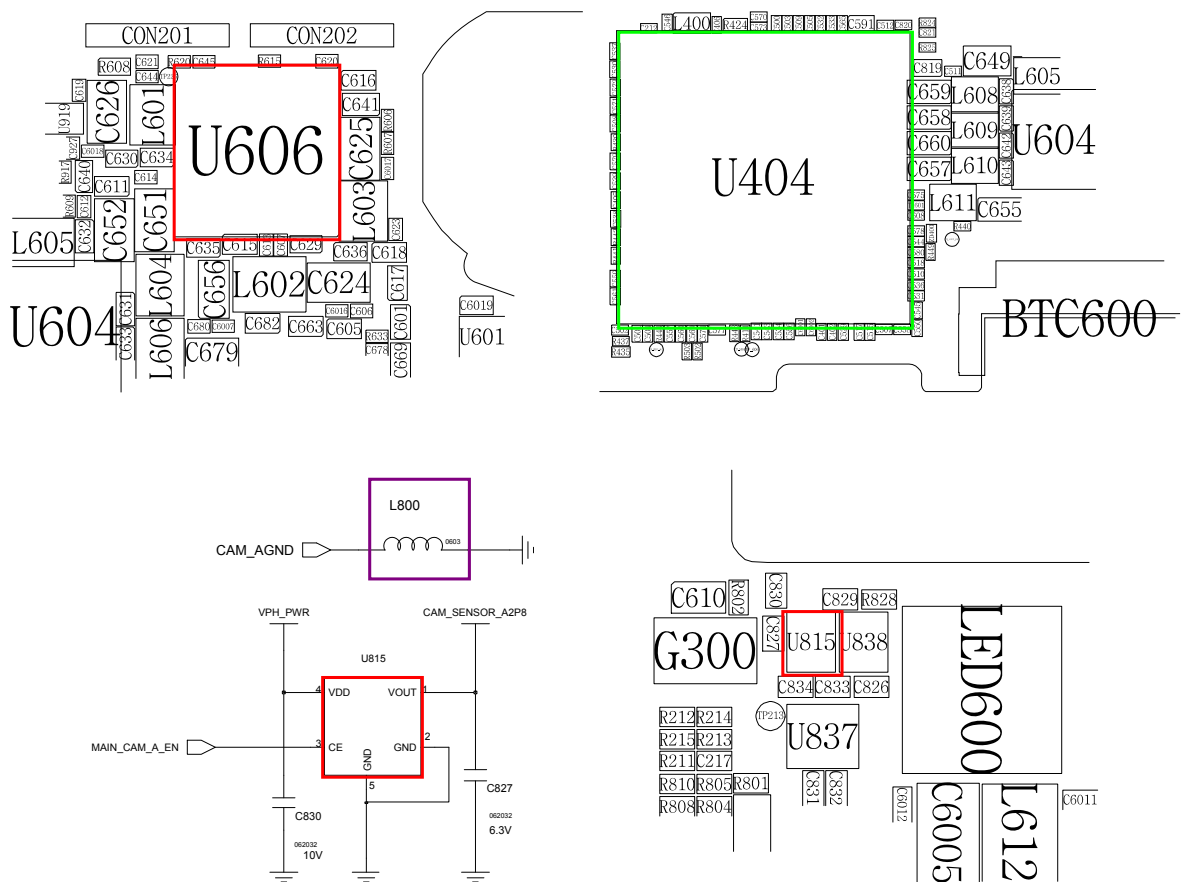




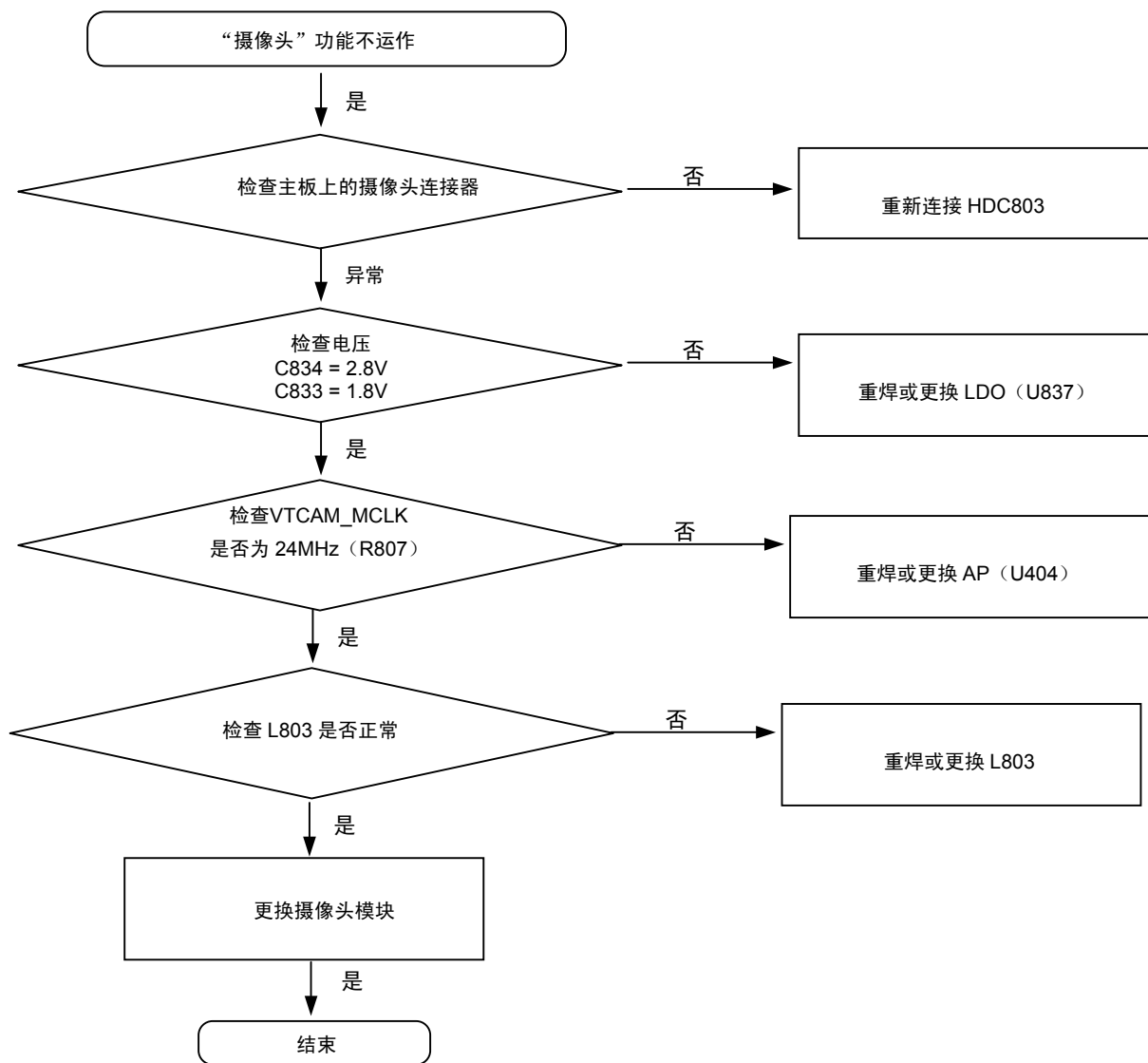
8-3-15.13M 摄像头

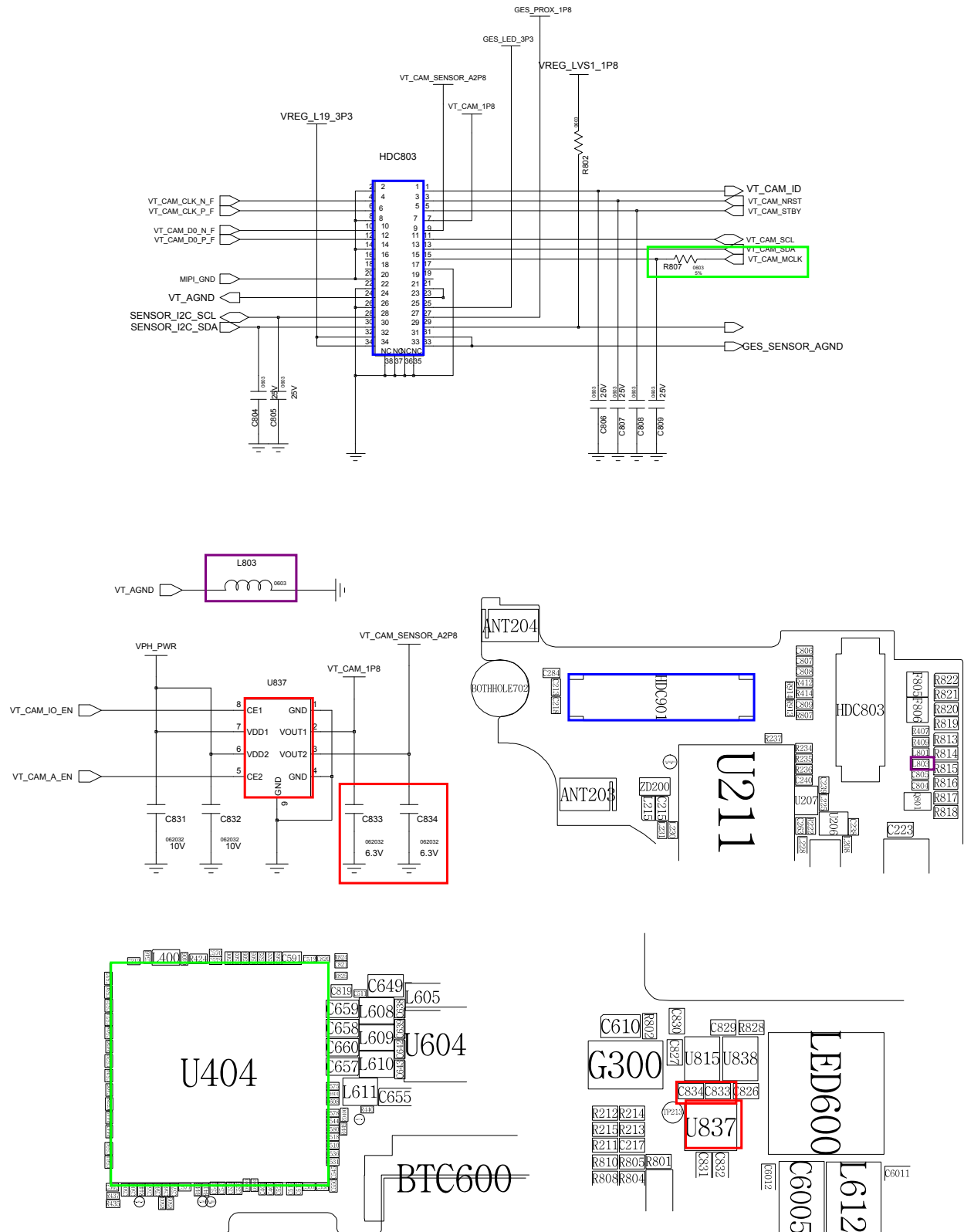




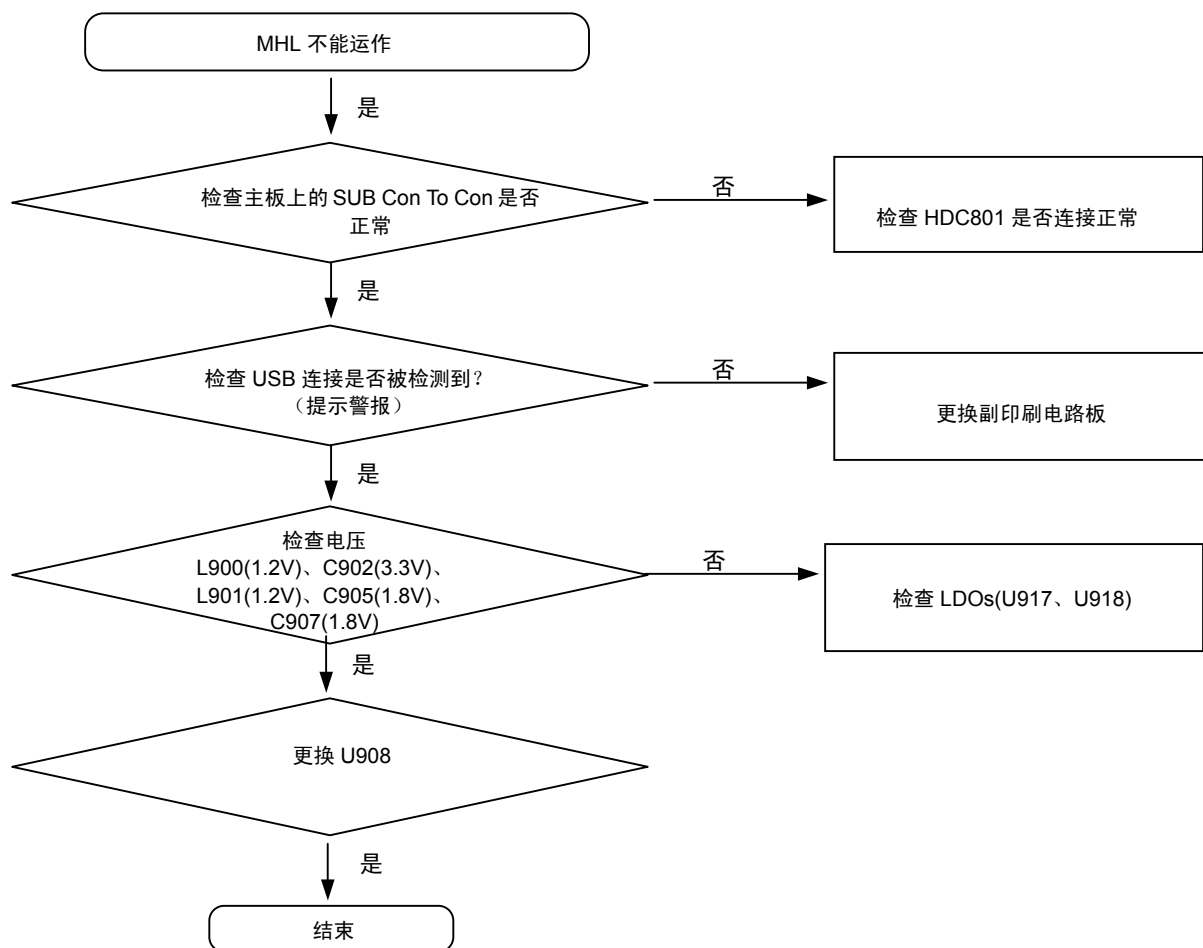


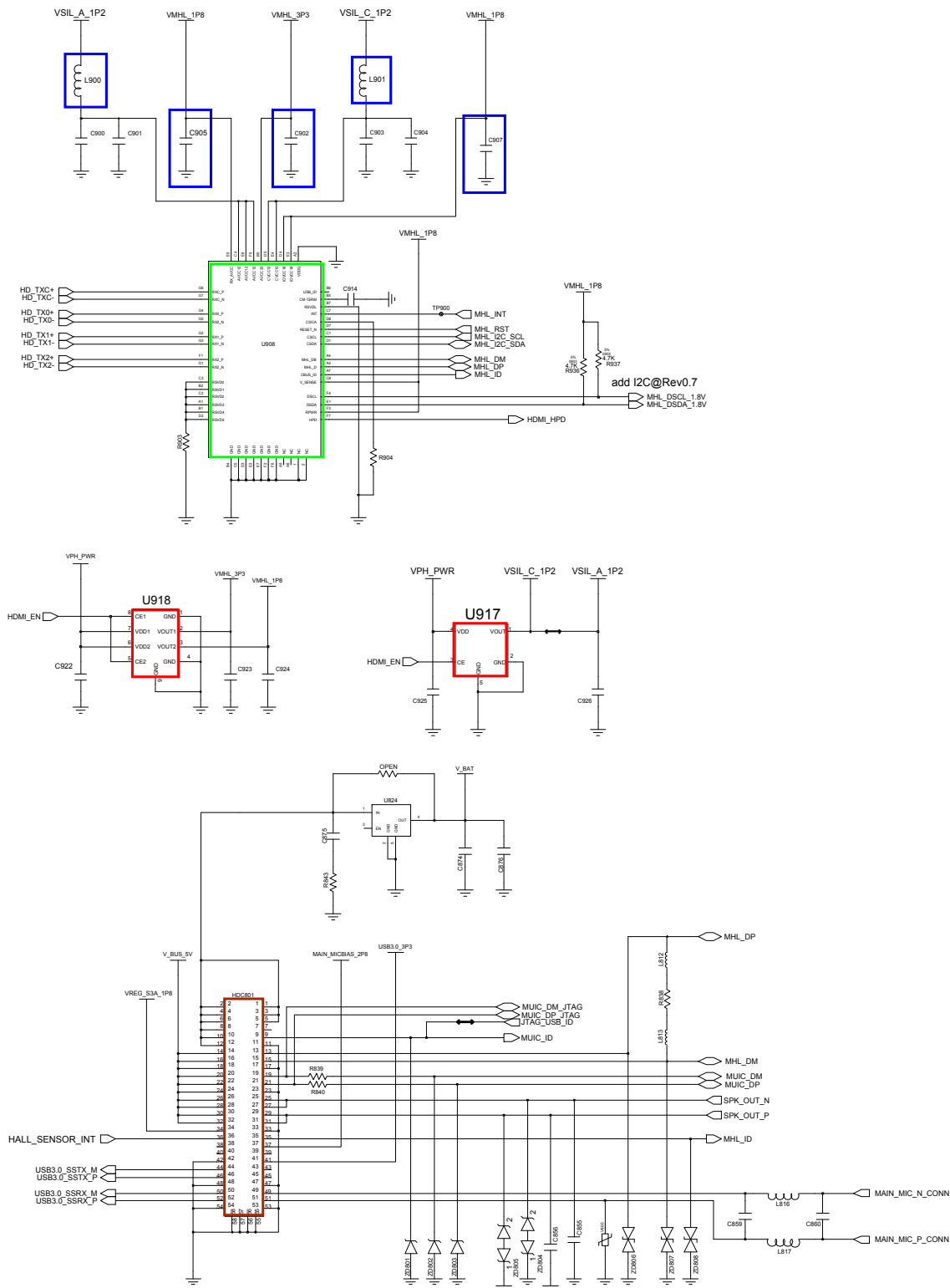
8-3-16.VT 摄像头

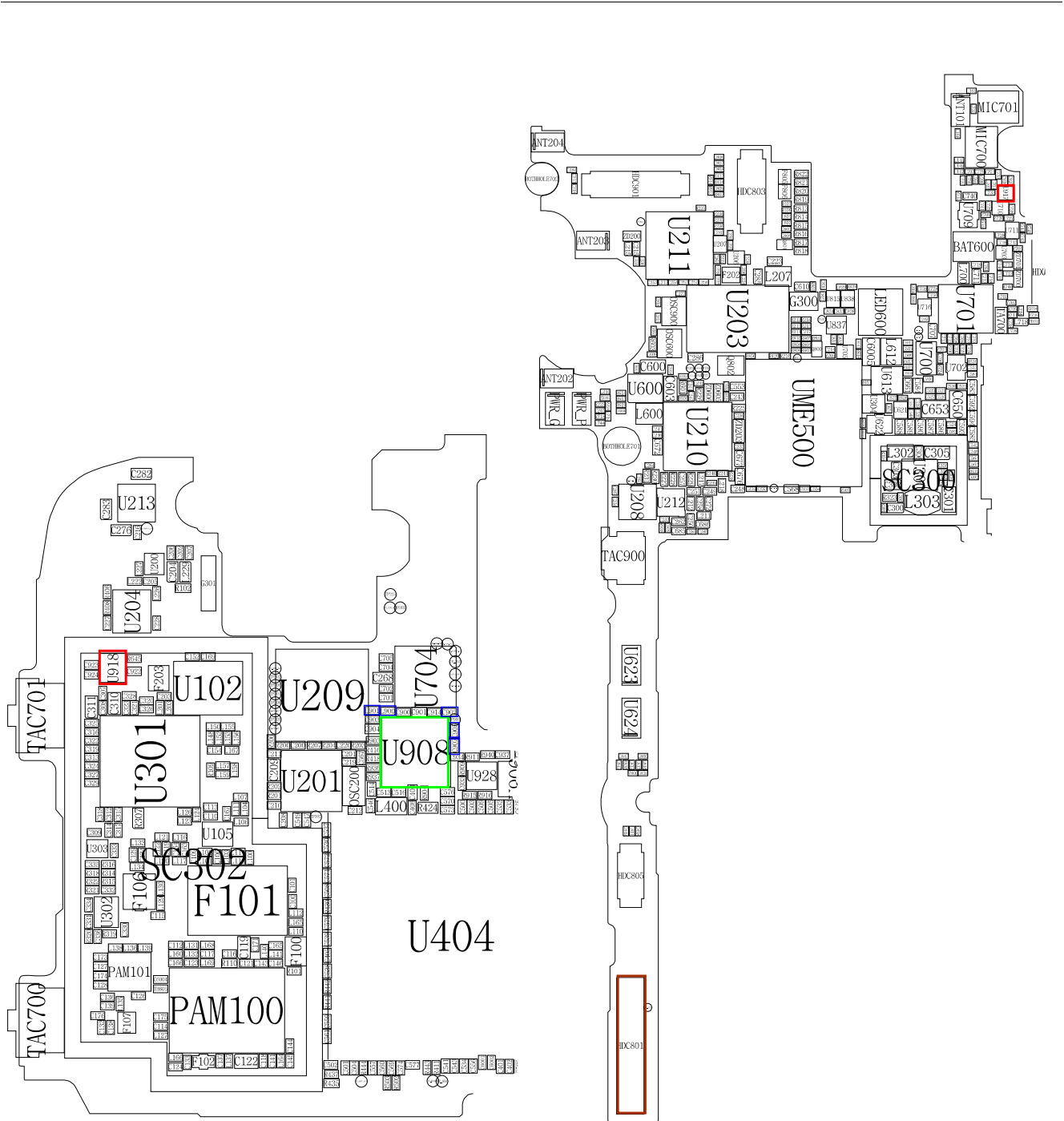




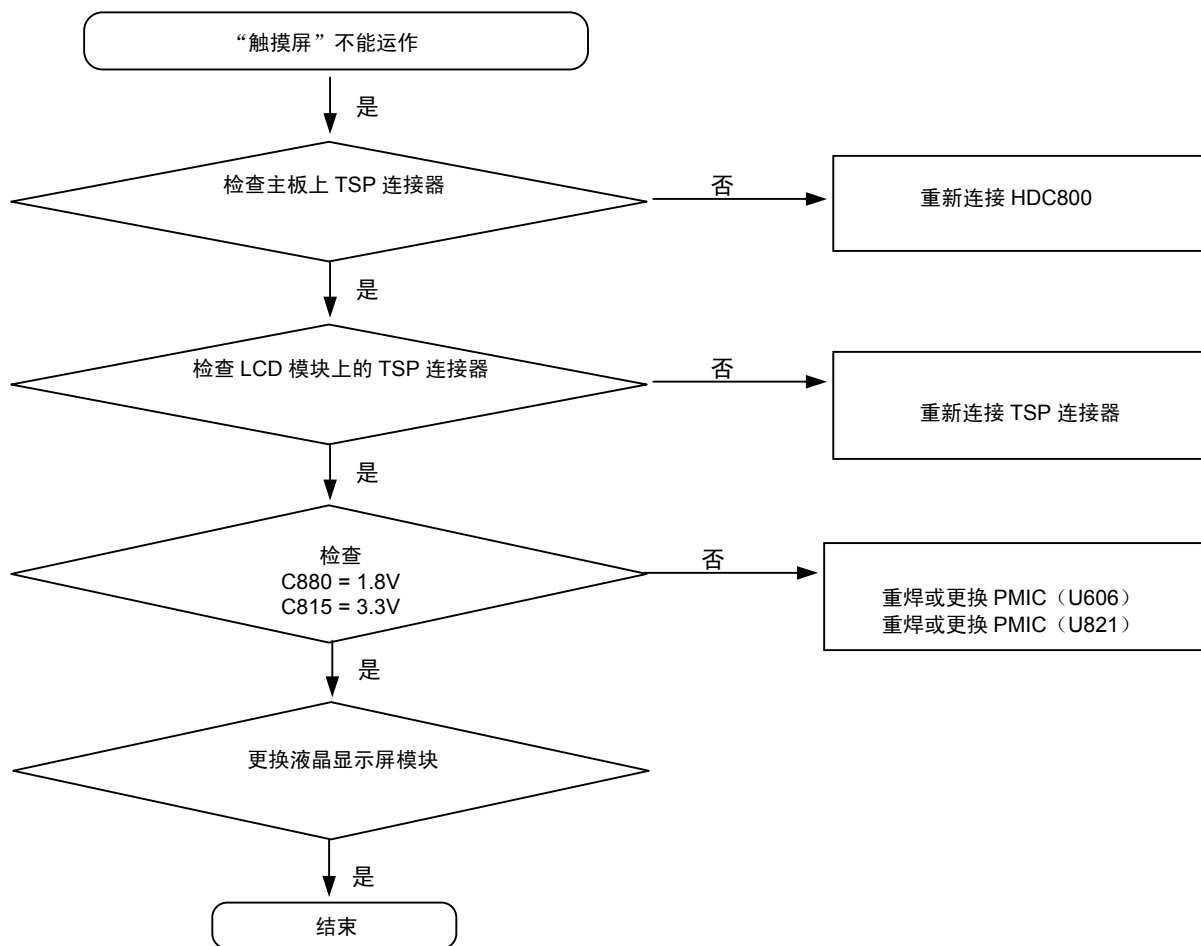
8-3-17.MHL

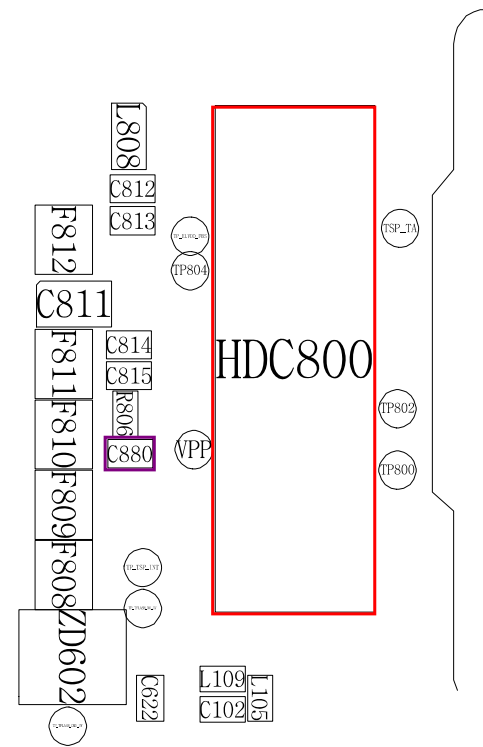
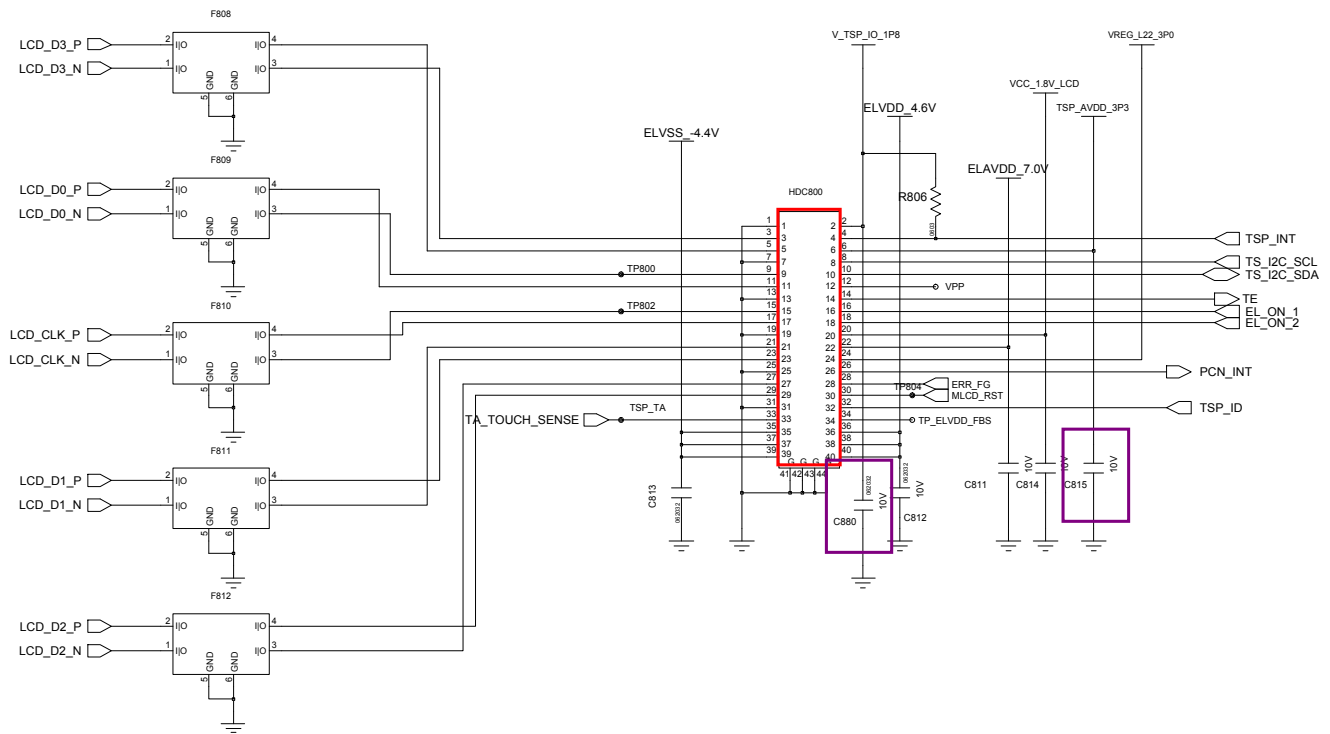


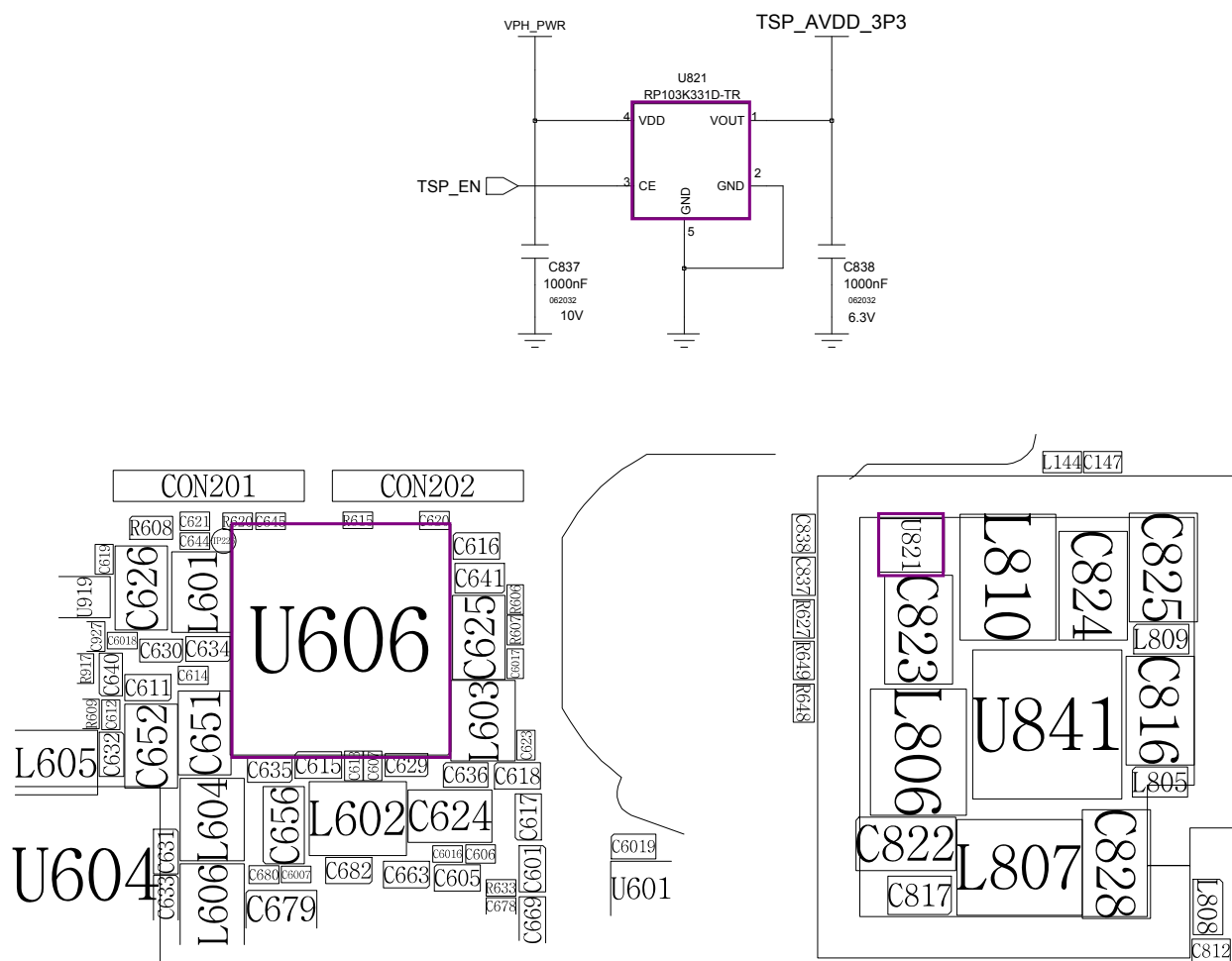




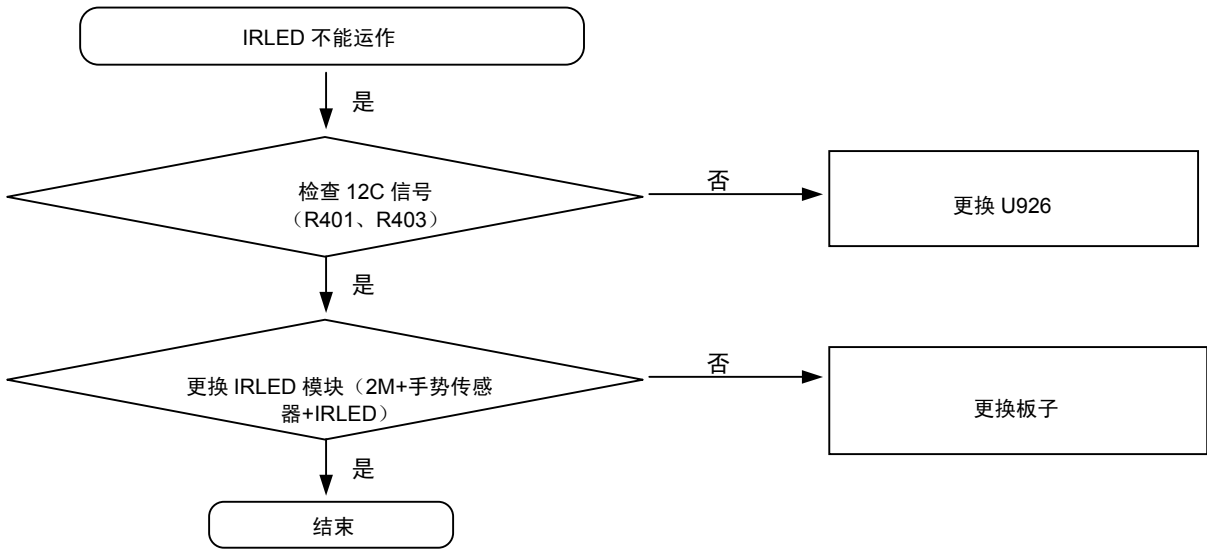
8-3-18.TSP

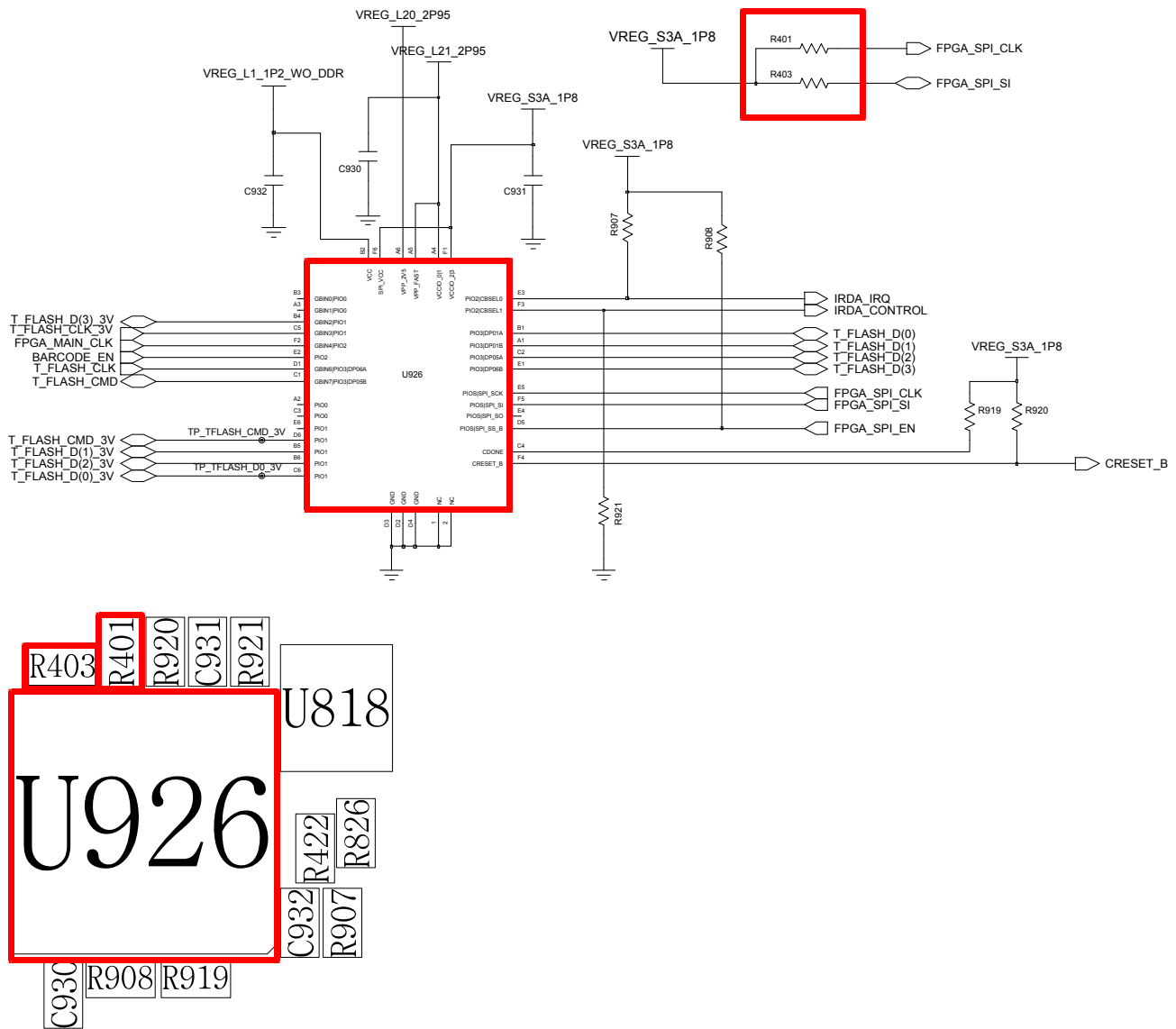




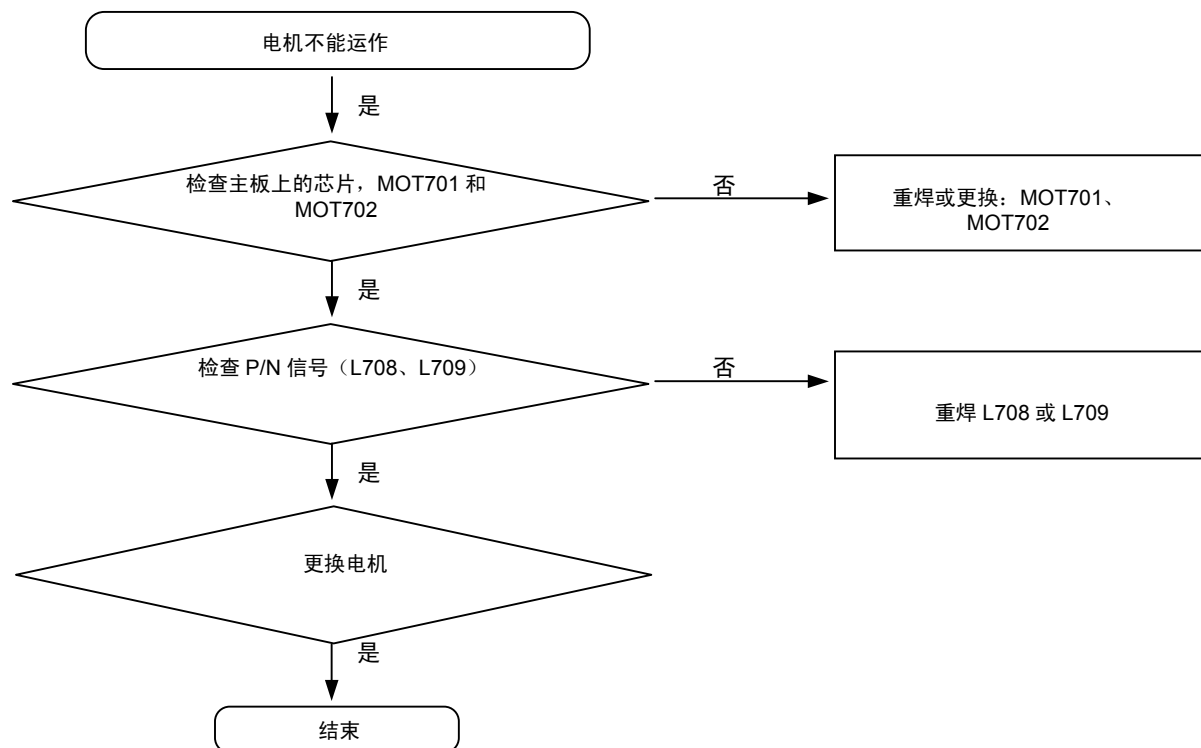


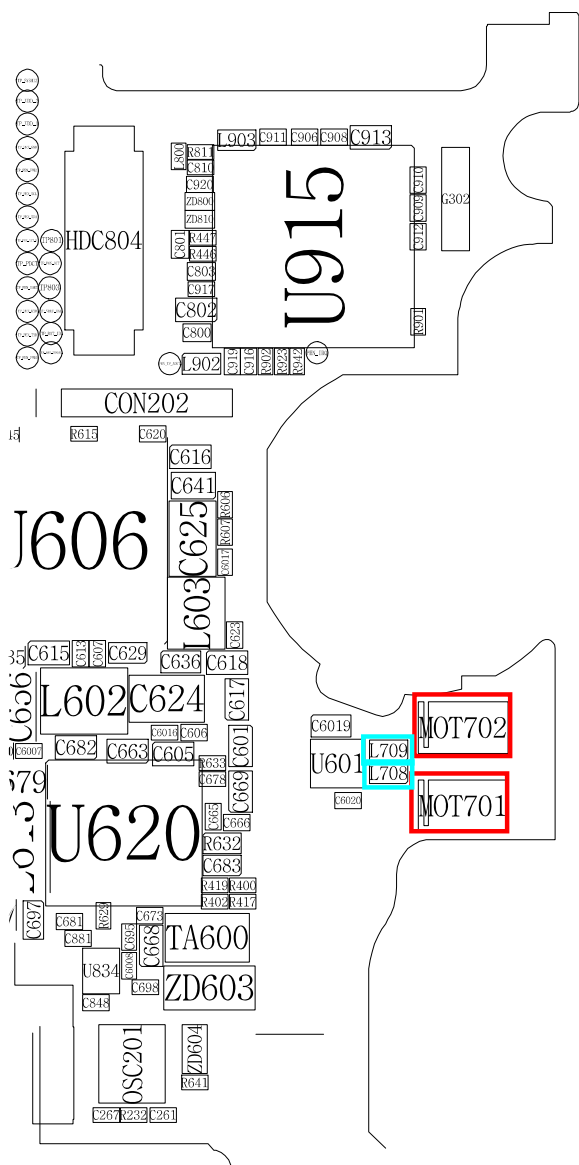
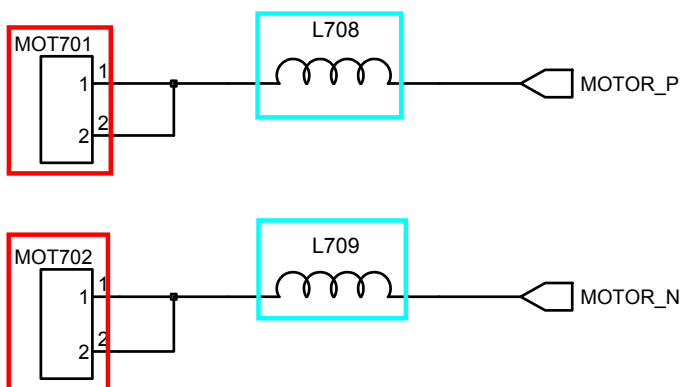
8-3-19.IRLED



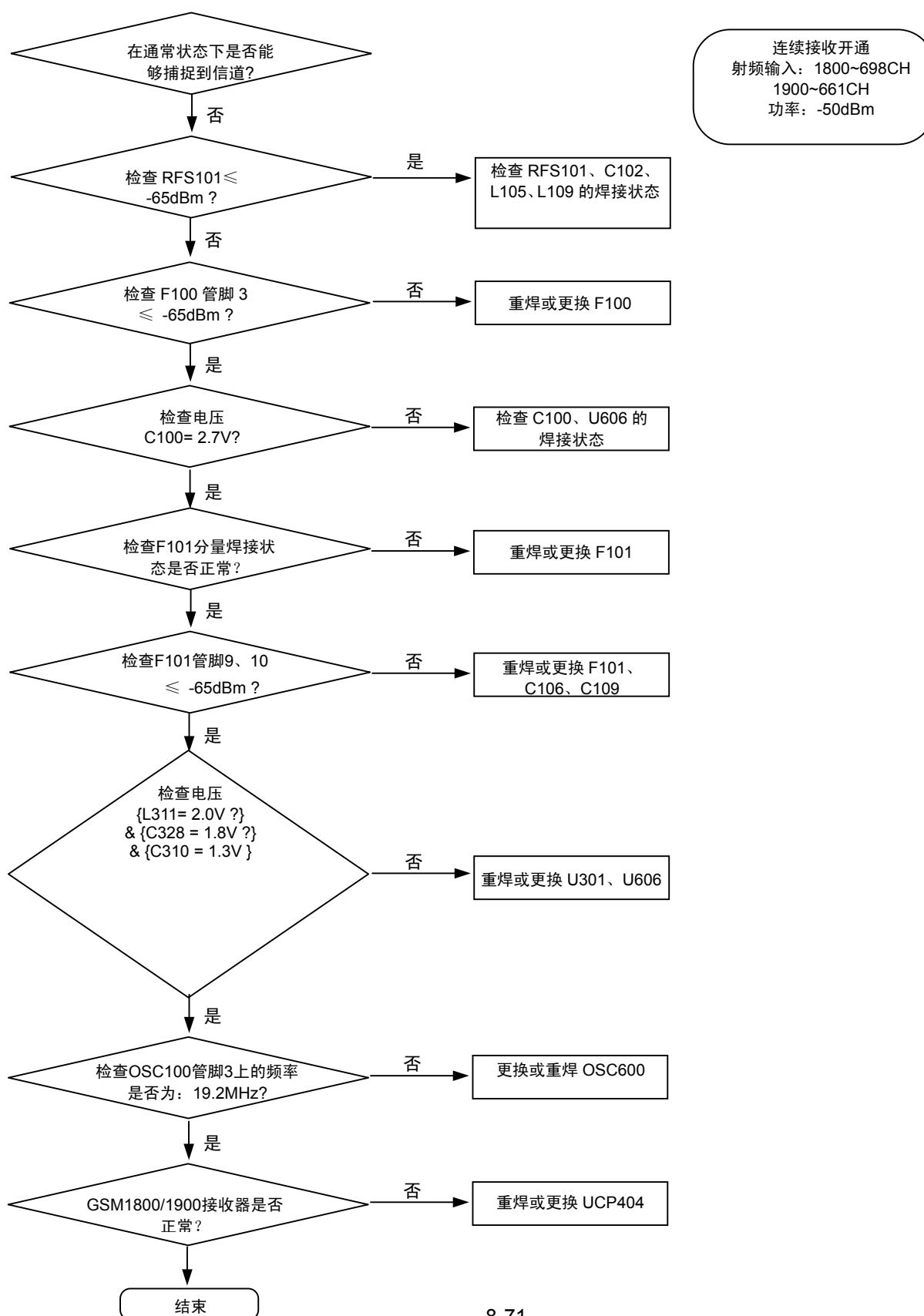


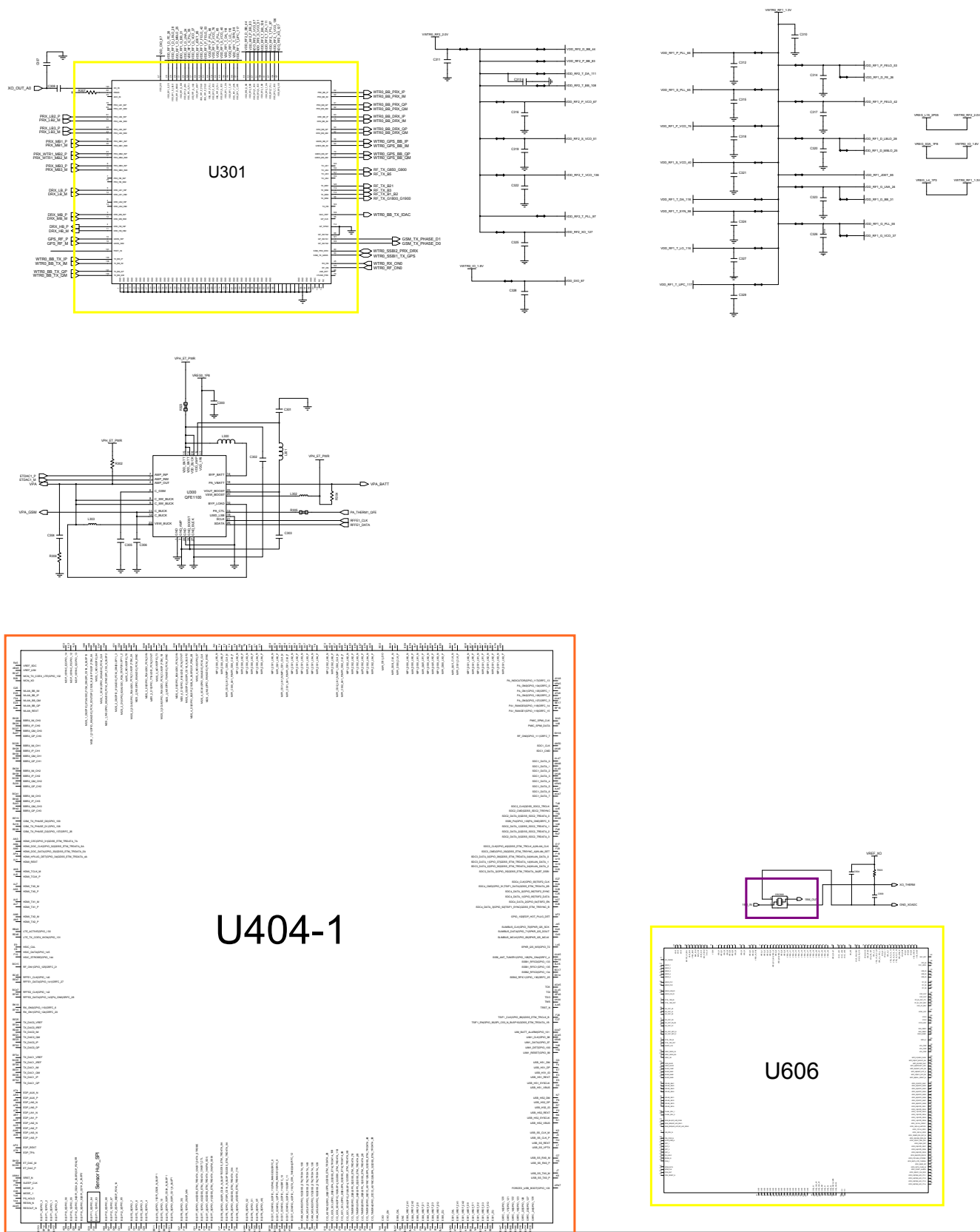
8-3-20.电机

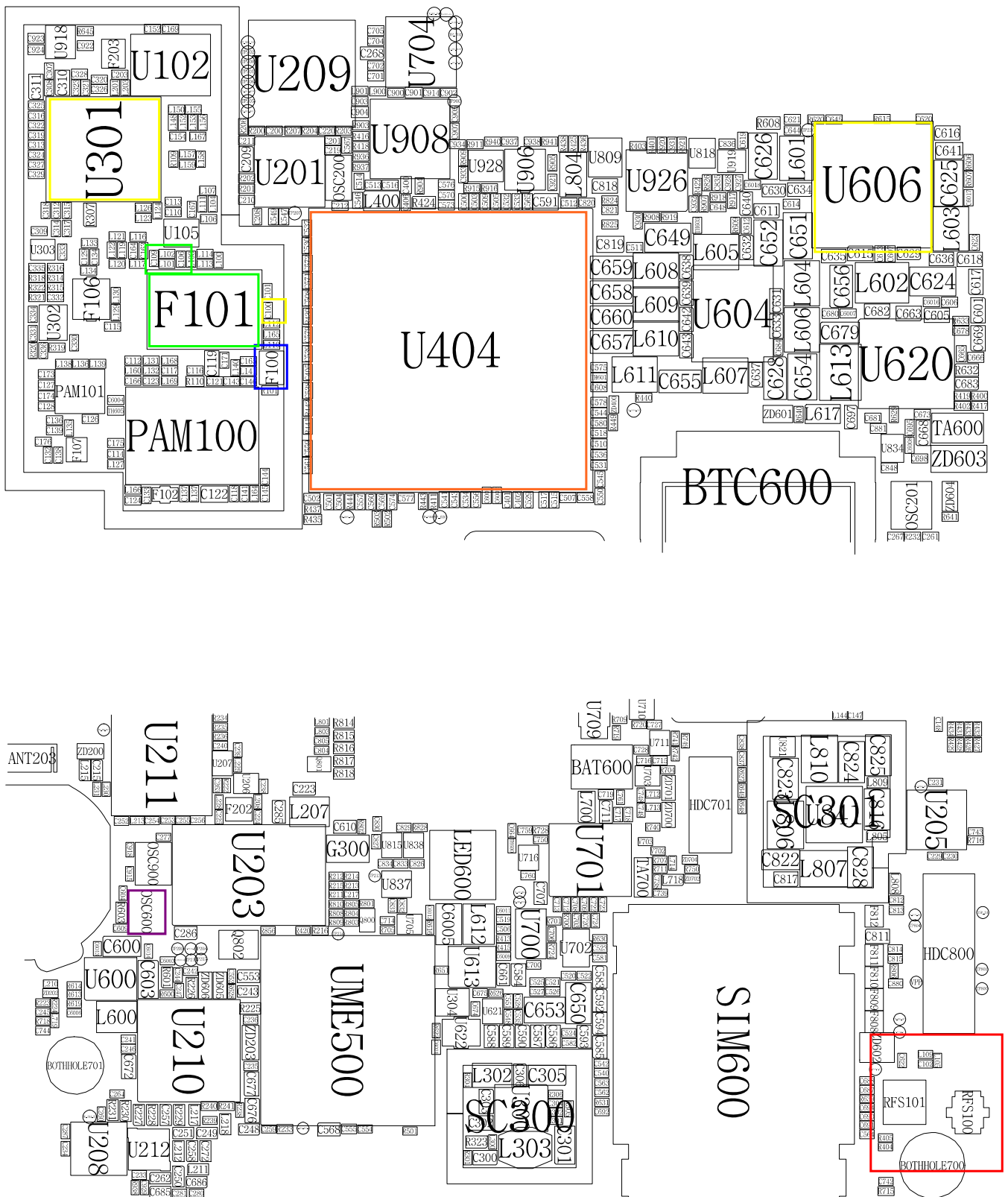




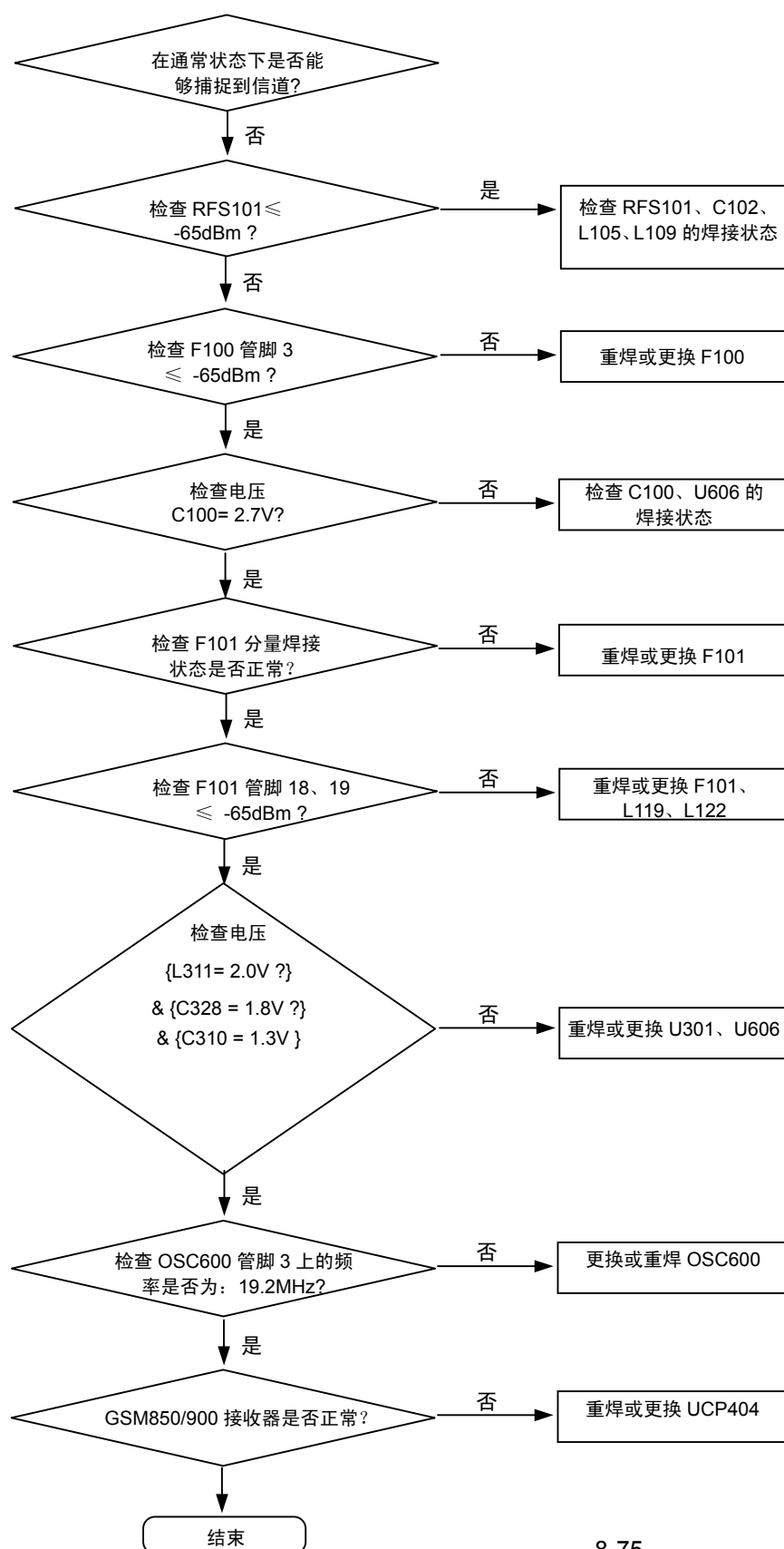
8-3-21.GSM1800/1900 接收器



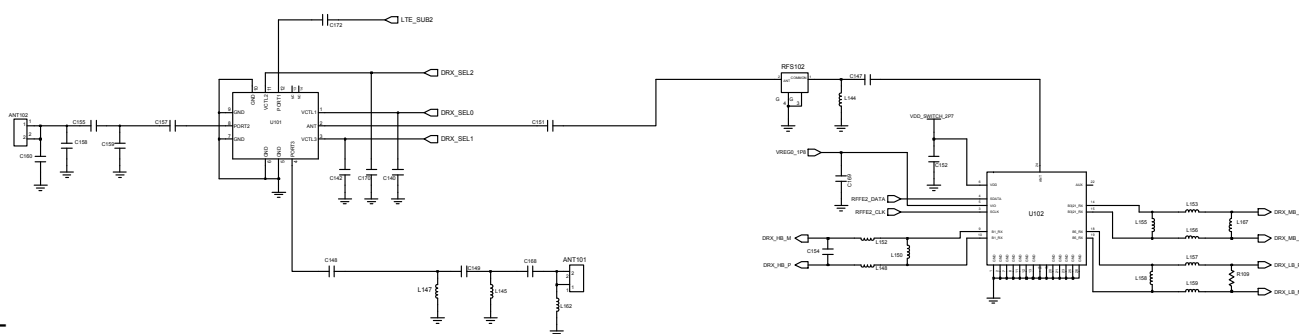
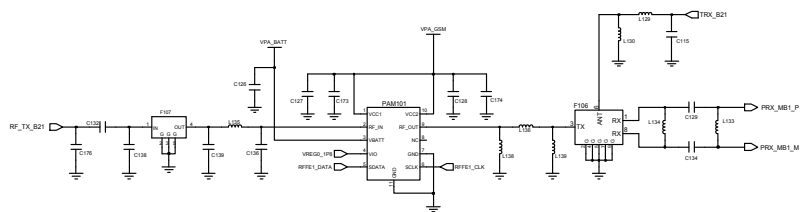
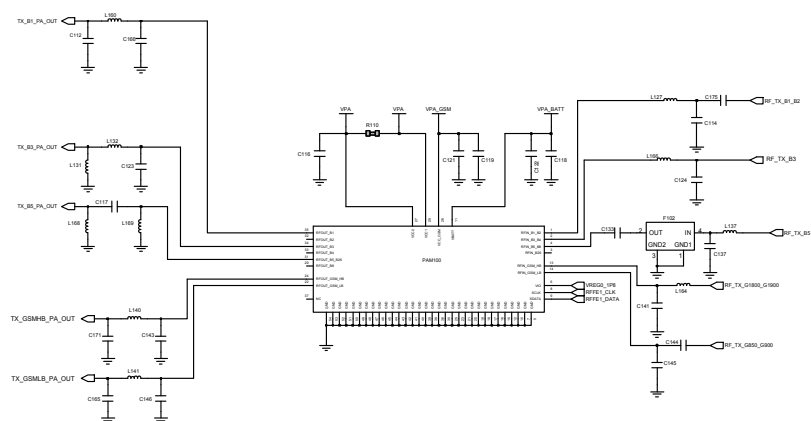


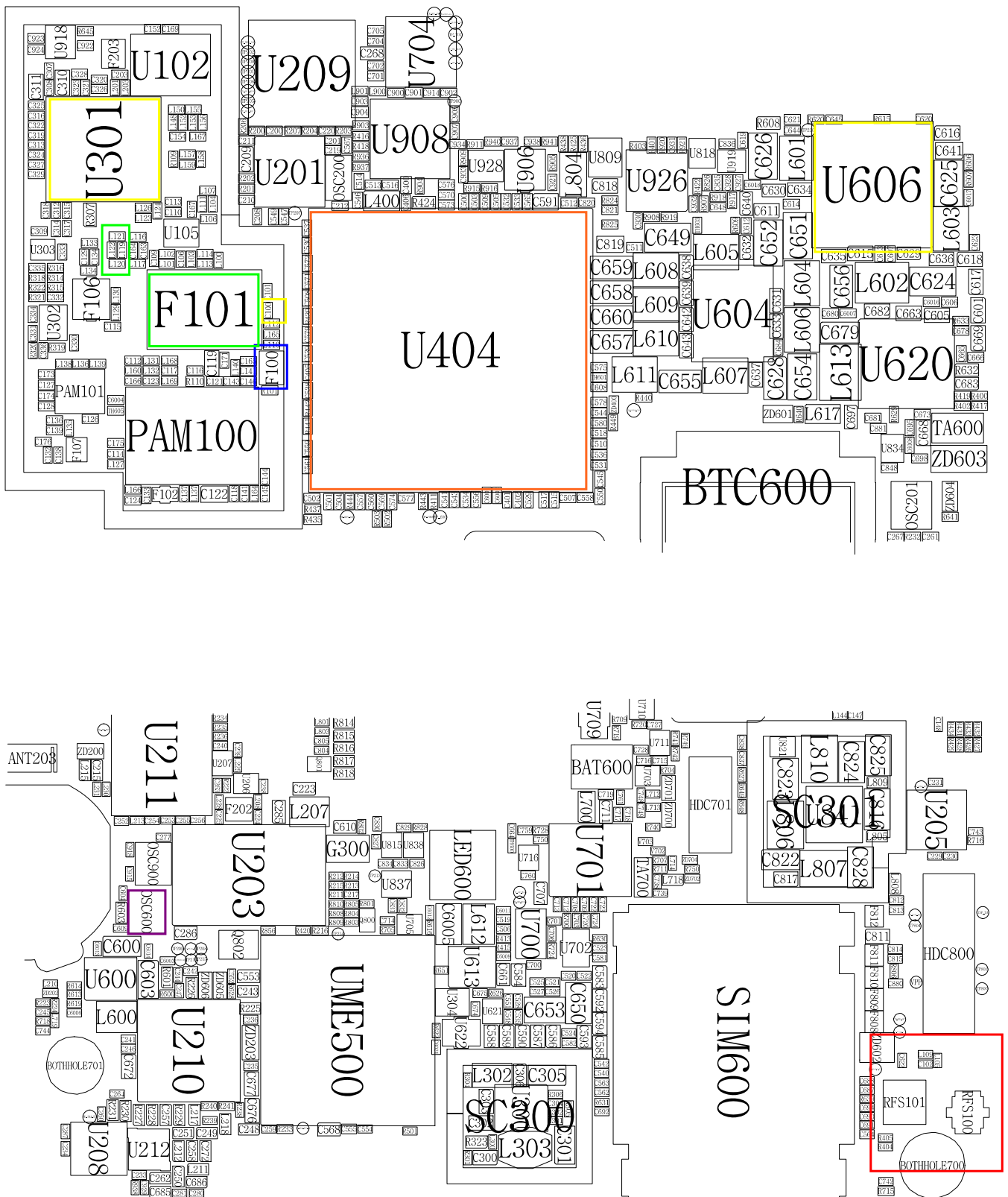


8-3-22.GSM850/900 接收器

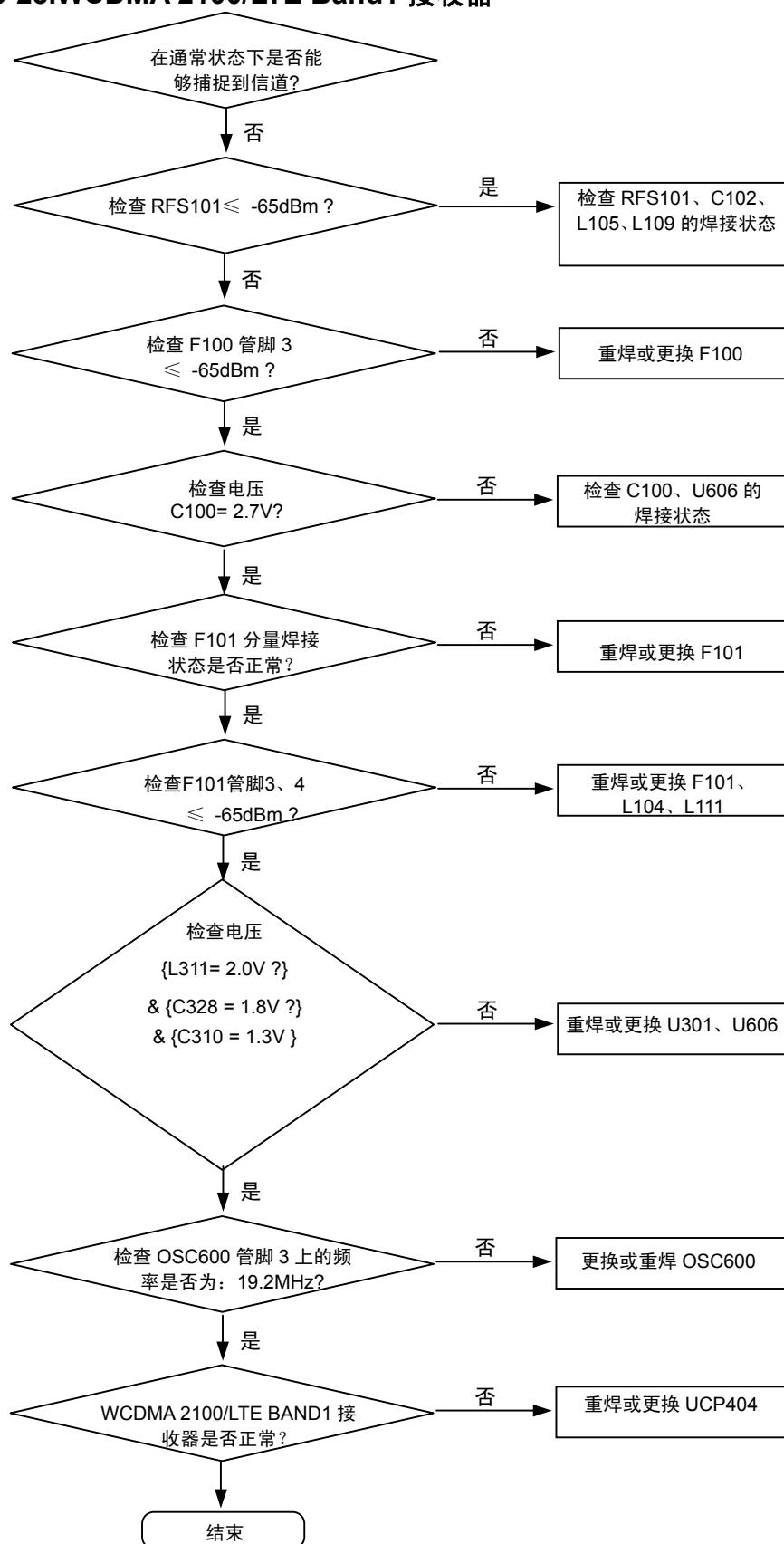


连续接收开通
射频输入：850~190CH
900~37CH
功率：-50dBm

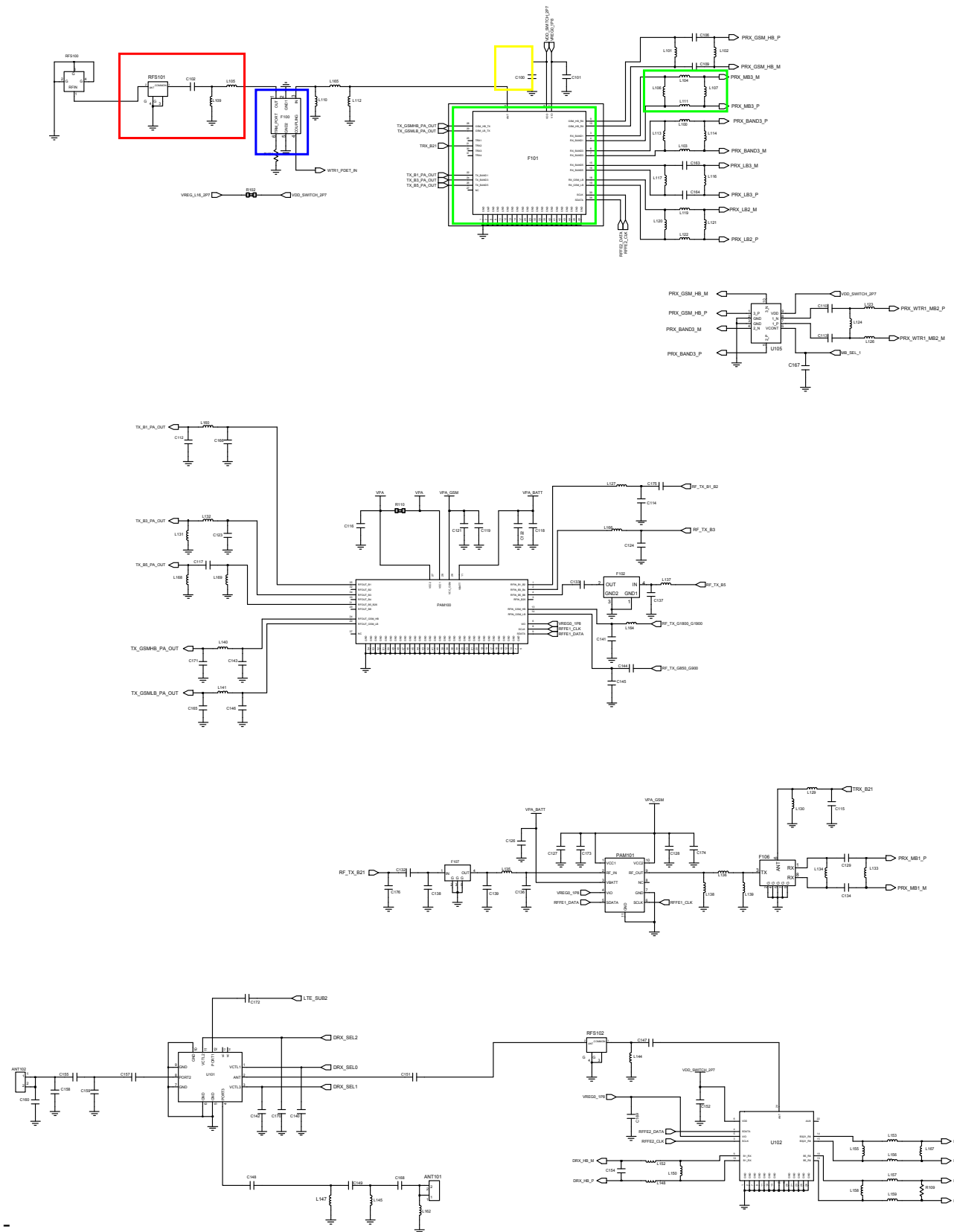


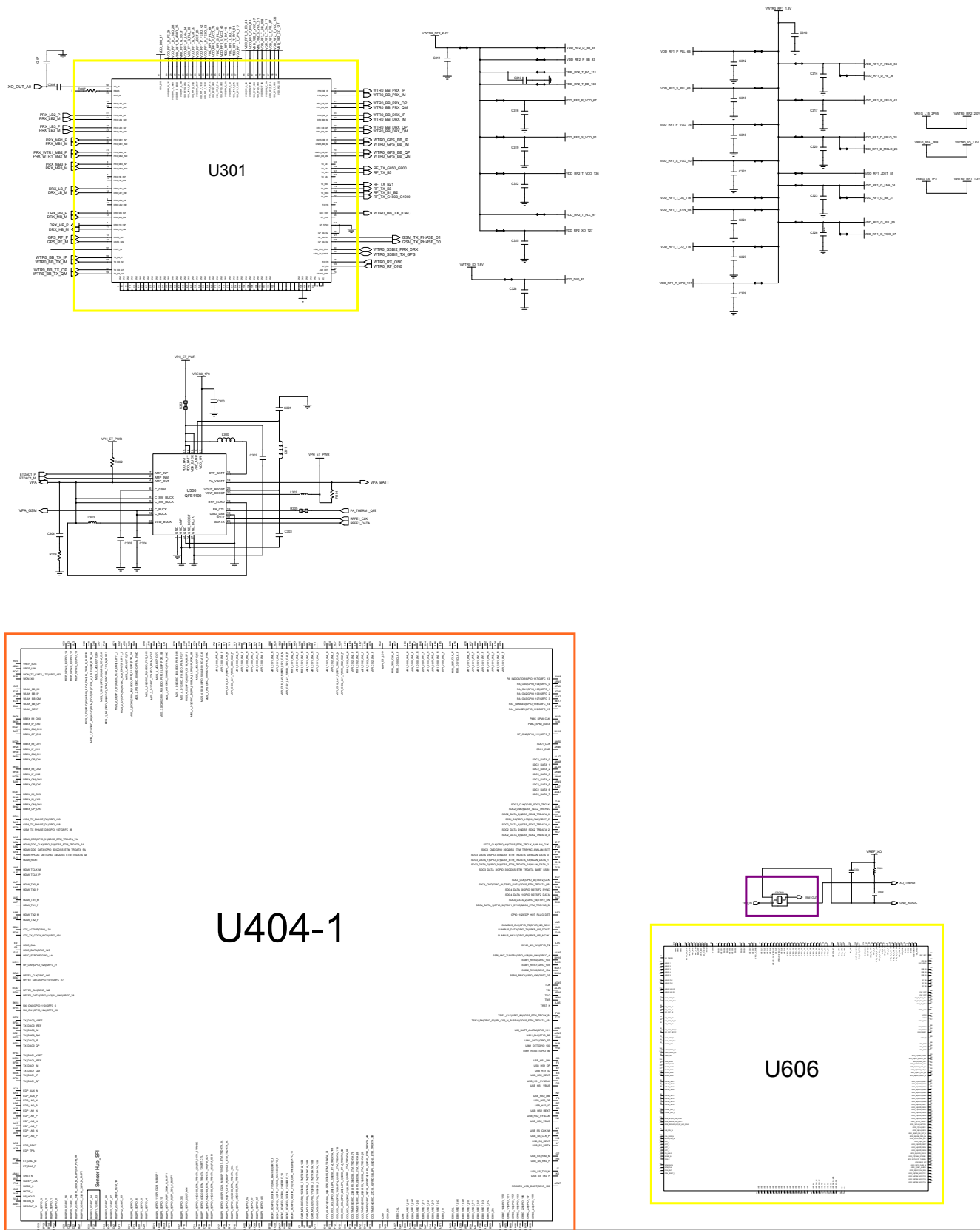


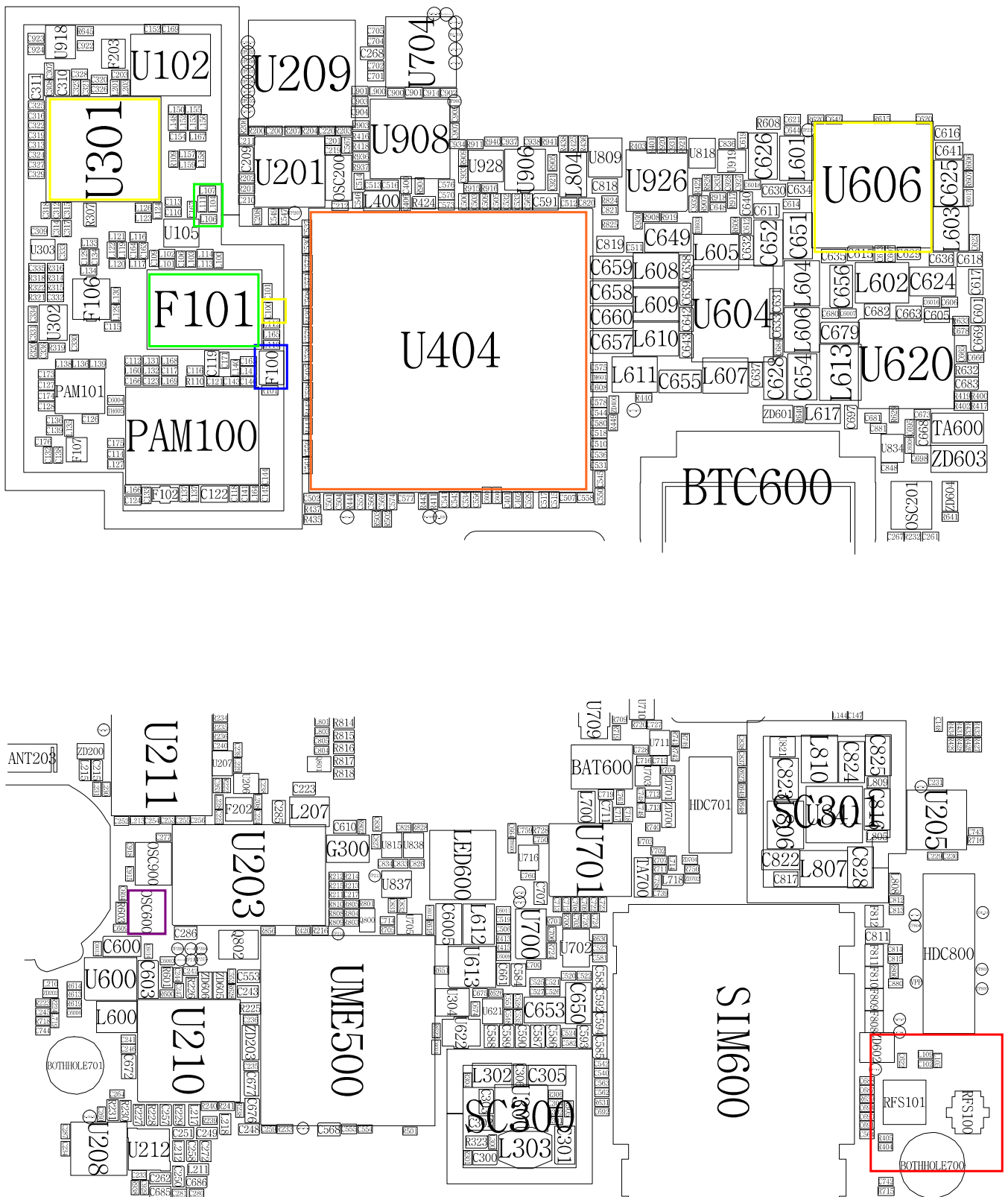
8-3-23.WCDMA 2100/LTE Band1 接收器



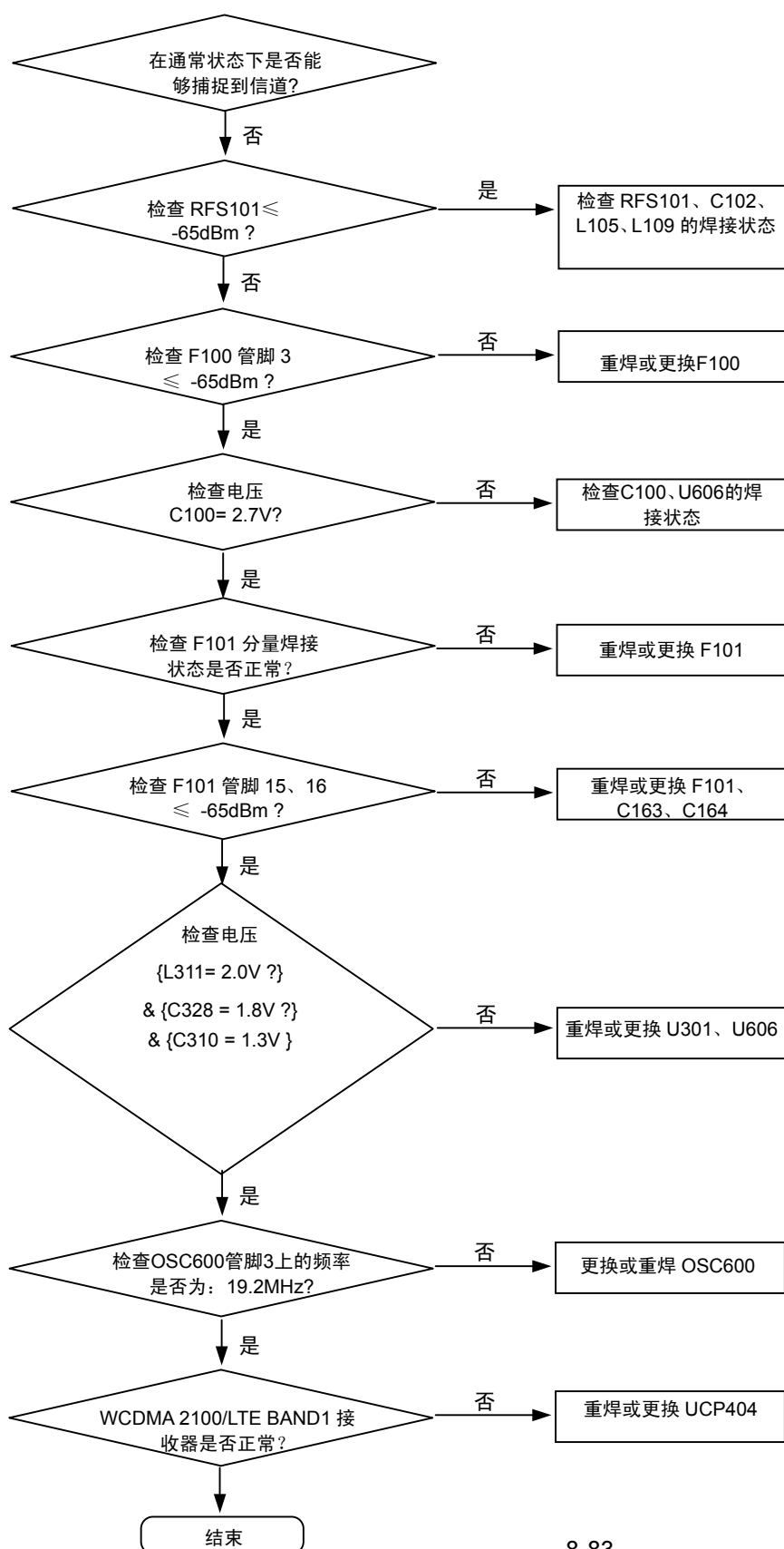
连续接收开通
射频输入：W2100~10700CH
LTE B1~300CH
功率：-50dBm



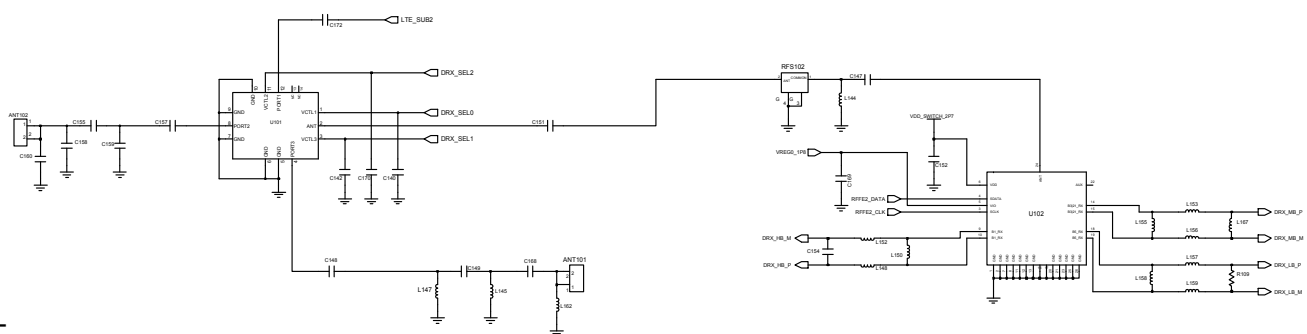
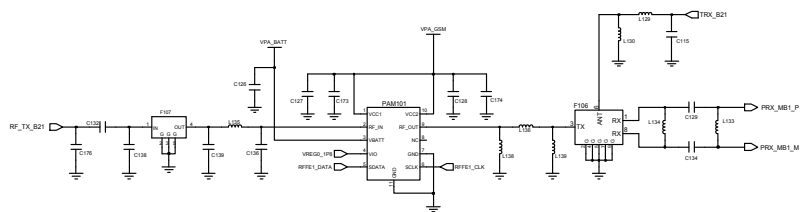
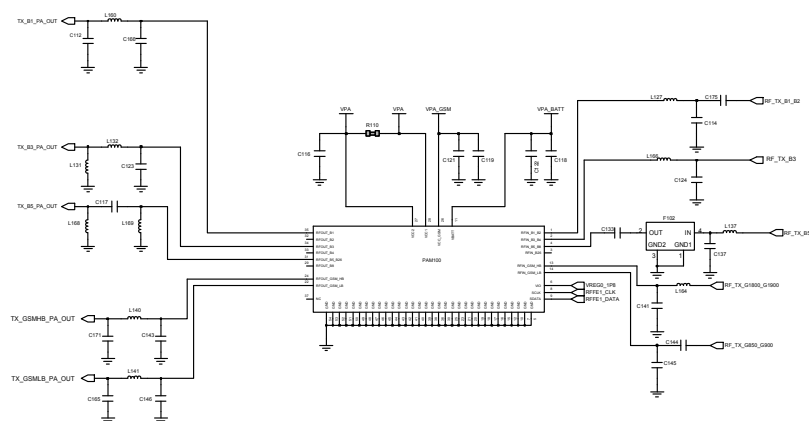


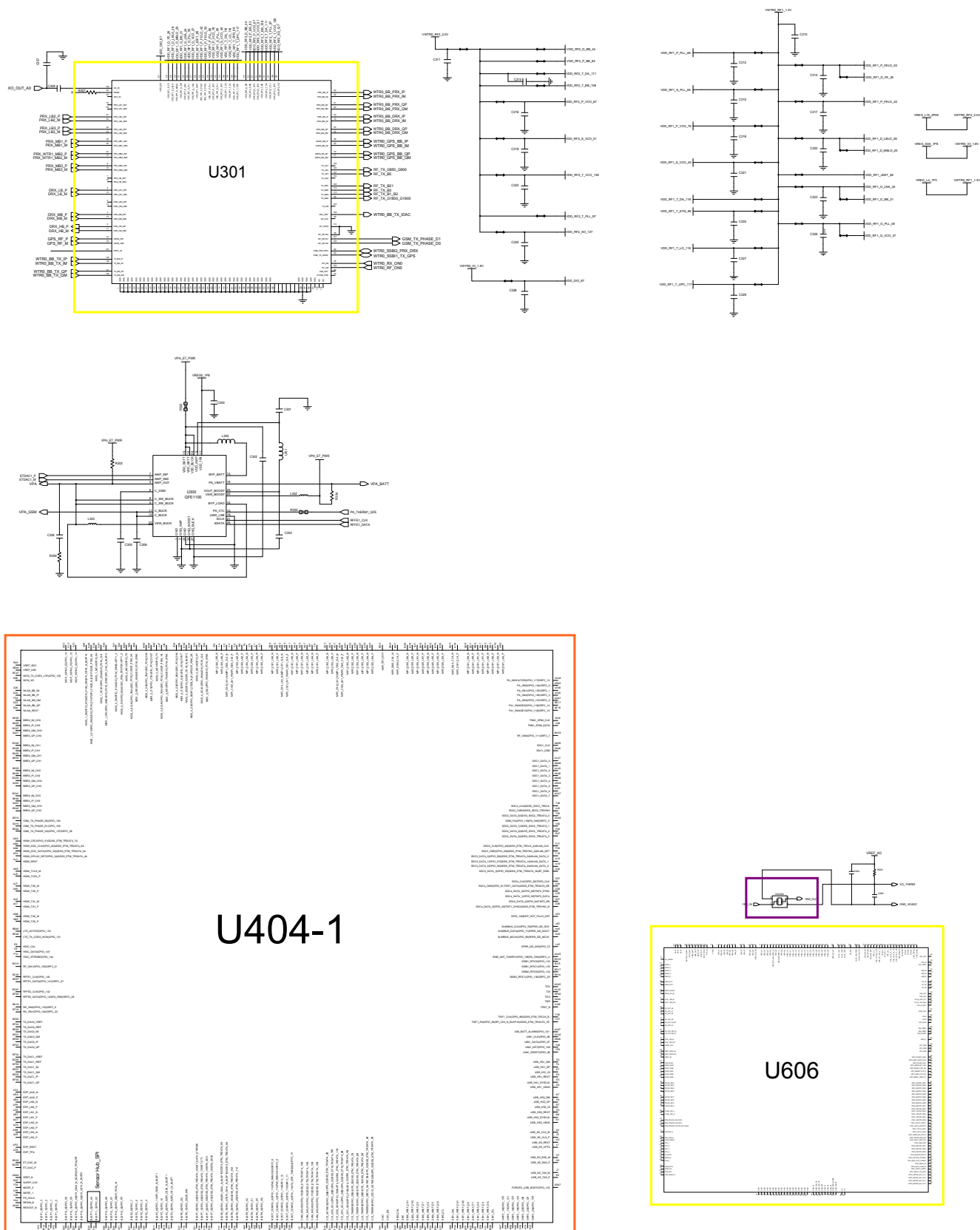


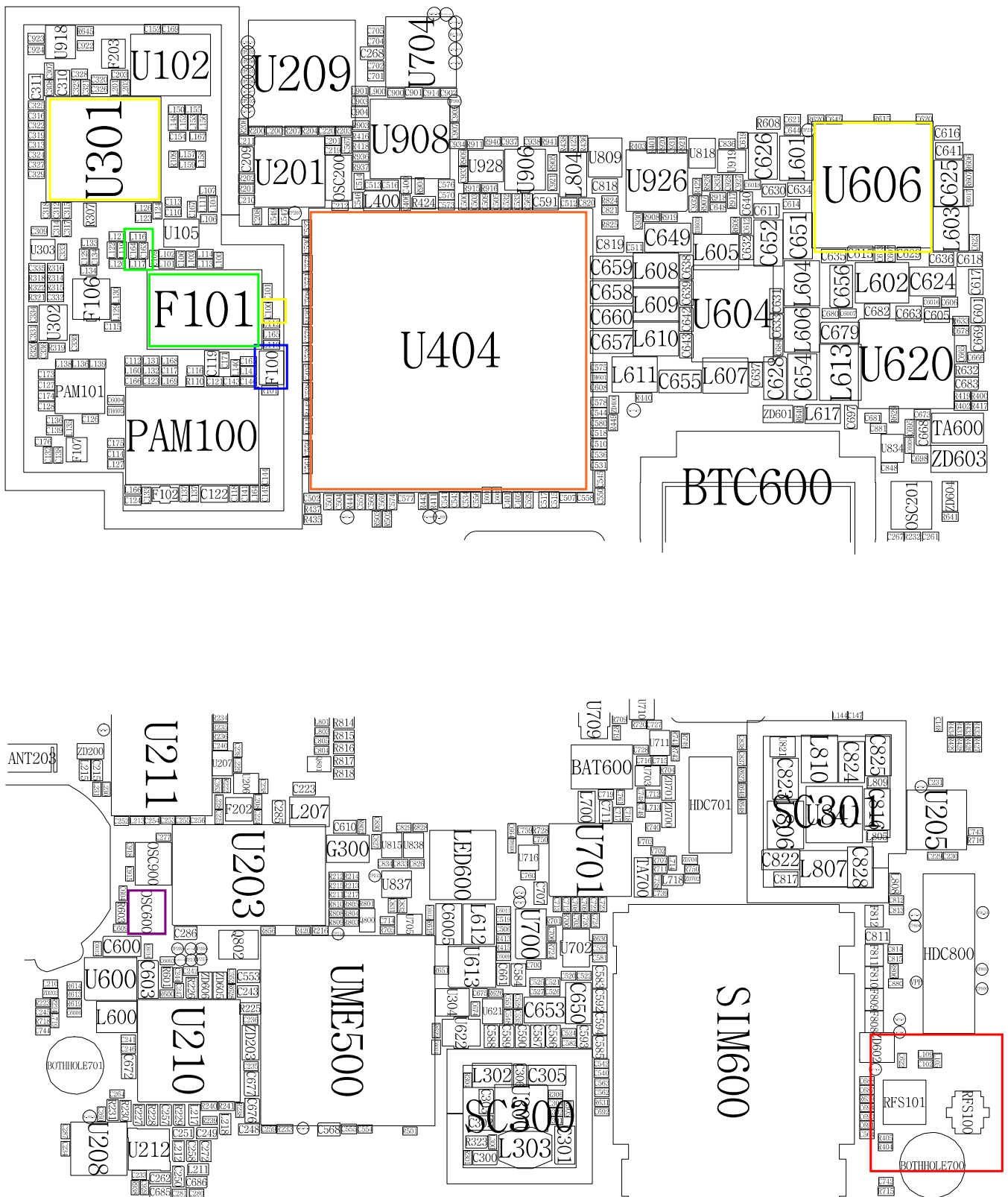
8-3-24.WCDMA 800/850/NEW800/LTE Band19 接收器



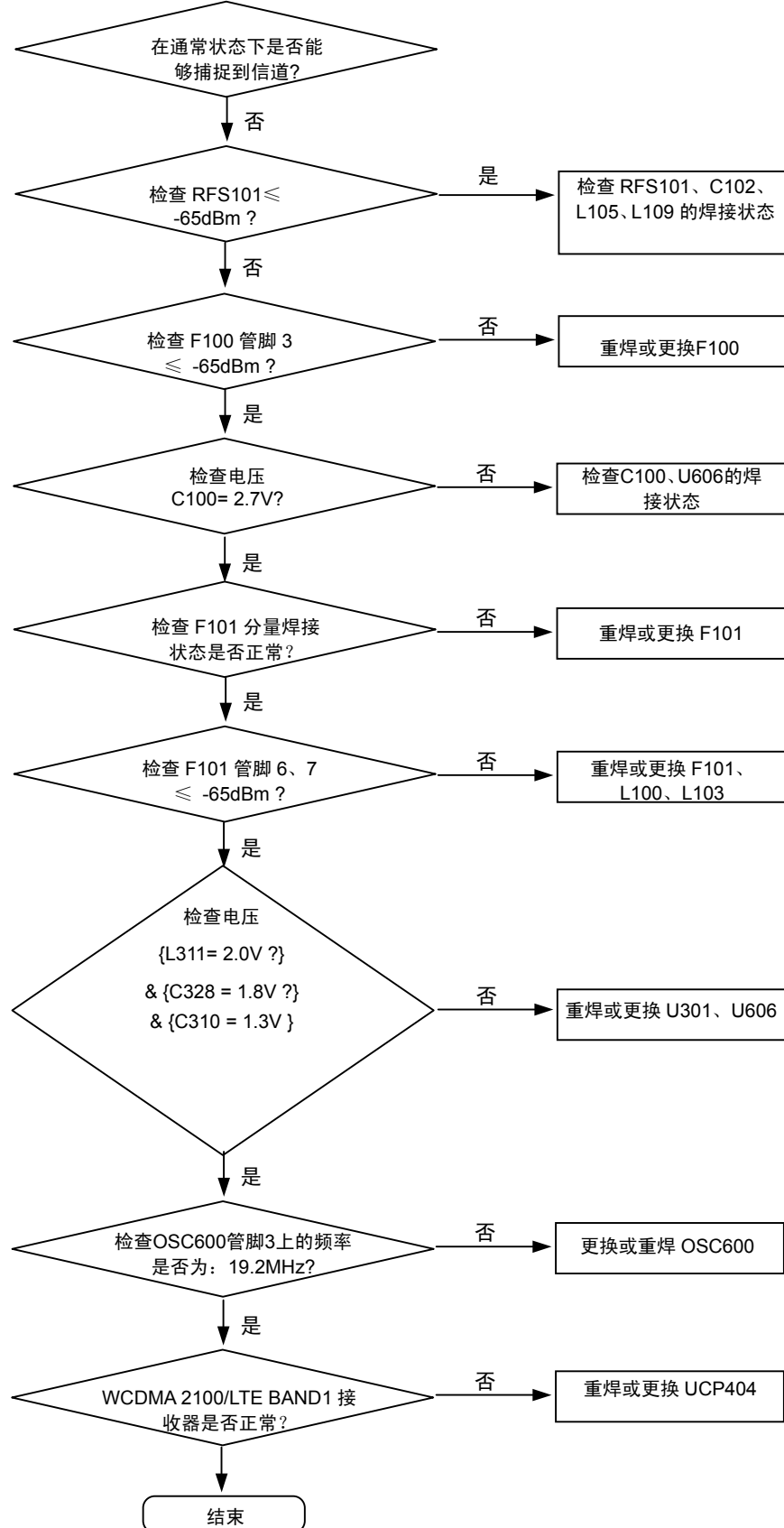
连续接收开通
射频输入：WCDMA~4400CH
LTE B19~6075CH
功率：-50dBm



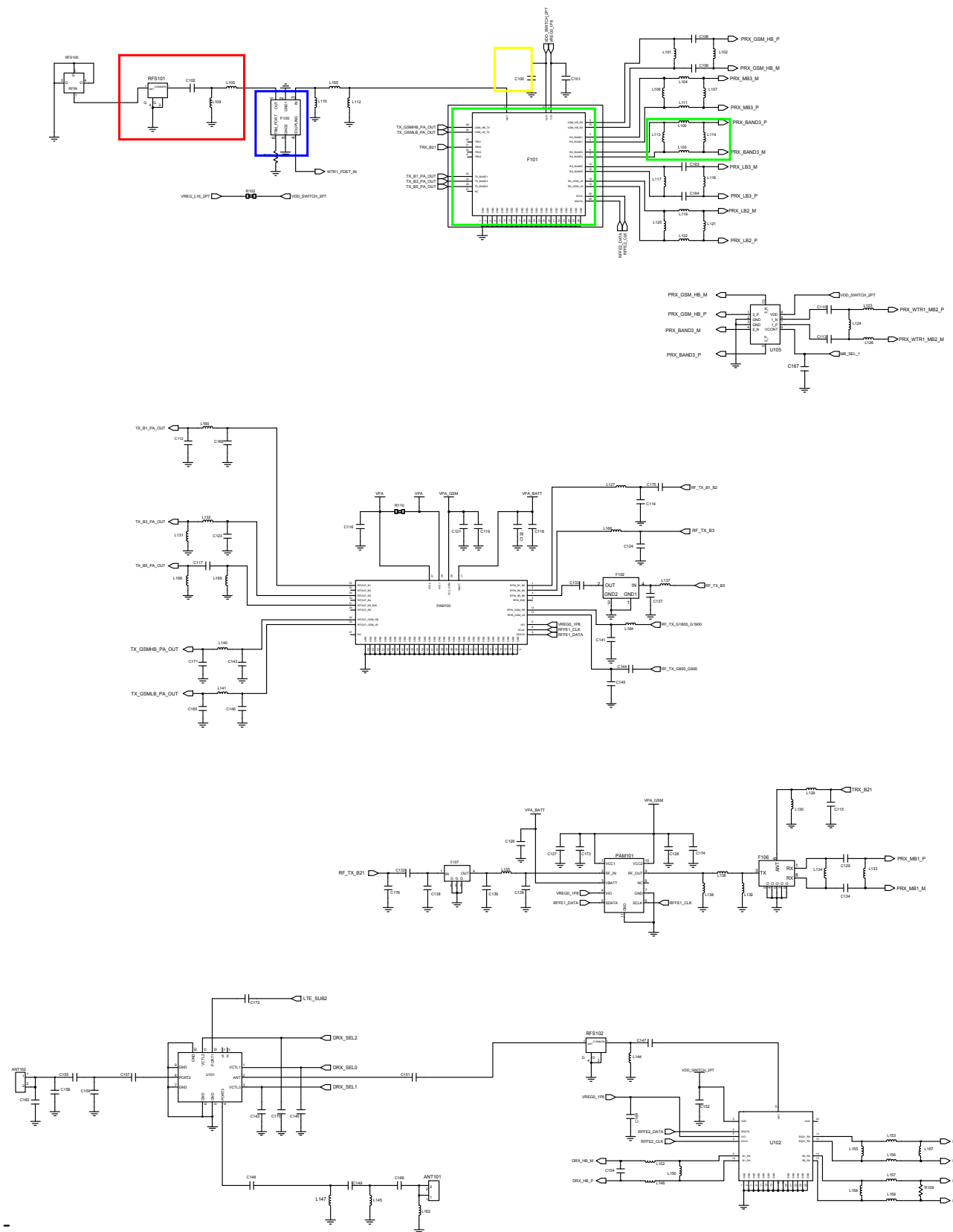


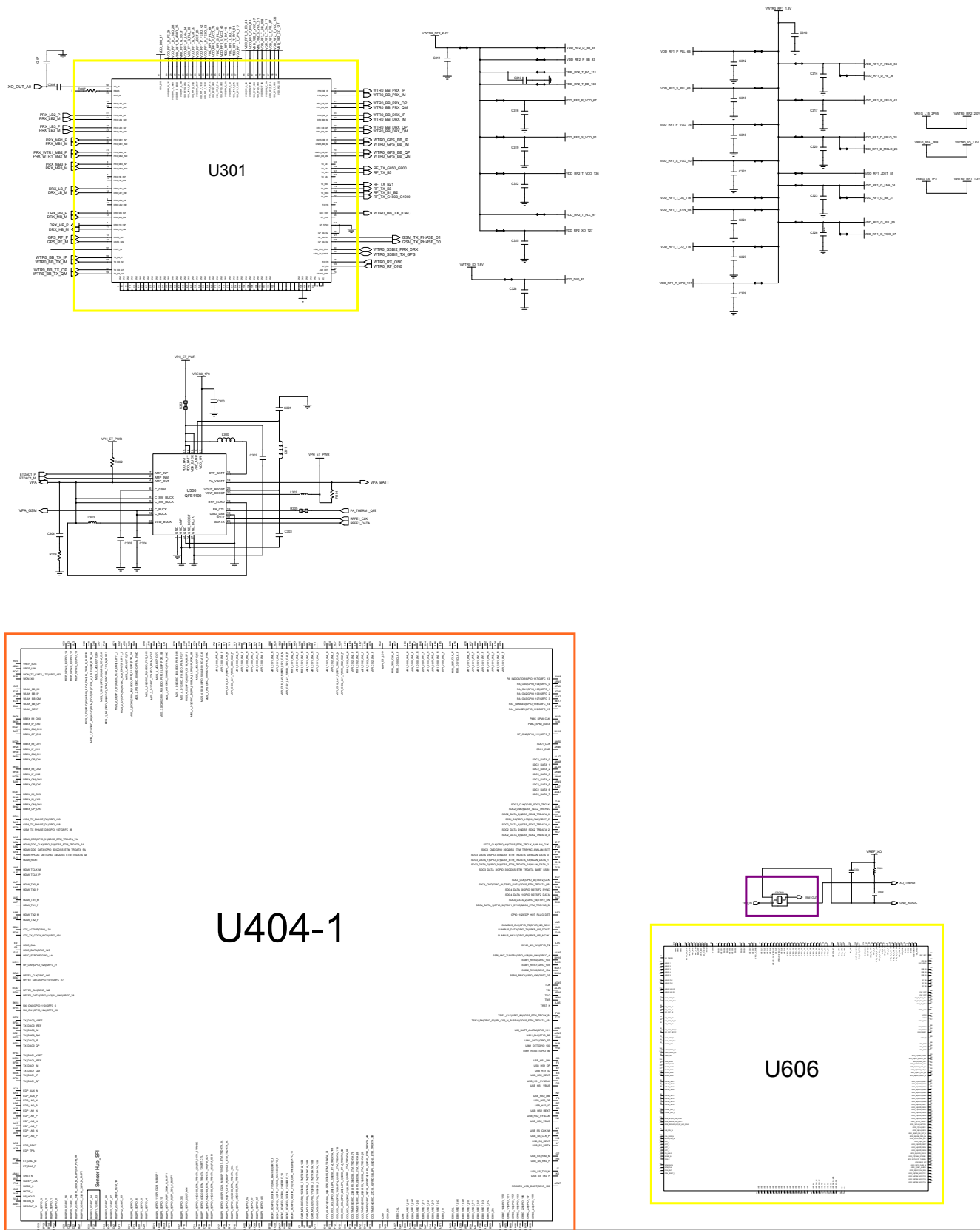


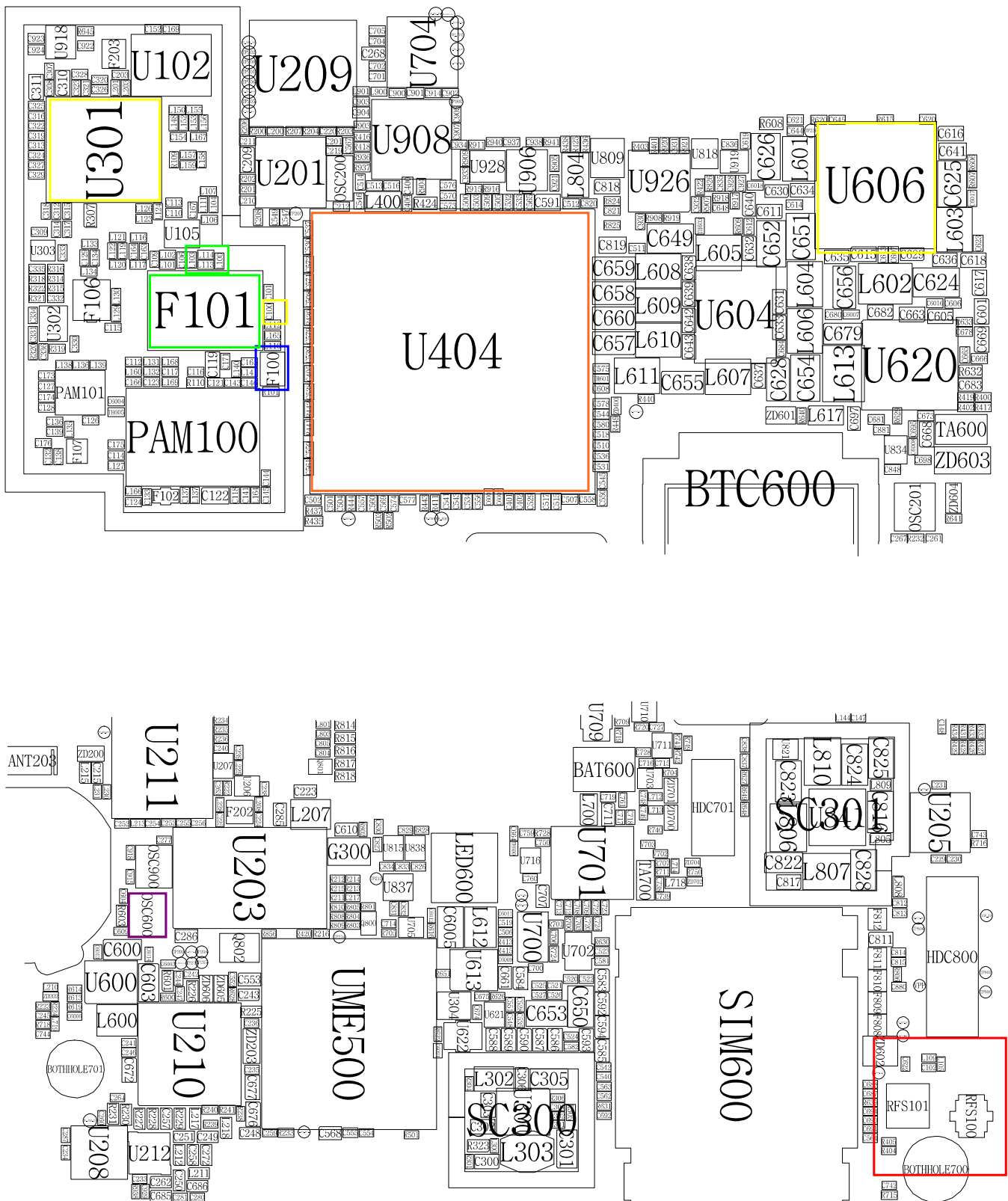
8-3-25.LTE Band3 接收器



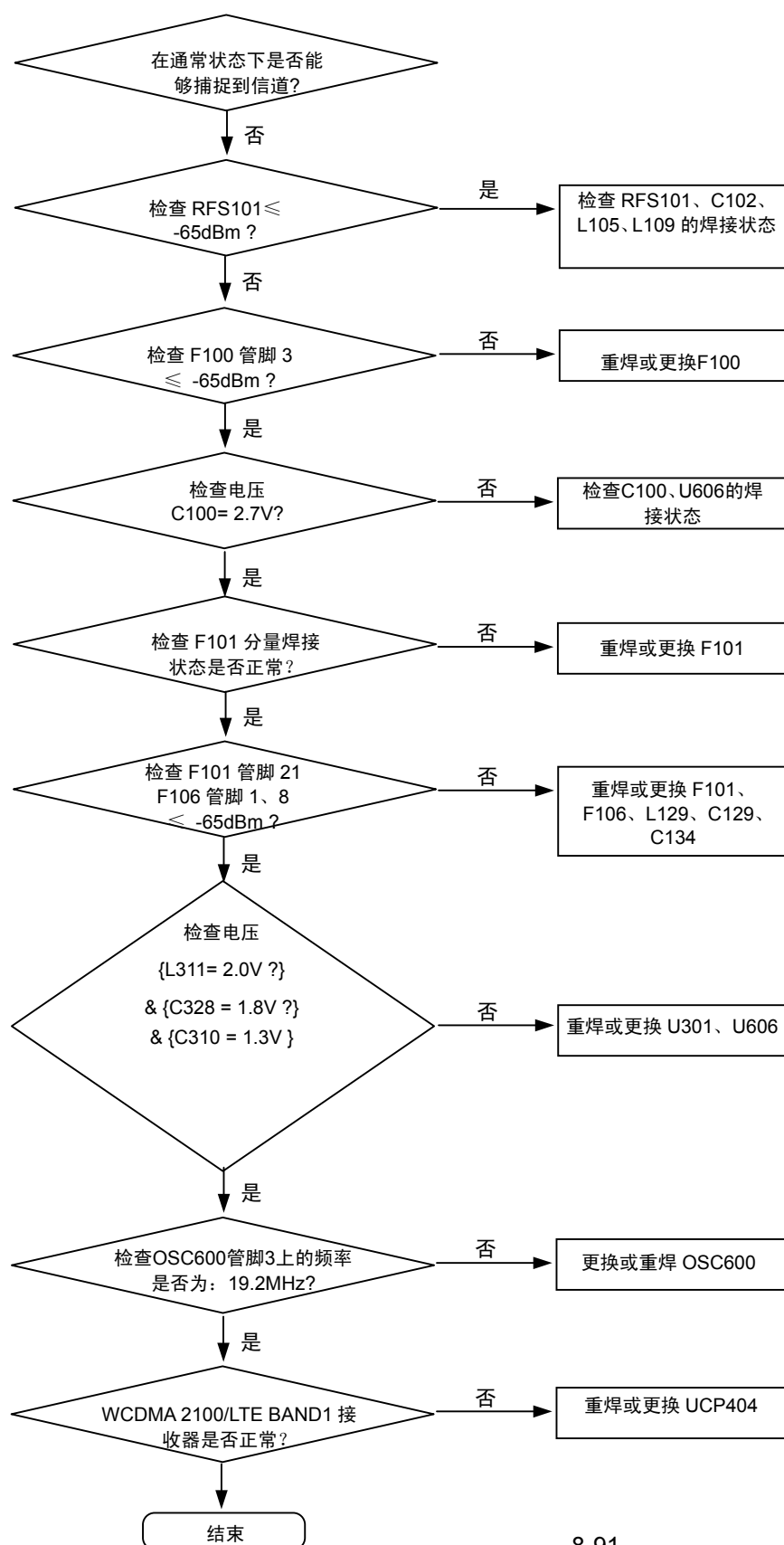
连续接收开通
射频输入：1575CH
功率：-50dBm



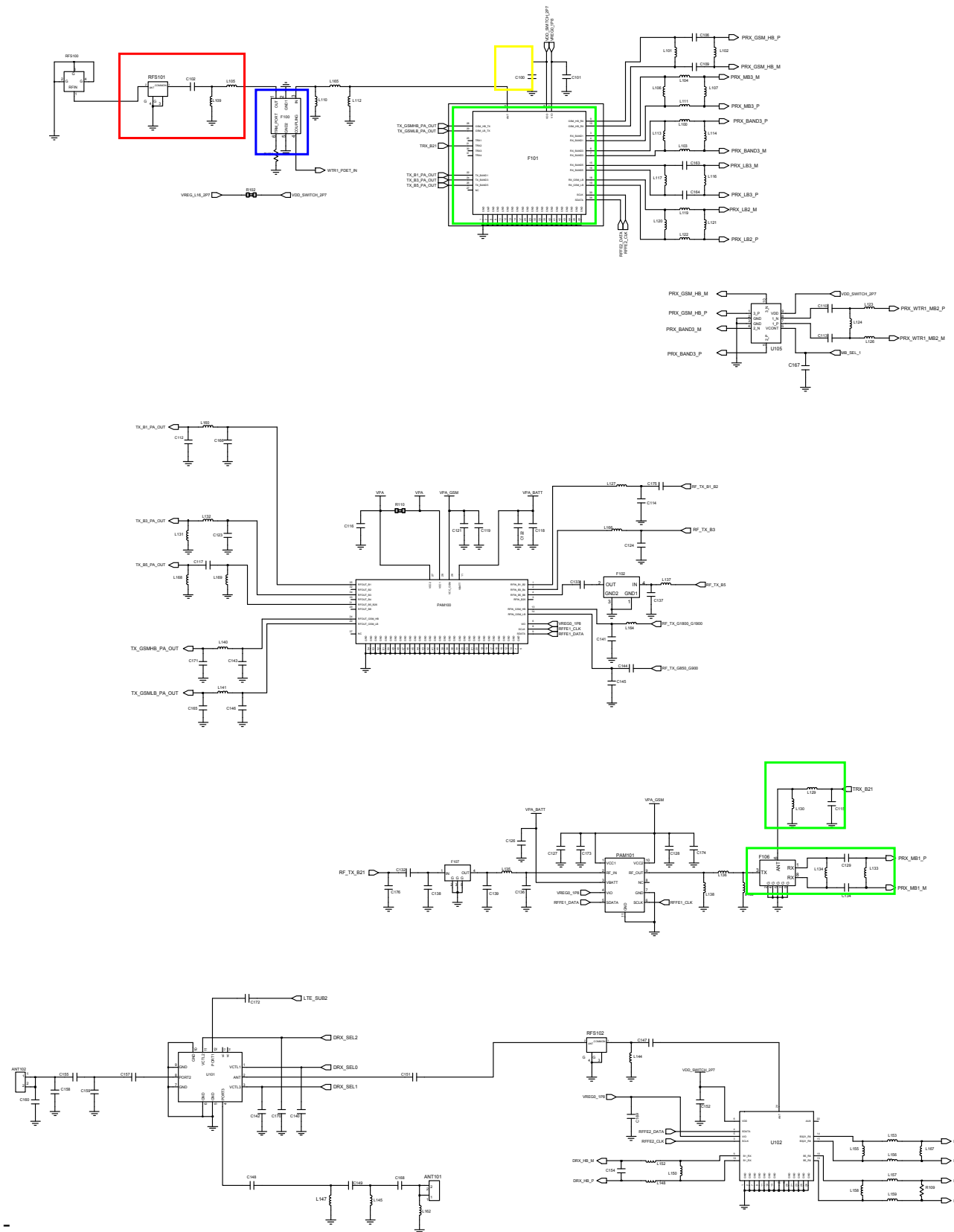


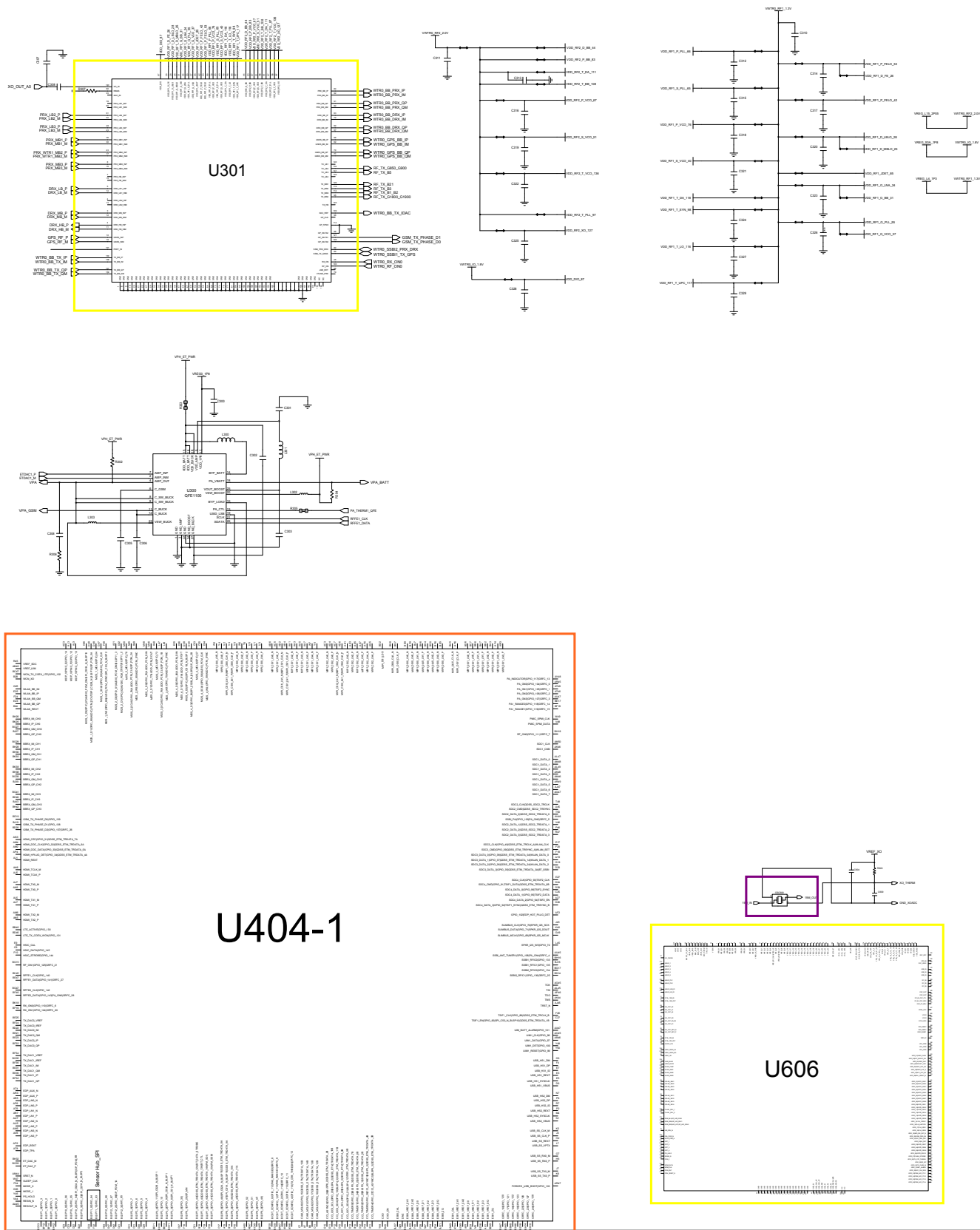


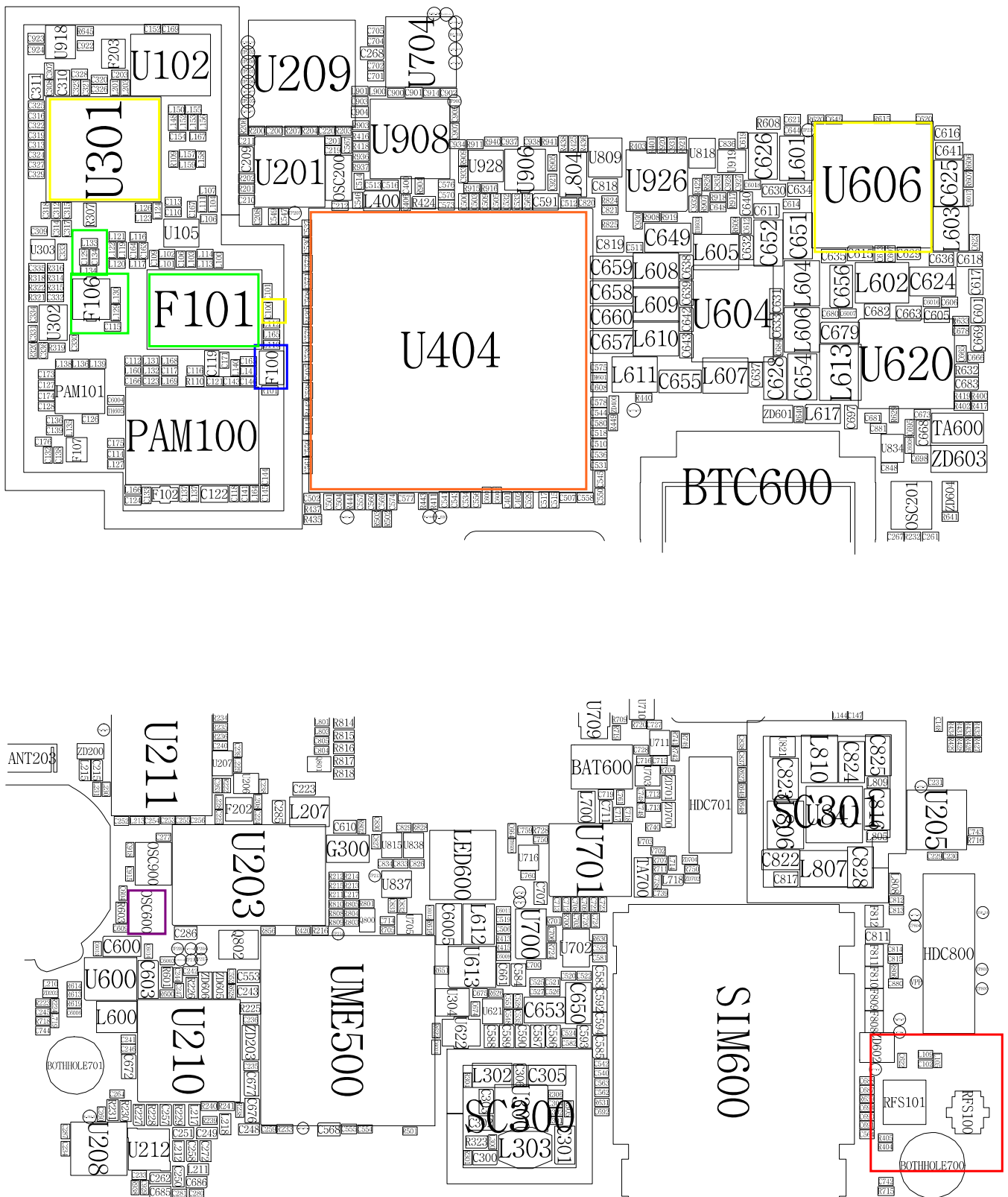
8-3-26.LTE Band21 接收器



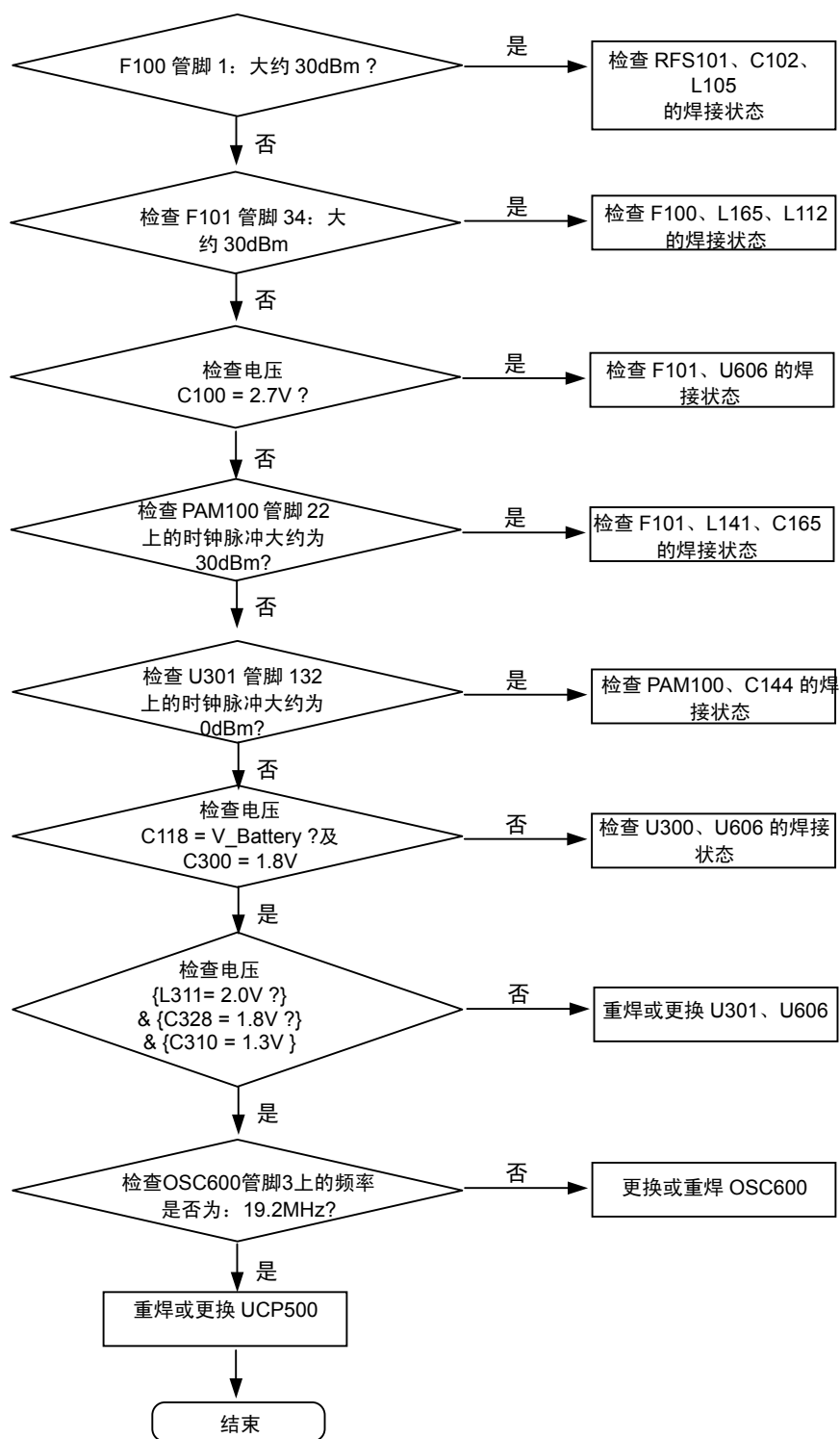
连续接收开通
射频输入：6525CH
功率：-50dBm



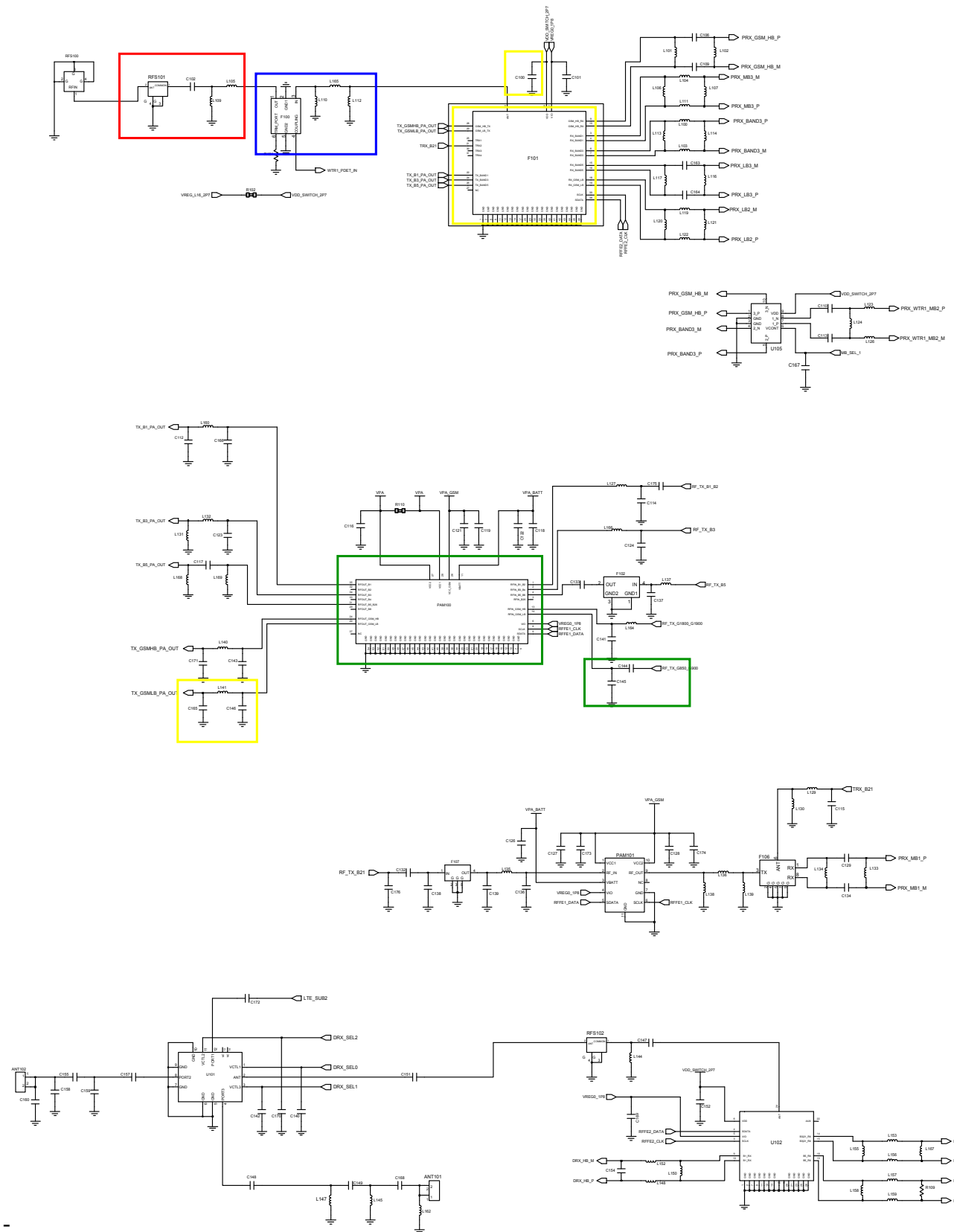


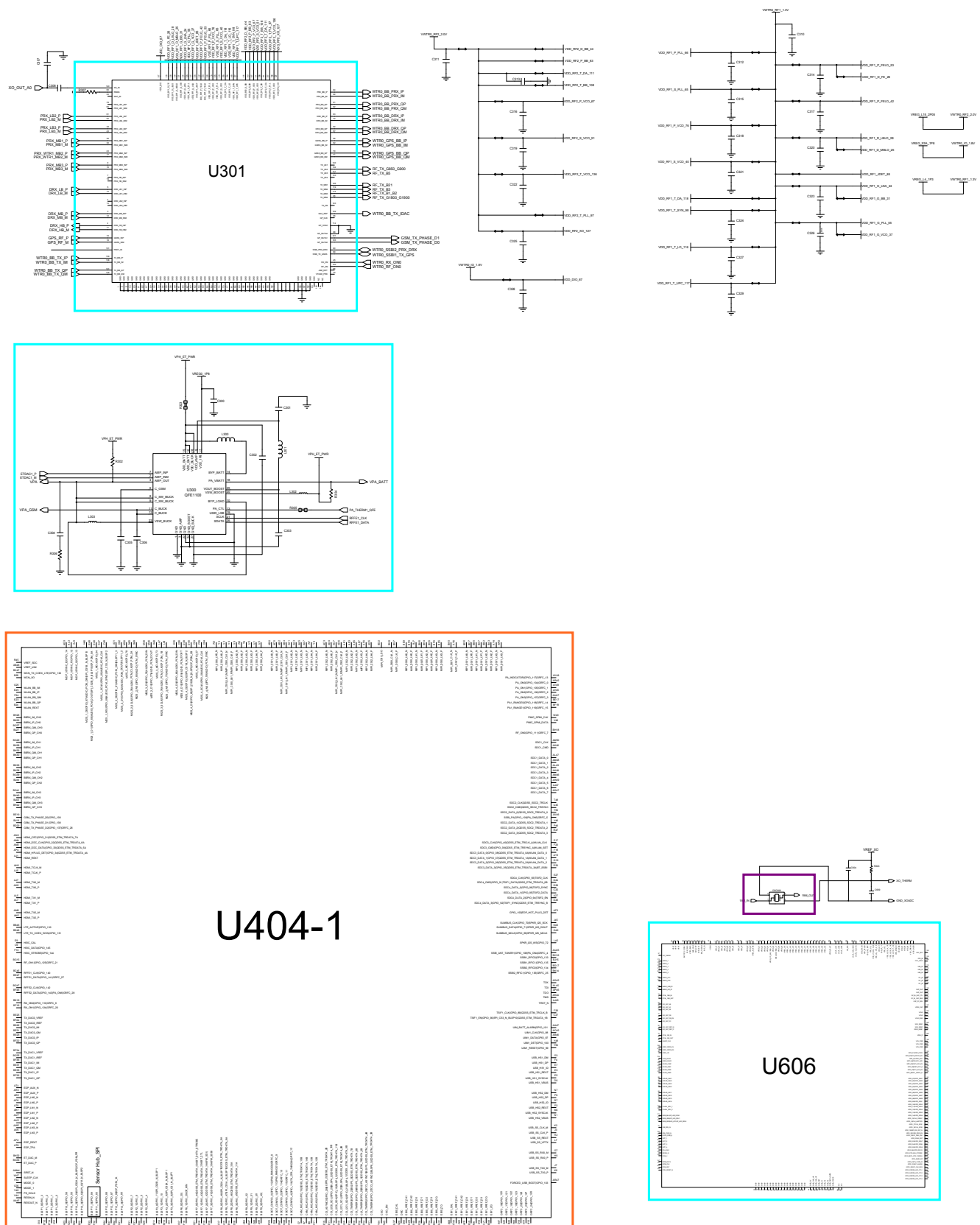


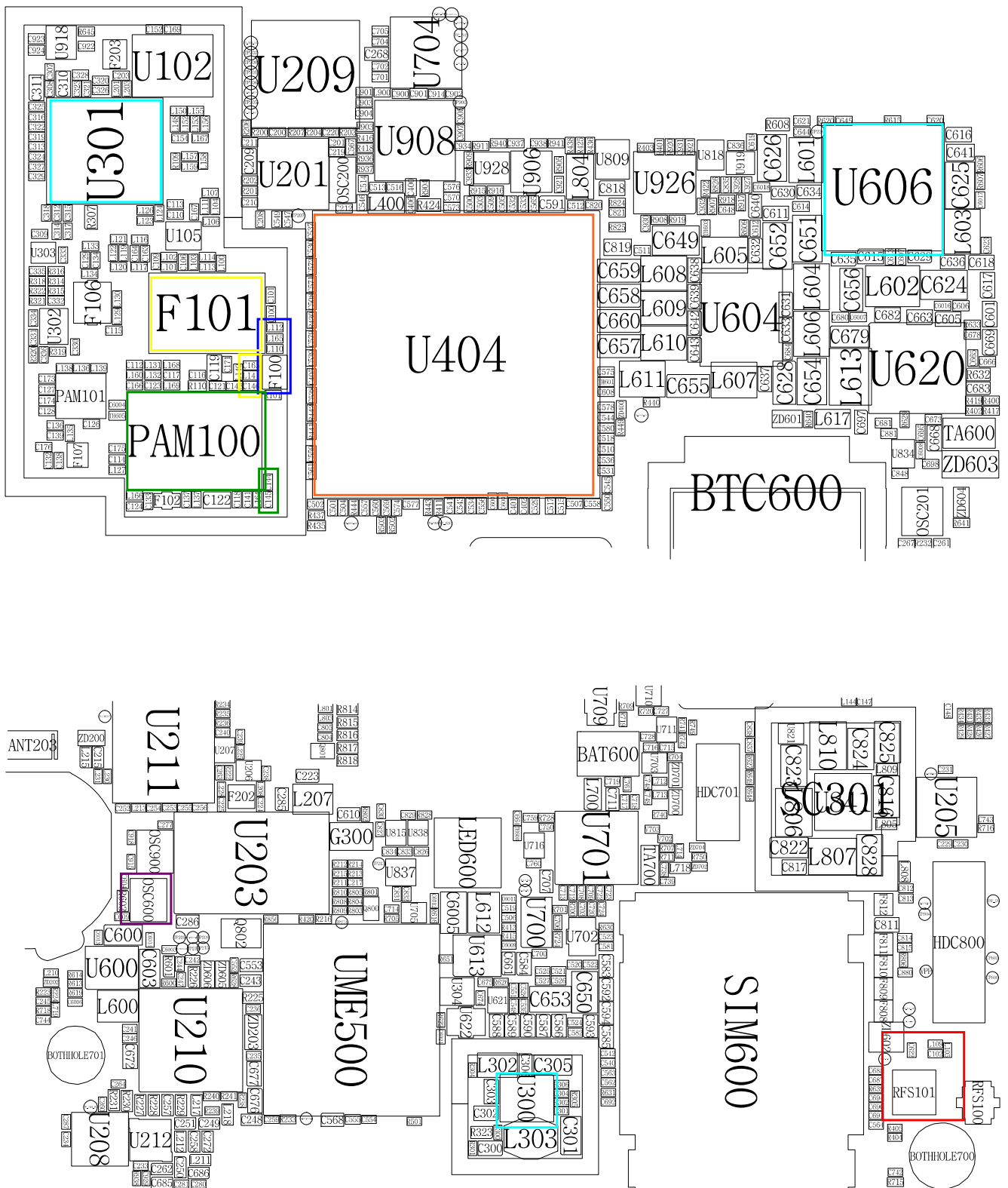
8-3-27.GSM850/GSM900 发射器



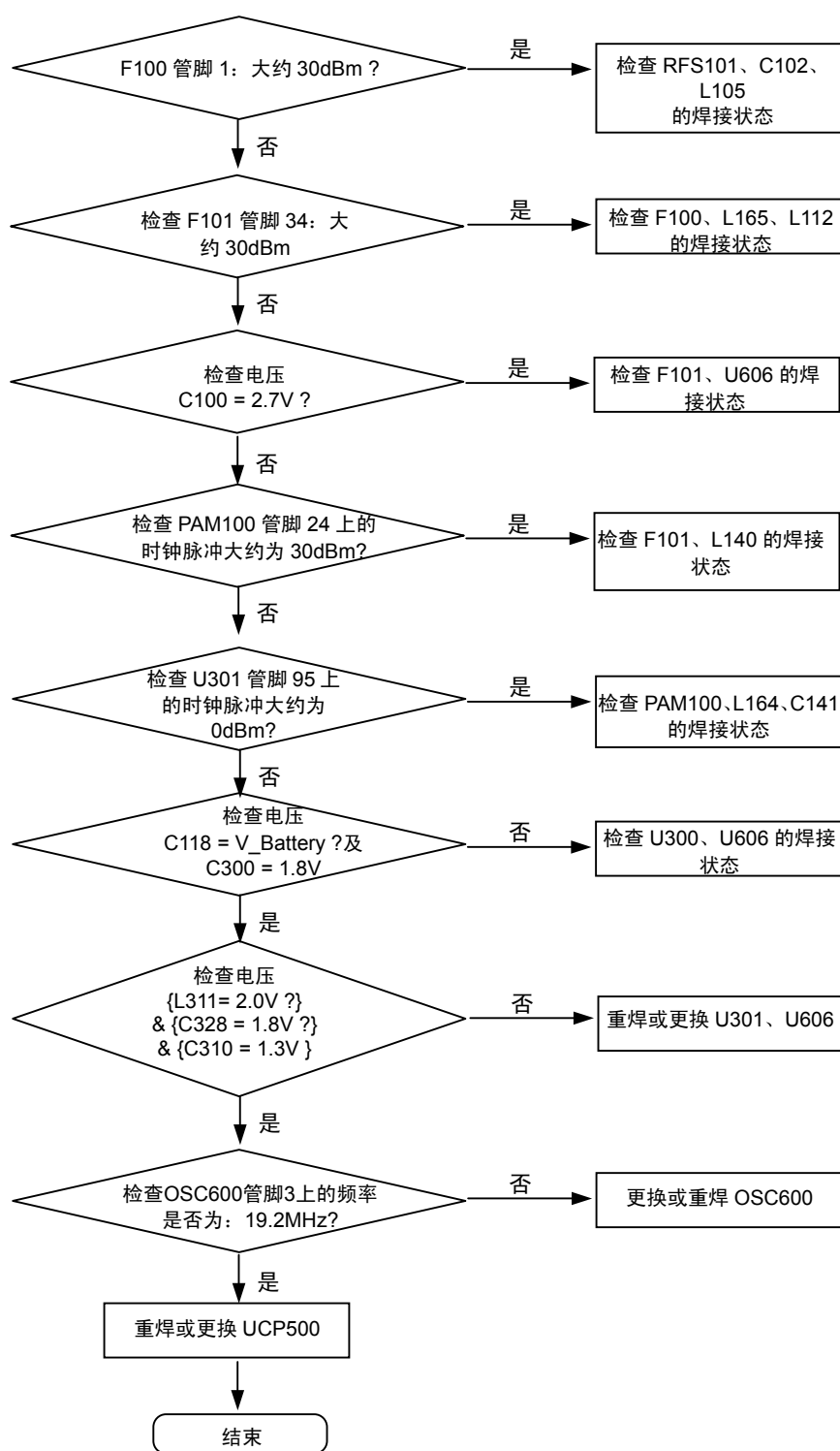
持续发射开通条件
 DAC 发射功率: 应用代码
 14500
 GSM850 CH: 190
 GSM900 CH: 37
 RBW: 100KHz
 VBW: 100KHz
 SPAN: 10MHz
 REF LEV.: 10dBm
 ATT.: 20dB



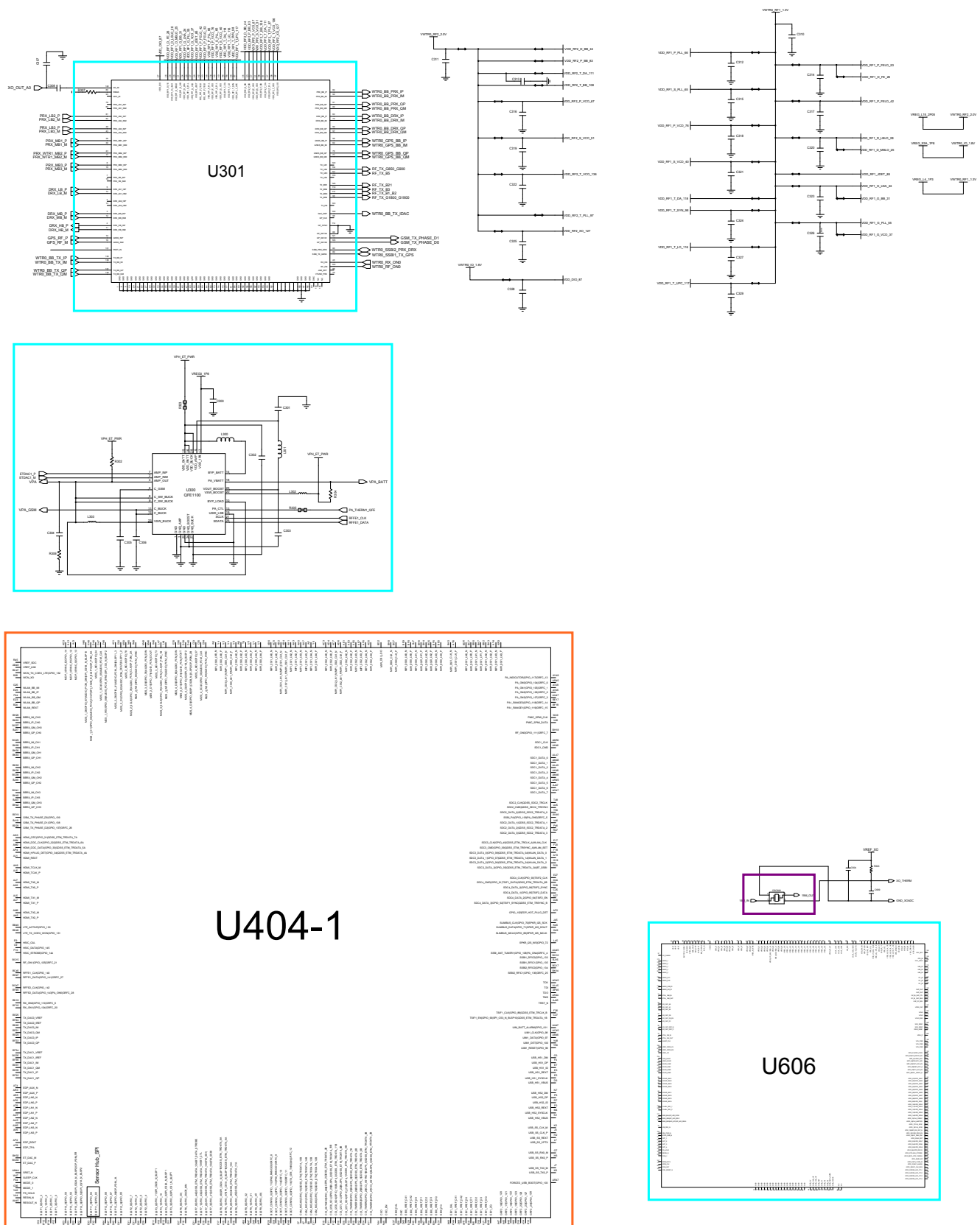


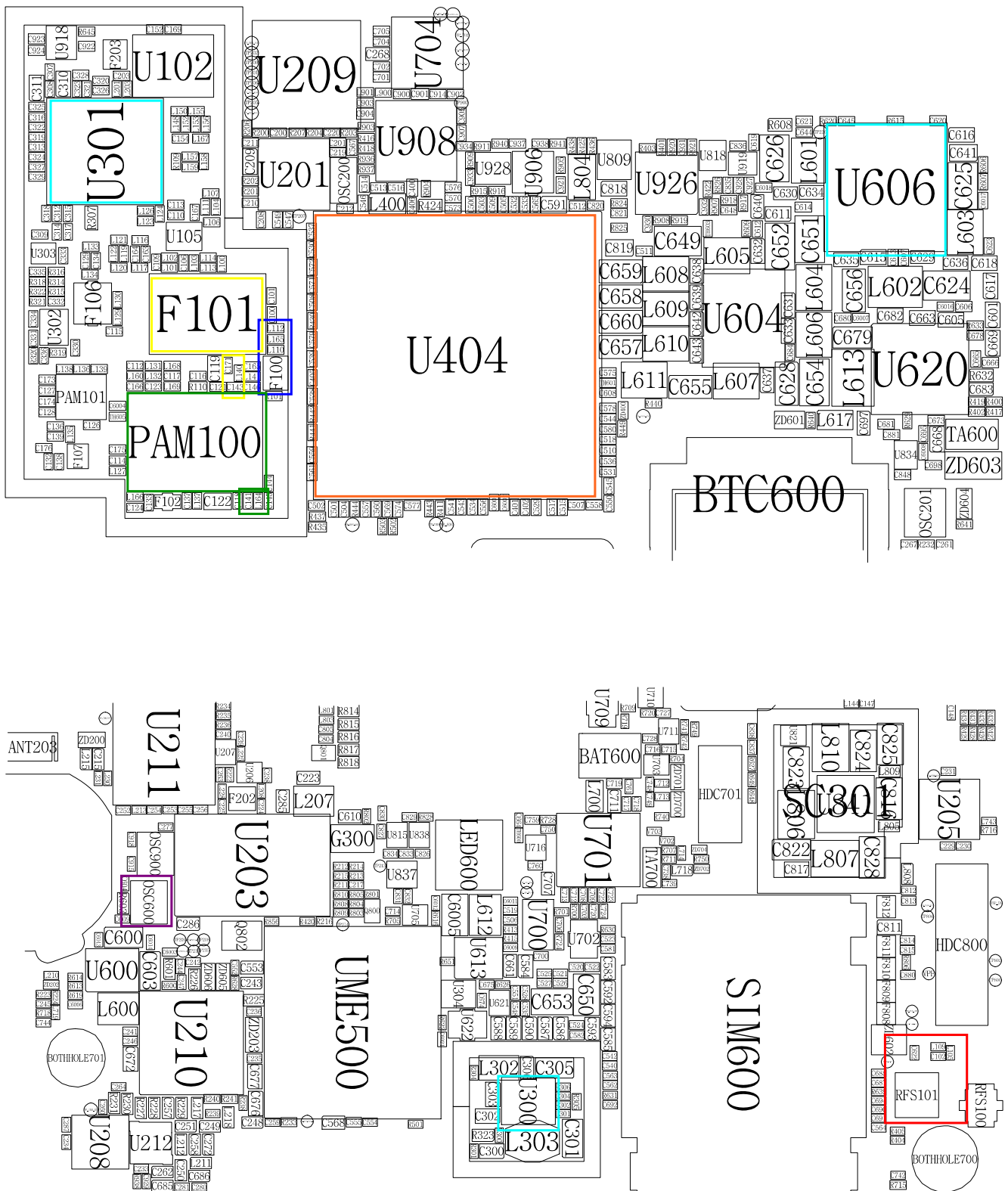


8-3-28.GSM1800/1900 发射器

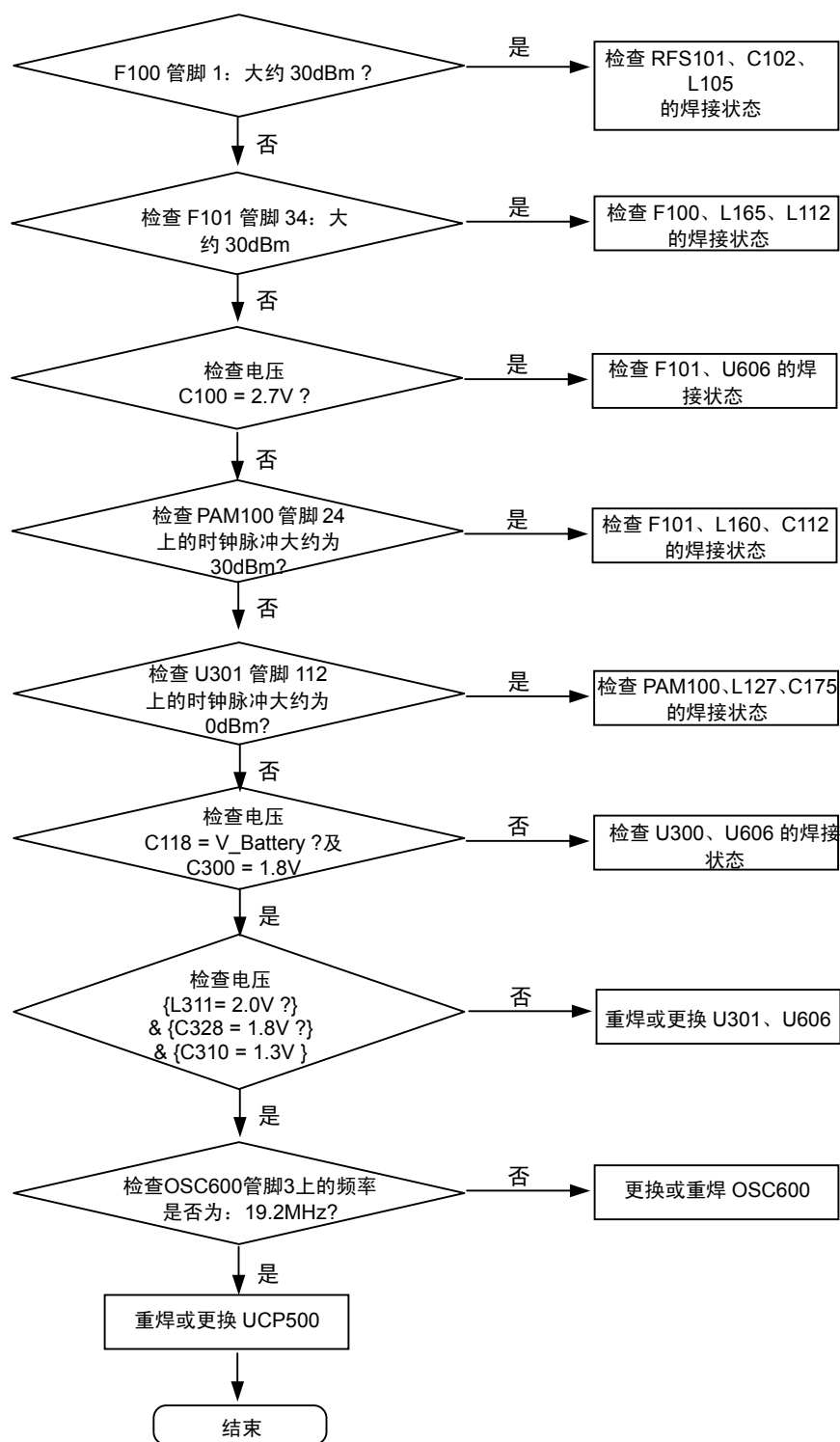


持续发射开通条件
 IDX1 发射功率: 应用代码
 GSM1800 CH: 698
 GSM1900 CH: 661
 RBW: 100KHz
 VBW: 100KHz
 SPAN: 10MHz
 REF LEV.: 10dBm
 ATT.: 20dB

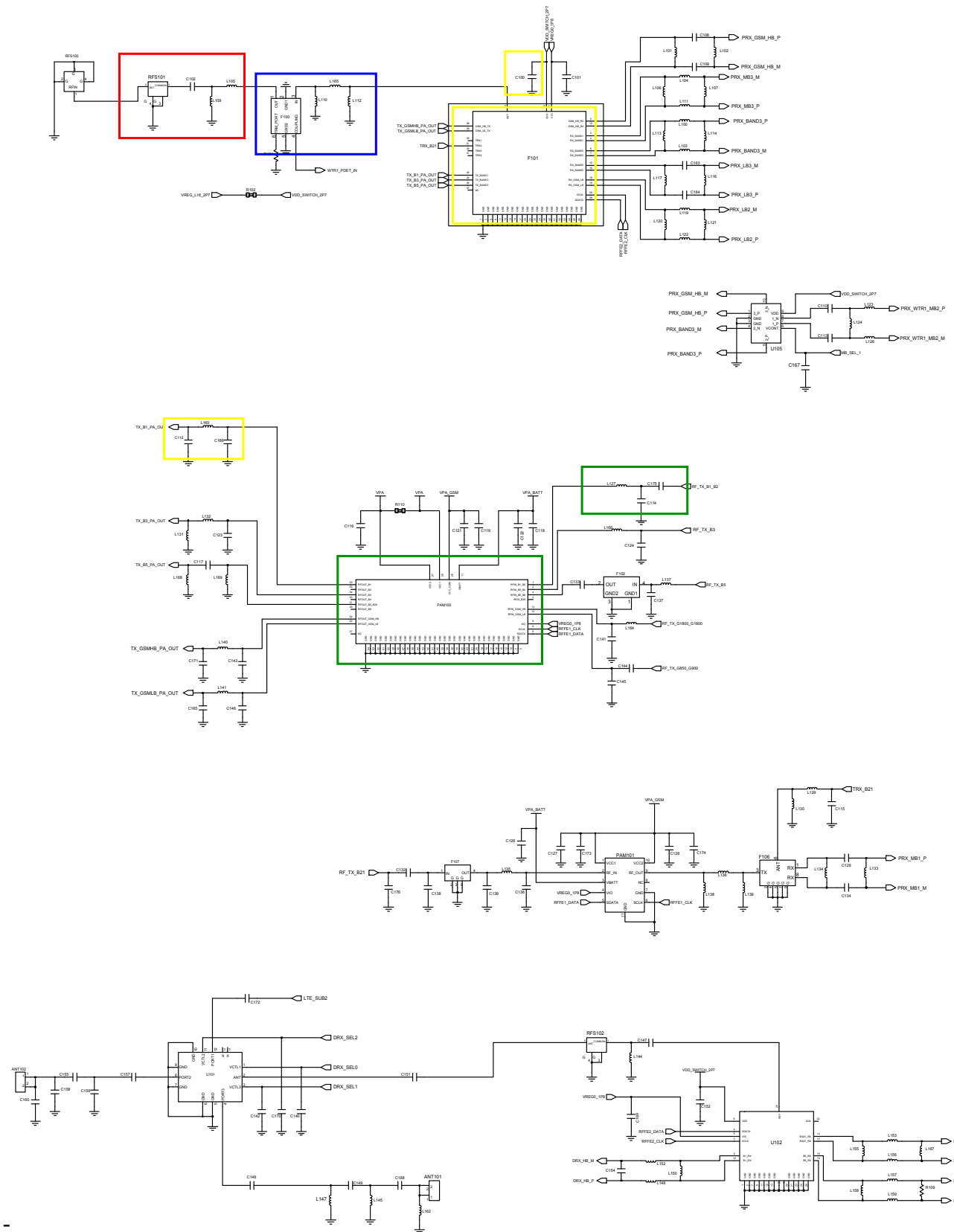


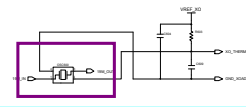
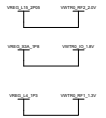


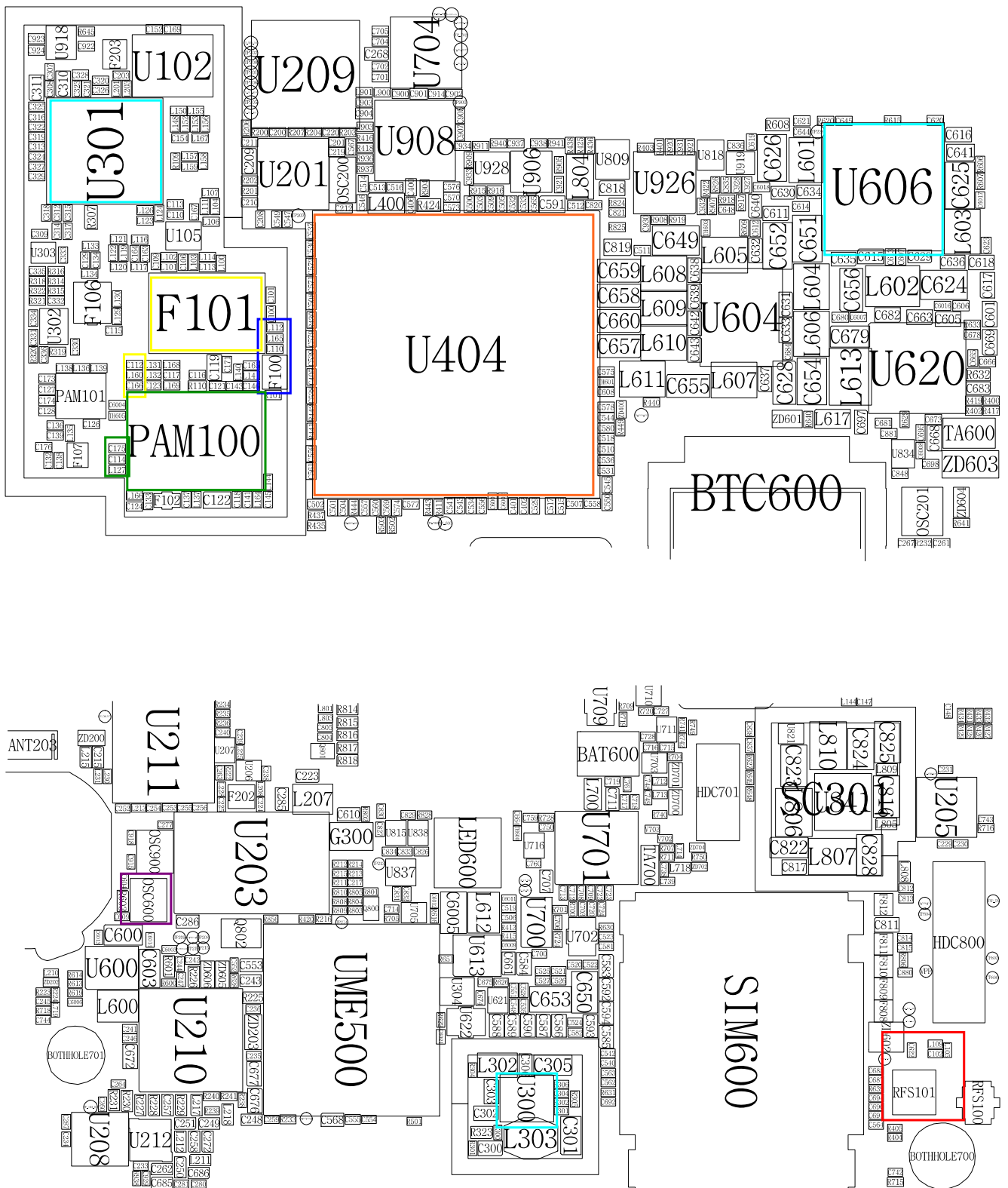
8-3-29.WCDMA 2100/LTE Band1 发射器



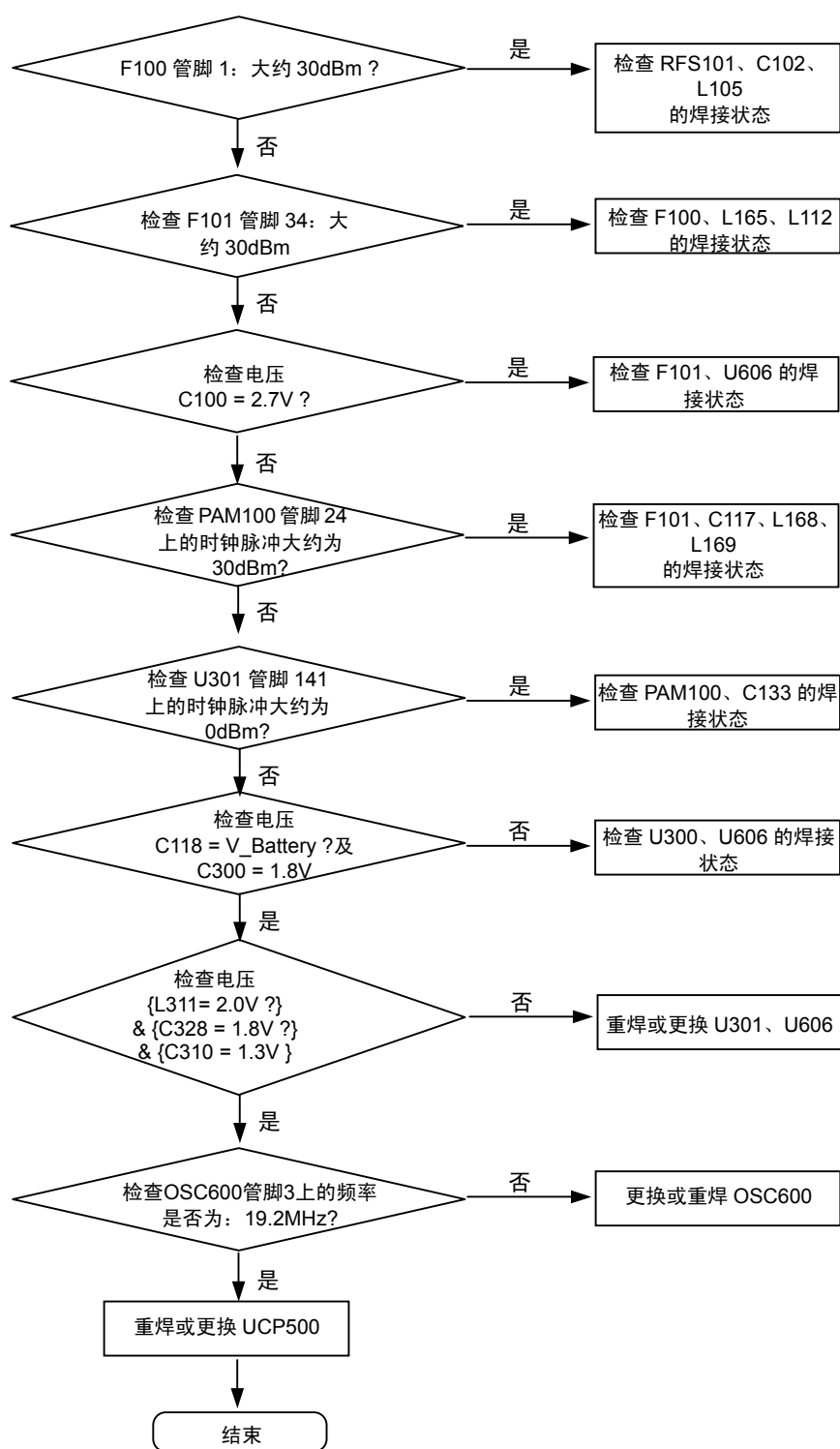
持续发射开通条件
 IDX1 发射功率: 应用代码
 WCDMA 2100 CH: 10700
 LTE Band1 CH: 300
 RBW: 100KHz
 VBW: 100KHz
 SPAN: 10MHz
 REF LEV.: 10dBm
 ATT.: 20dB



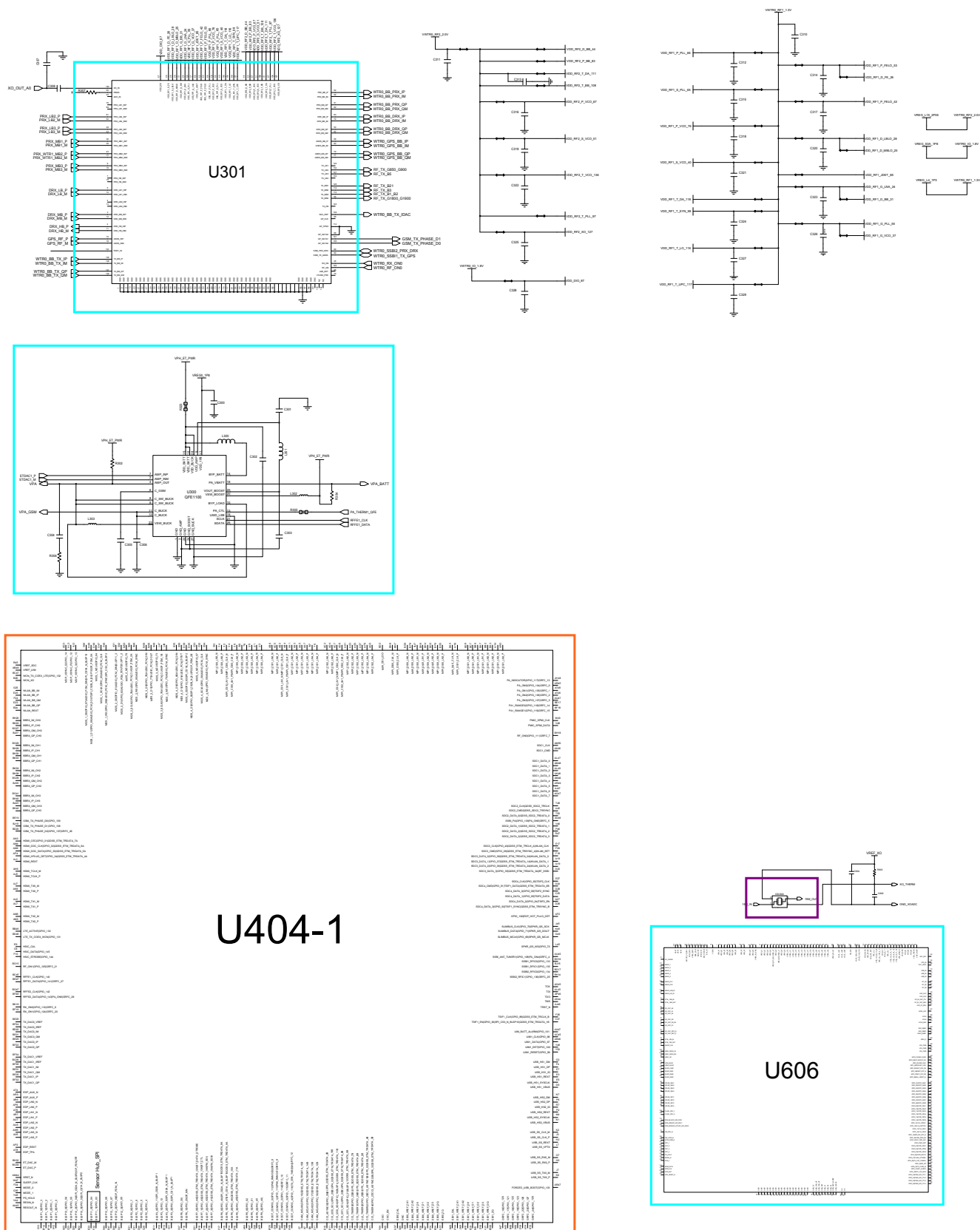


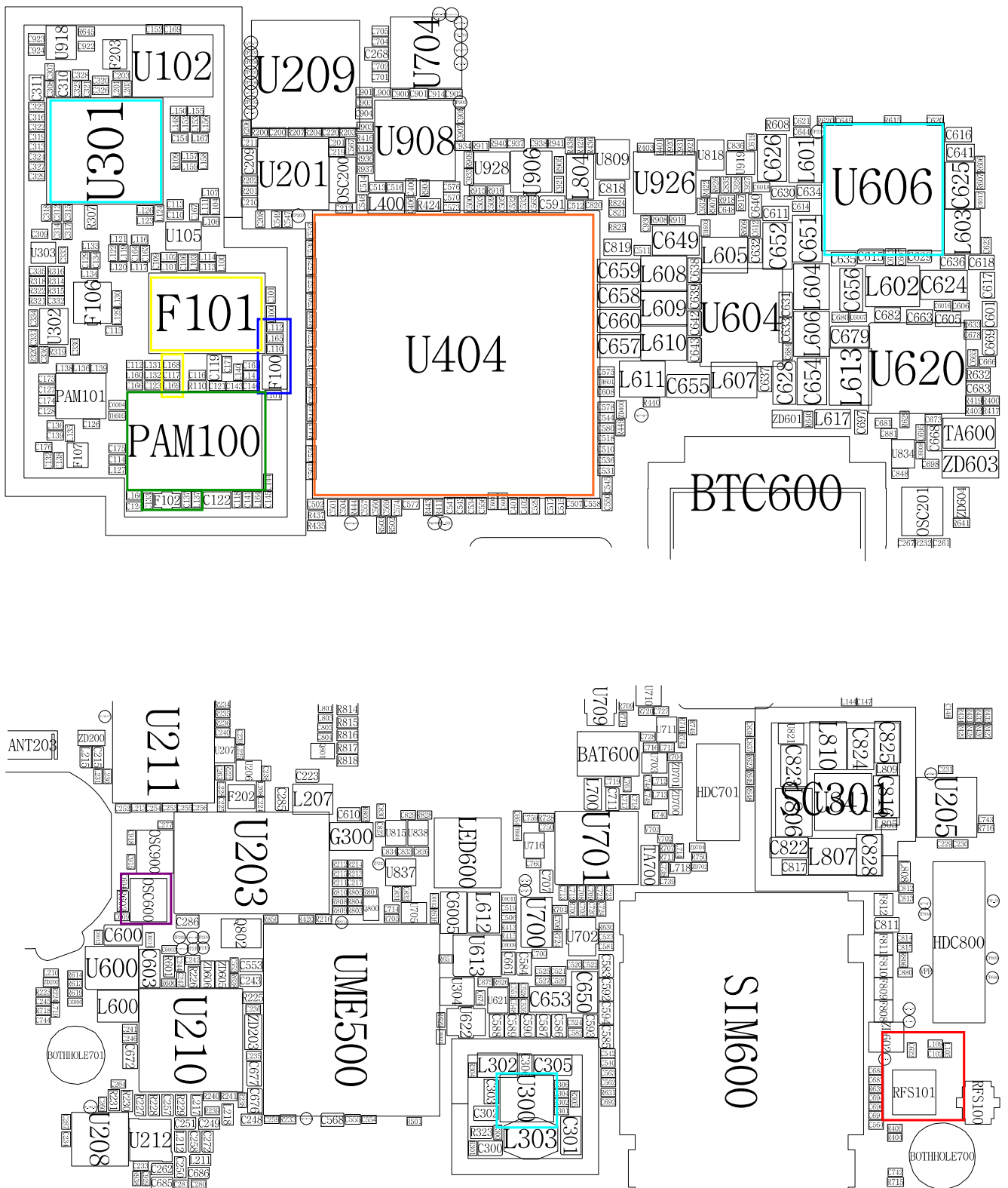


8-3-30.WCDMA800/850/NEW800/LTE Band19 发射器

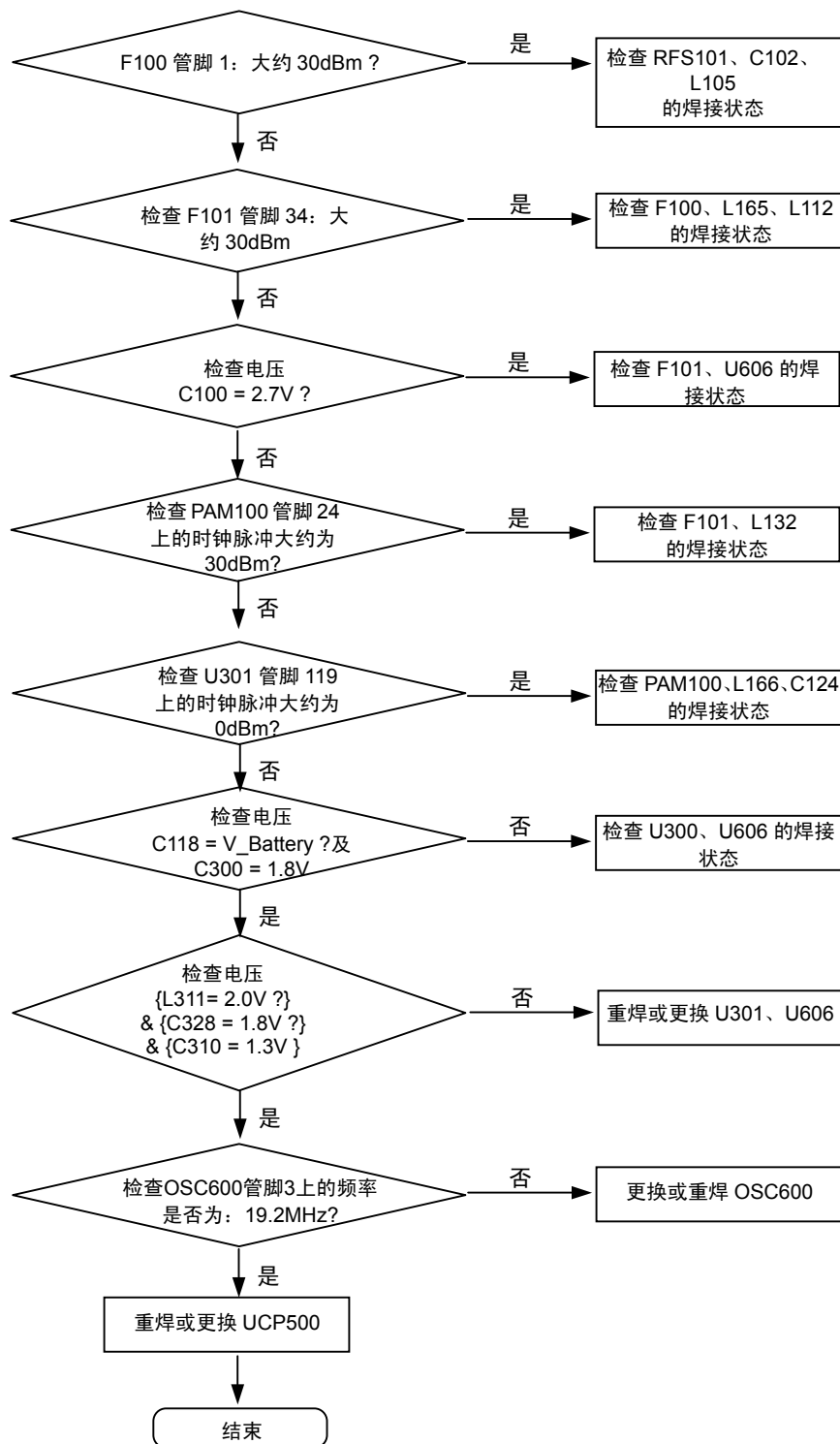


持续发射开通条件
 IDX1 发射功率: 应用代码
 WCDMA 800 CH: 4400
 LTE Band 19 CH: 6075
 RBW: 100KHz
 VBW: 100KHz
 SPAN: 10MHz
 REF LEV.: 10dBm
 ATT.: 20dB

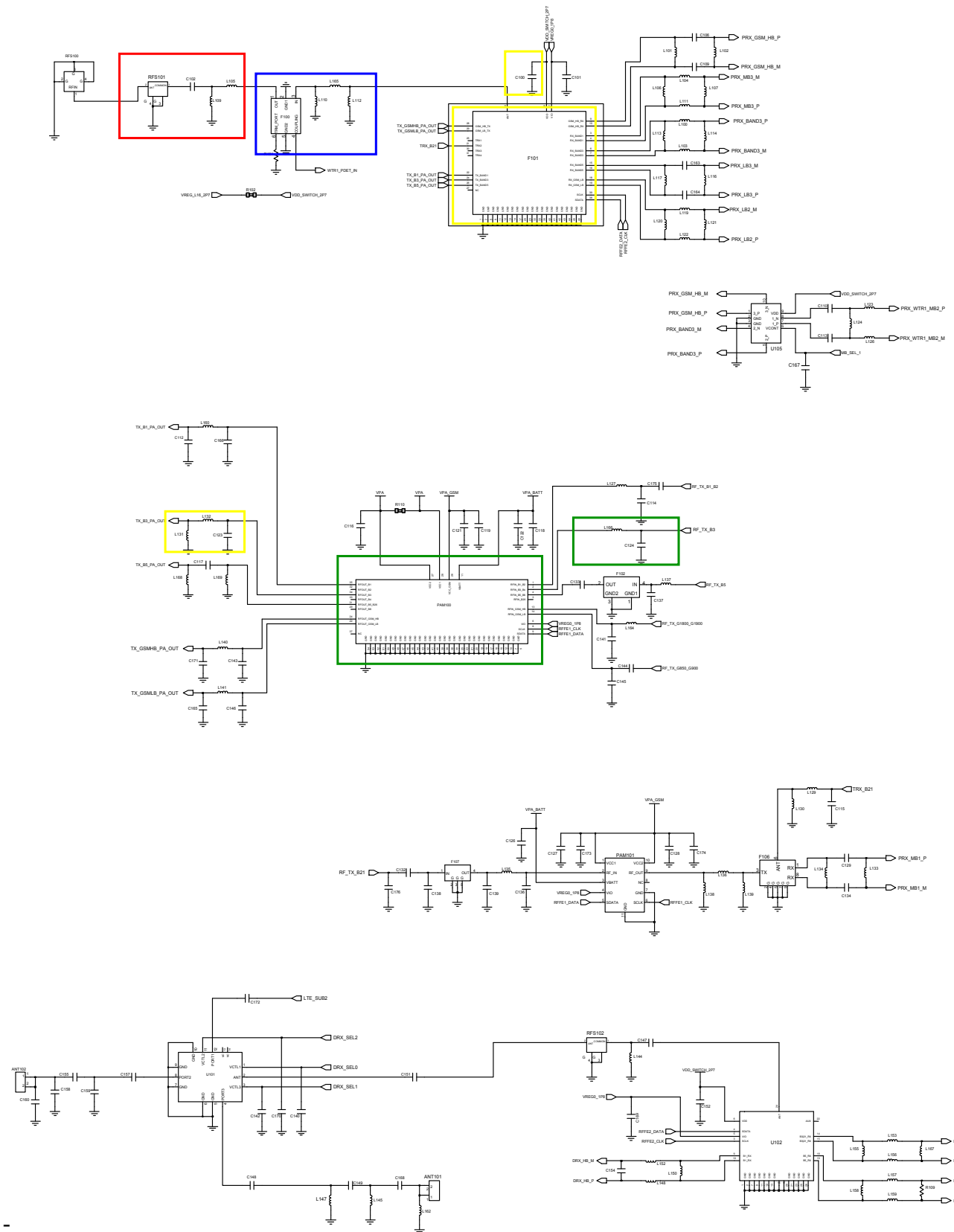


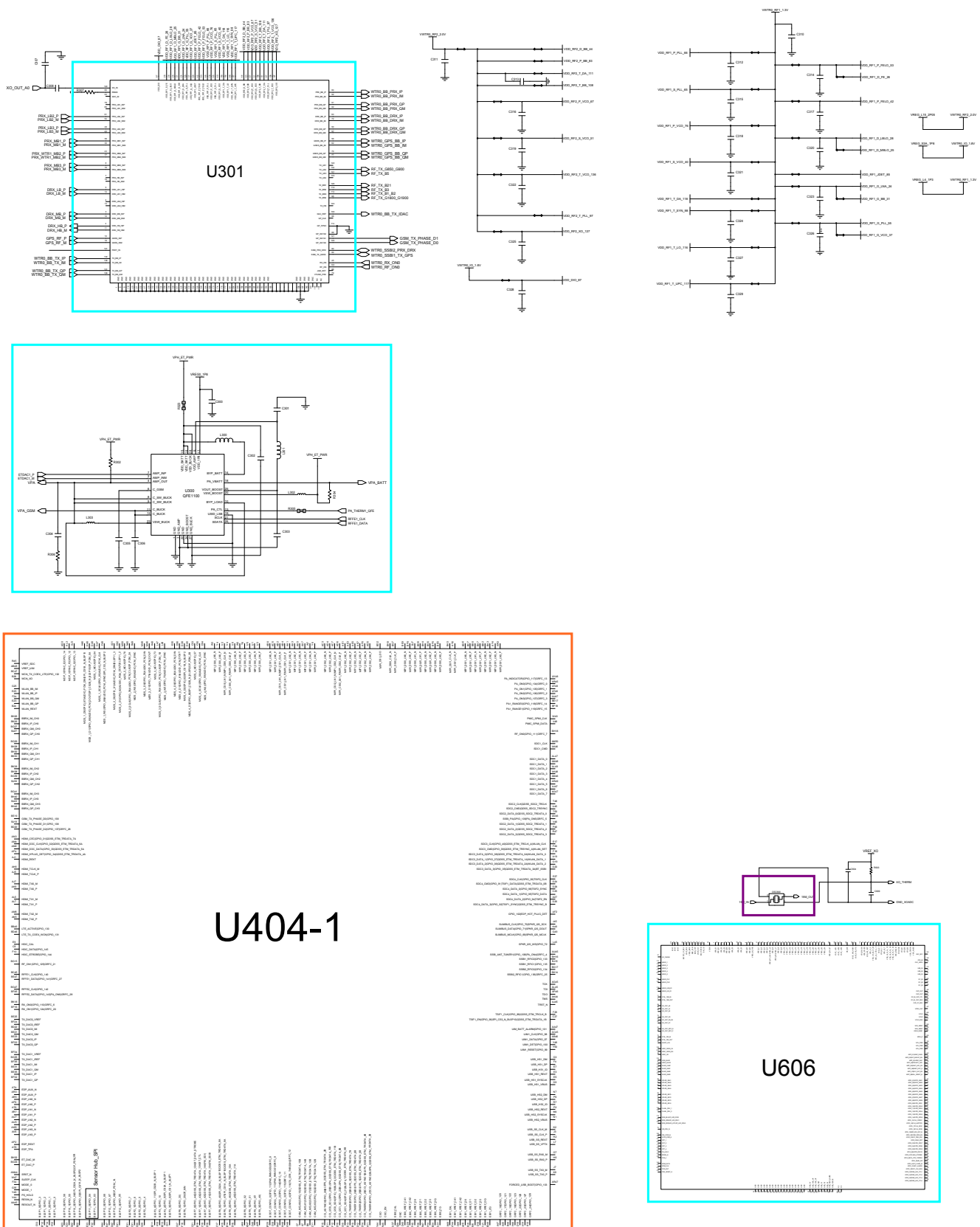


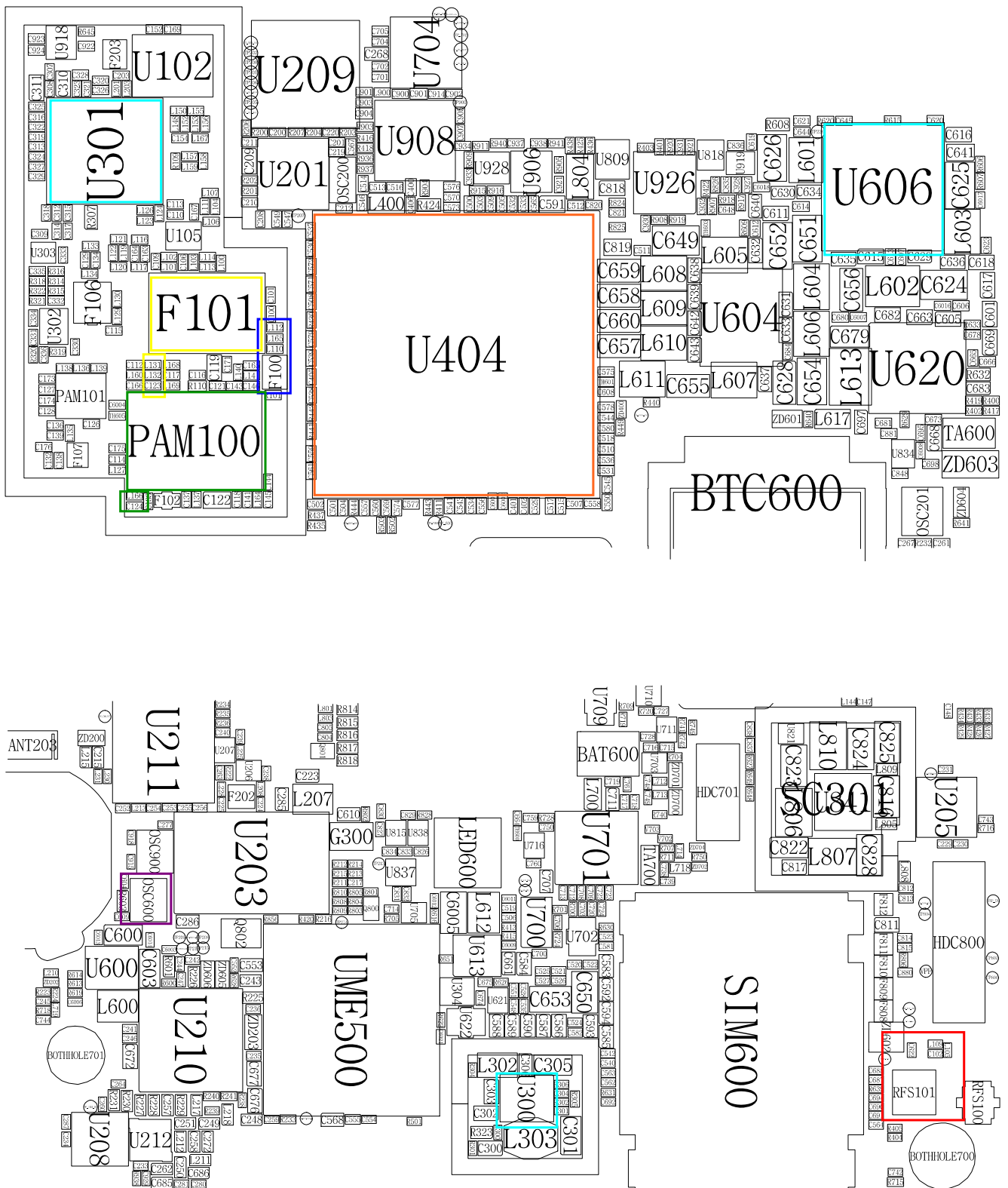
8-3-31.LTE Band3 发射器



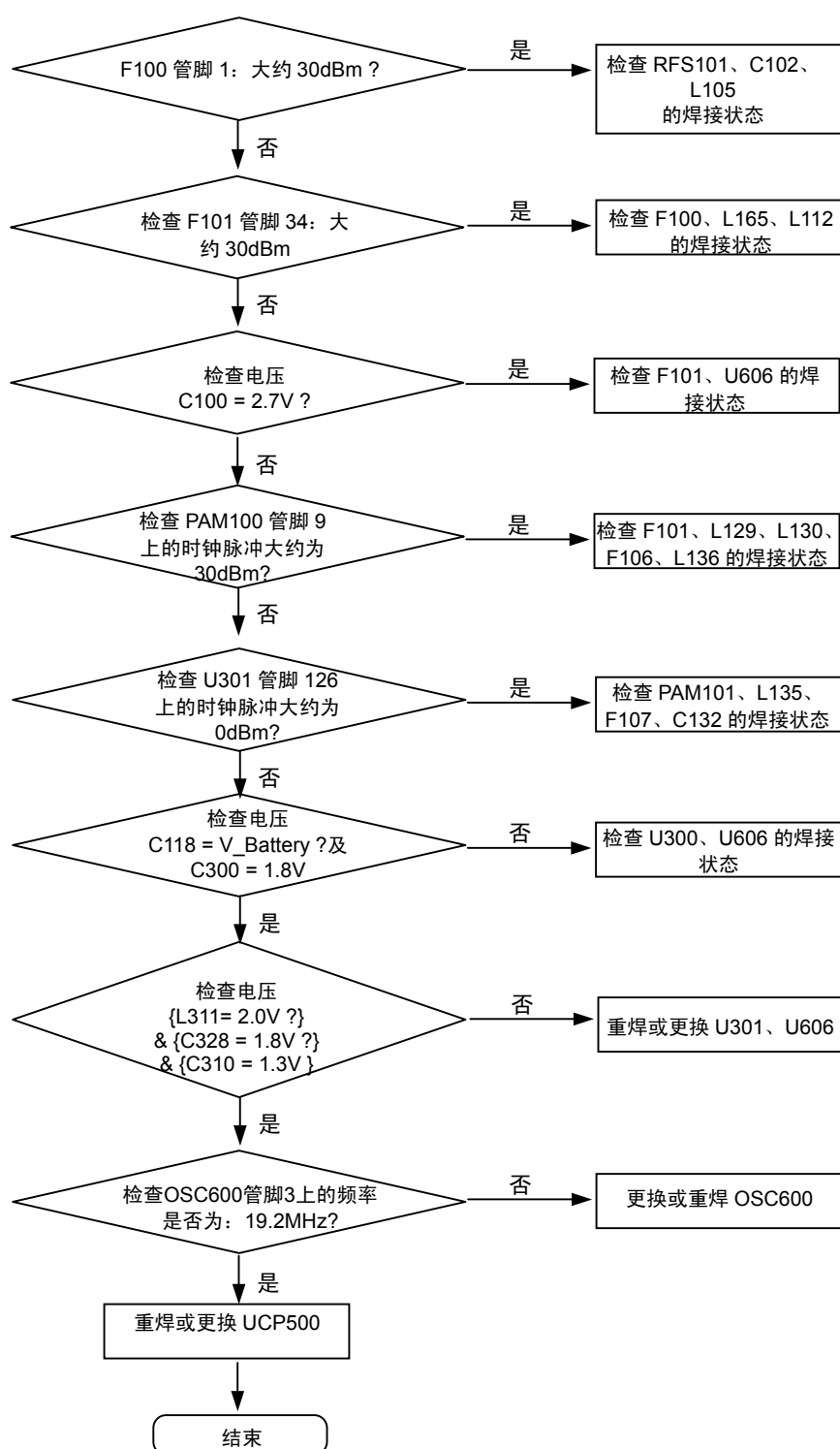
持续发射开通条件
 IDX1 发射功率: 应用代码
 LTE Band 3 CH: 1575
 RBW: 100KHz
 VBW: 100KHz
 SPAN: 10MHz
 REF LEV.: 10dBm
 ATT.: 20dB



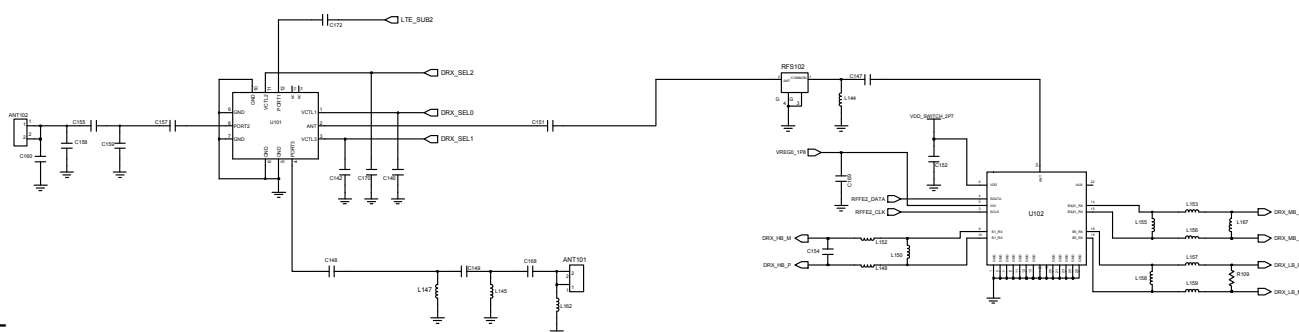
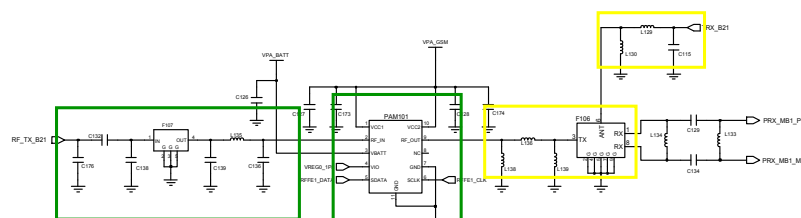
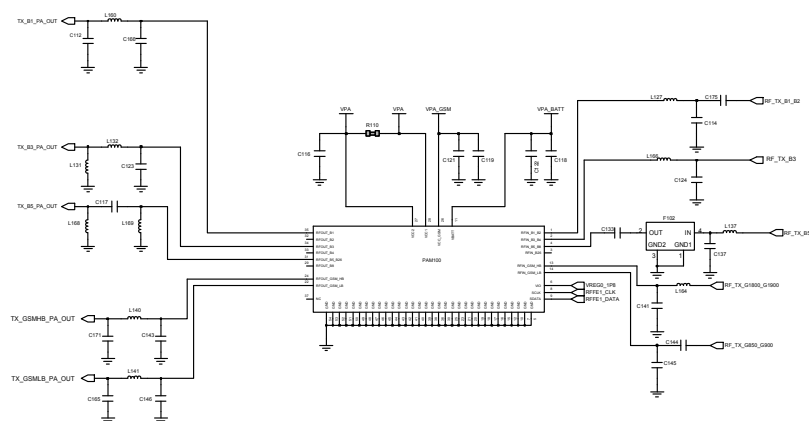


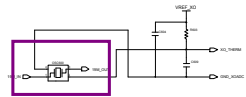
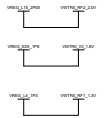


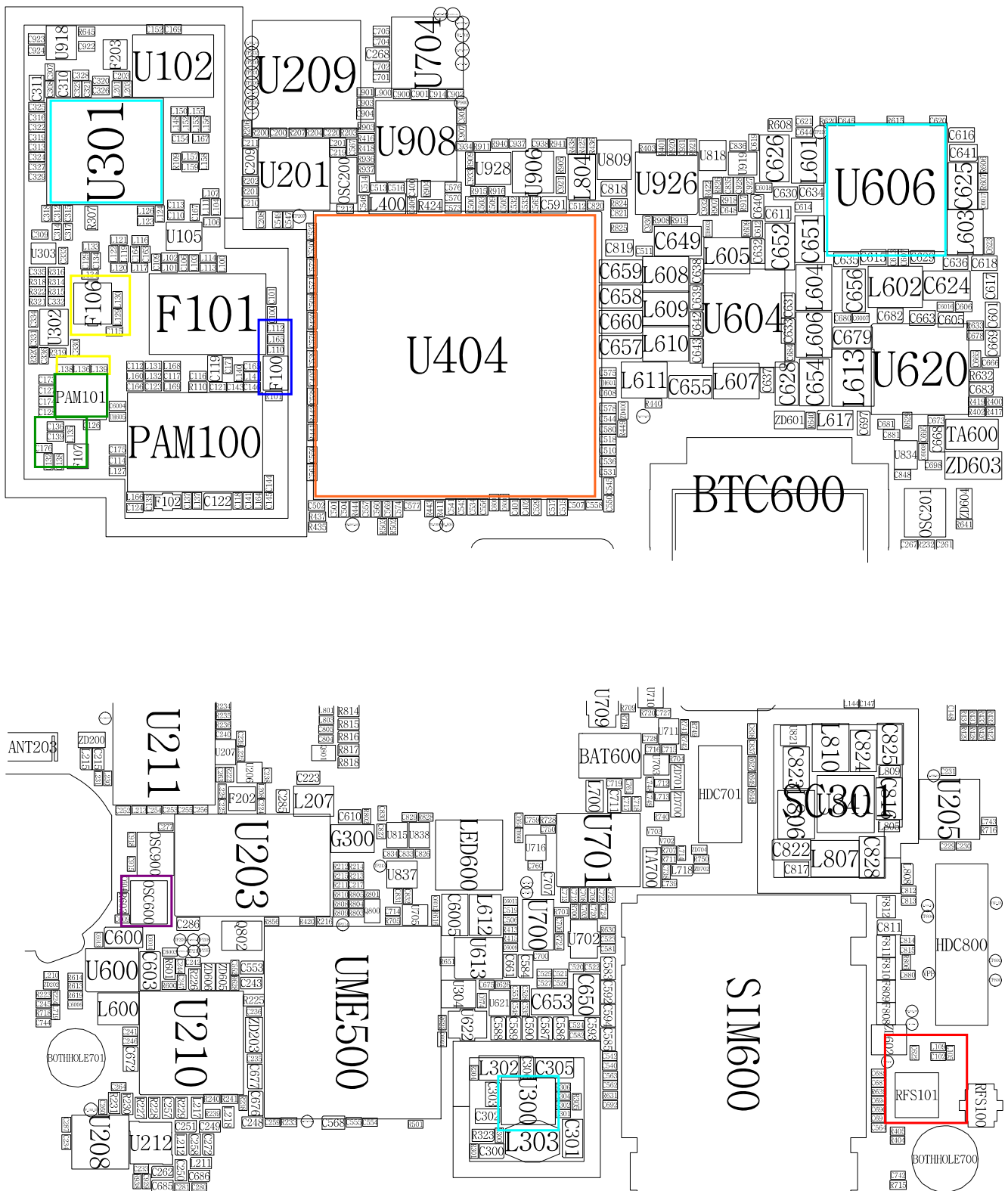
8-3-32.LTE Band21 发射器



持续发射开通条件
 IDX1 发射功率: 应用代码
 LTE Band21 CH: 6525
 RBW: 100KHz
 VBW: 100KHz
 SPAN: 10MHz
 REF LEV.: 10dBm
 ATT.: 20dB



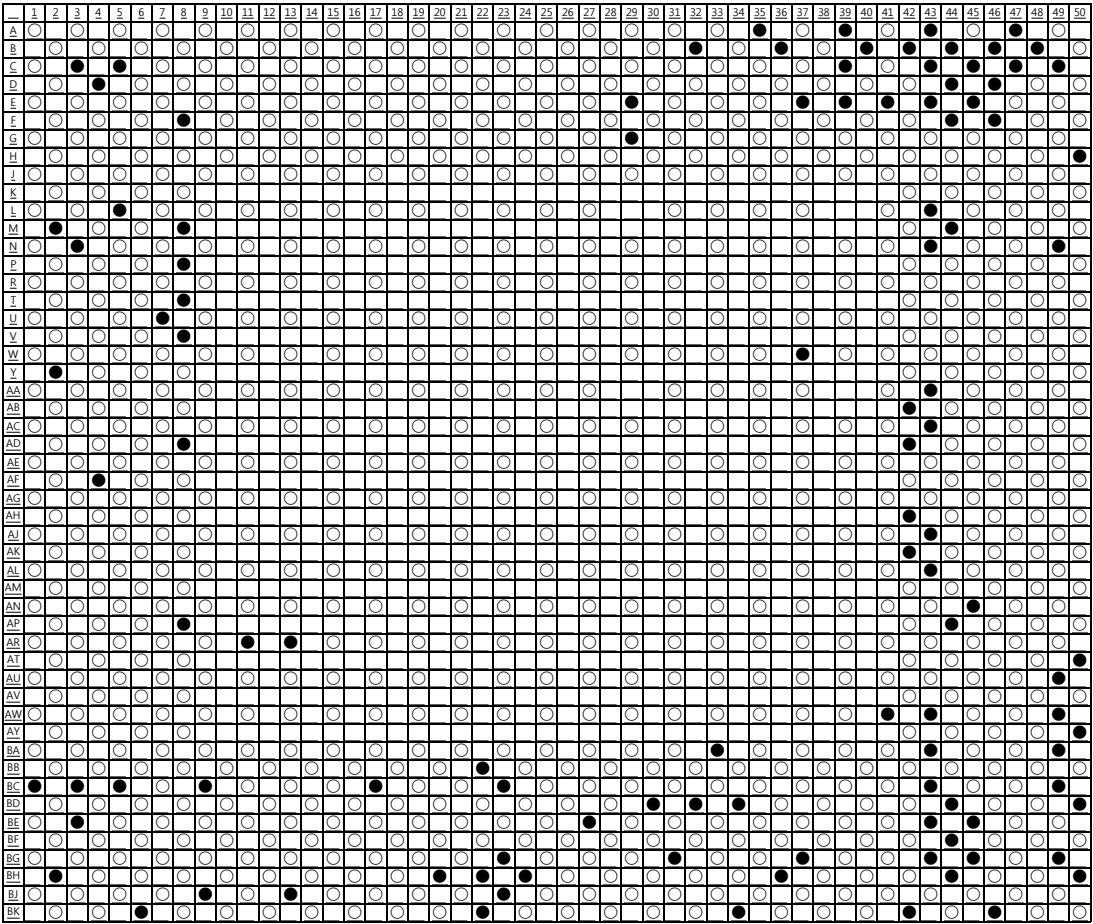




8-4.维修原理图

-故障处（顶部视图）

●:NC



UCP404